ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड
(भारत सरकार का उद्यम)
GRID CONTROLLER OF INDIA LIMITED
(A Government of India Enterprise)
[formerly Power System Operation Corporation Limited (POSOCO)]
राष्ट्रीय भार प्रेषण केन्द्र / National Load Despatch Centre
कार्यालय : बी-9, प्रथम एवं द्वितीय तल, कुतुब इंस्टीट्यूशनल एरिया, कटवारिया सराय, नई दिल्ली - 110016
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CIN : U40105DL2009GOI188682, Website : www.grid-india.in, E-mail : gridindiacc@grid-india.in, Tel.: 011- 42785855

संदर्भ संख्या: – GRID-INDIA/NLDC/MR/ दिनांक: 24.07.2024

सेवा में,

वितरण सूची के अनुसार

विषय:- निष्पादन रिपोर्ट – **जून 2024**

महोदय,

आई०ई०जी०सी०-2010 के प्रावधान के अनुसार, **जून 2024** माह की अखिल भारतीय प्रणाली की निष्पादन रिपोर्ट संलग्न है।

धन्यवाद

भवदीय

एस.सी. दंभारे
वरिष्ठ महाप्रबन्धक
प्रणाली प्रचालन, रा.भा.प्रे.कें.

संलग्नक: मासिक प्रणाली रिपोर्ट

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Executive Director, NRLDC,18/A,SJSS Marg, Katwaria Sarai, New Delhi-110016

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Executive Director, NERLDC, Dongtieh, Lower Nongrah, Laplang, Shillong-793006

GRID CONTROLLER OF INDIA LIMITED

Formerly Power System Operation Corporation Limited

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

NATIONAL LOAD DESPATCH CENTRE –राष्ट्रीय भार प्रेषण केन्द्र

मासिक ऑपरेशन रिपोर्ट

Monthly Operation Report



जून 2024

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड
राष्ट्रीय भार प्रेषण केंद्र, नई दिल्ली

GRID CONTROLLER OF INDIA LIMITED
Formerly Power System Operation Corporation Limited

NATIONAL LOAD DESPATCH CENTRE, NEW DELHI

ग्रिड-इंडिया
GRID-INDIA

माह जून 2024 के लिए प्रचालन निष्पादन रिपोर्ट

OPERATIONAL PERFORMANCE REPORT FOR THE MONTH OF JUNE -2024

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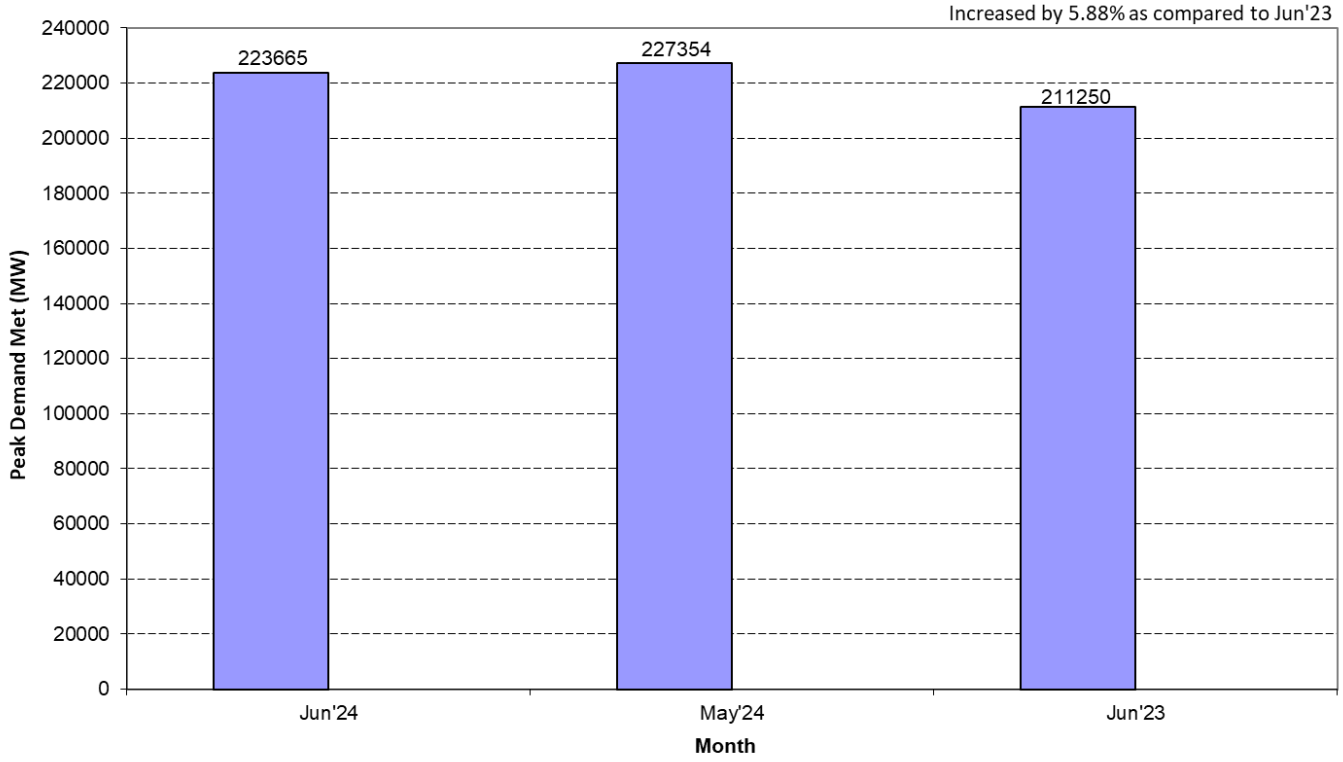
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1. SUMMARY OF REPORT FOR THE MONTH OF JUNE-2024

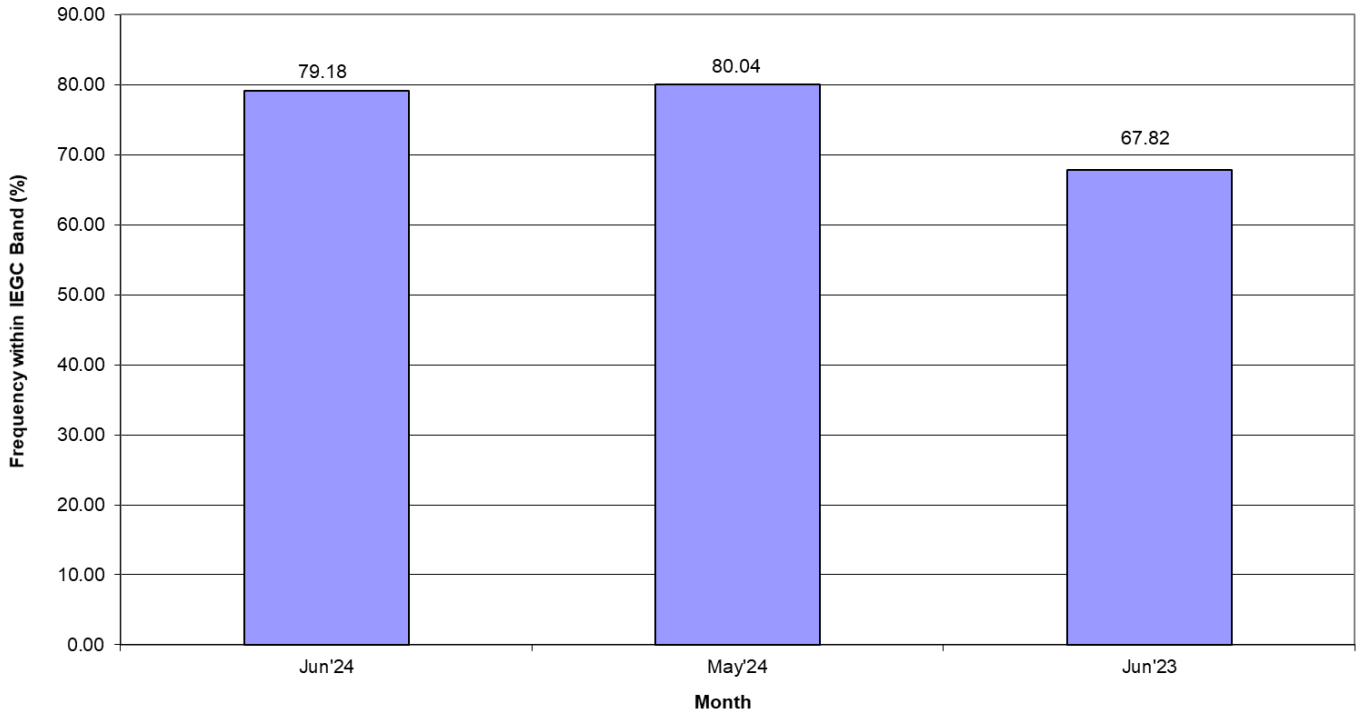
EVENING PEAK HOUR (at 2000hrs) DEMAND MET AT NATIONAL LEVEL (MW)

Decreased by 1.62% as compared to May'24

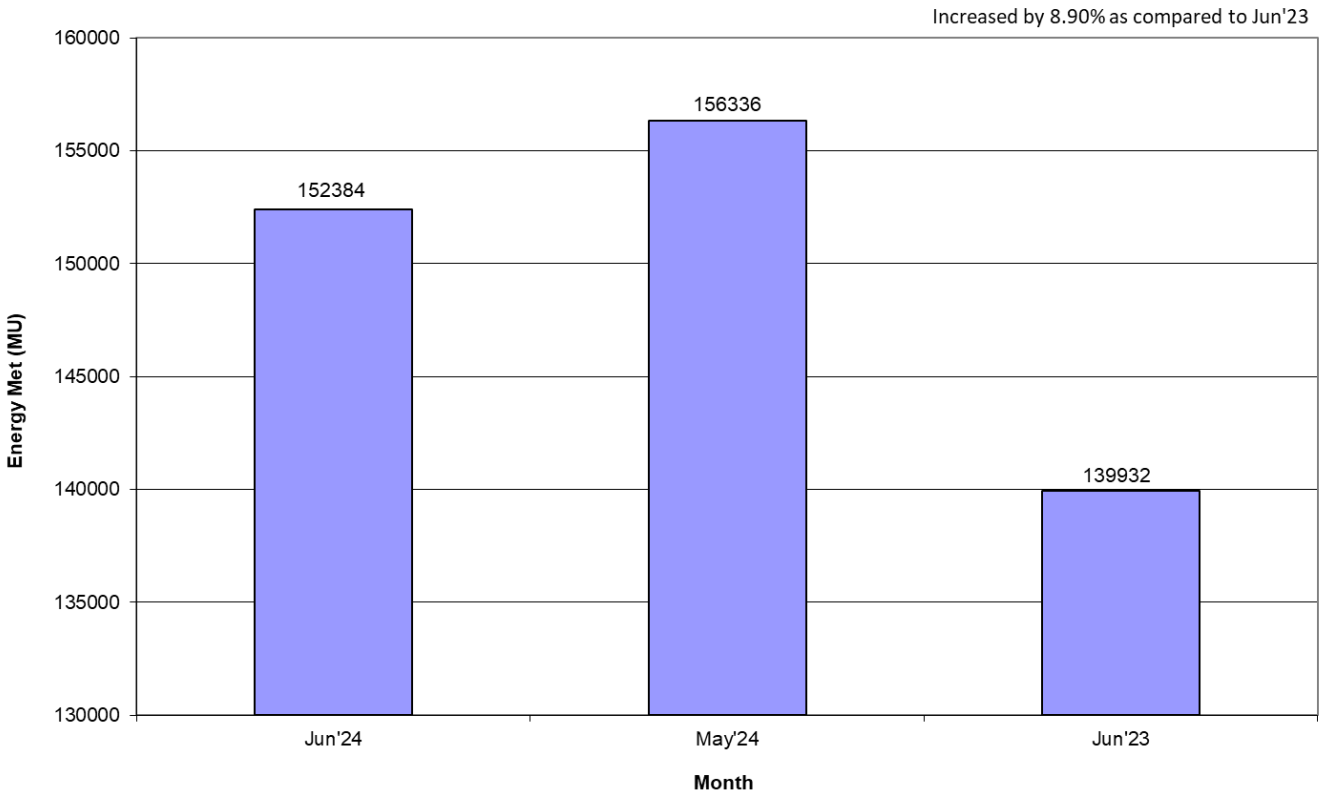
Increased by 5.88% as compared to Jun'23



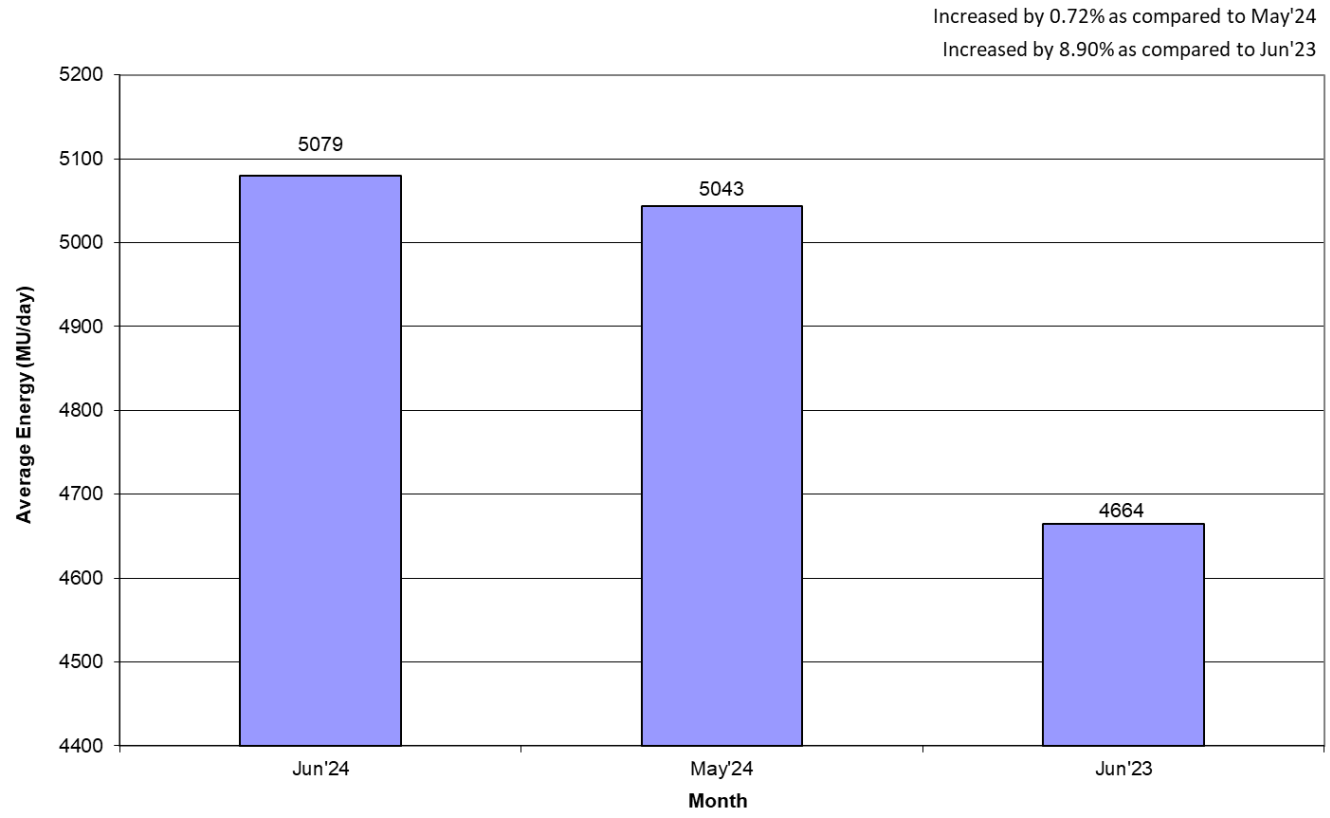
Frequency within IEGC Band



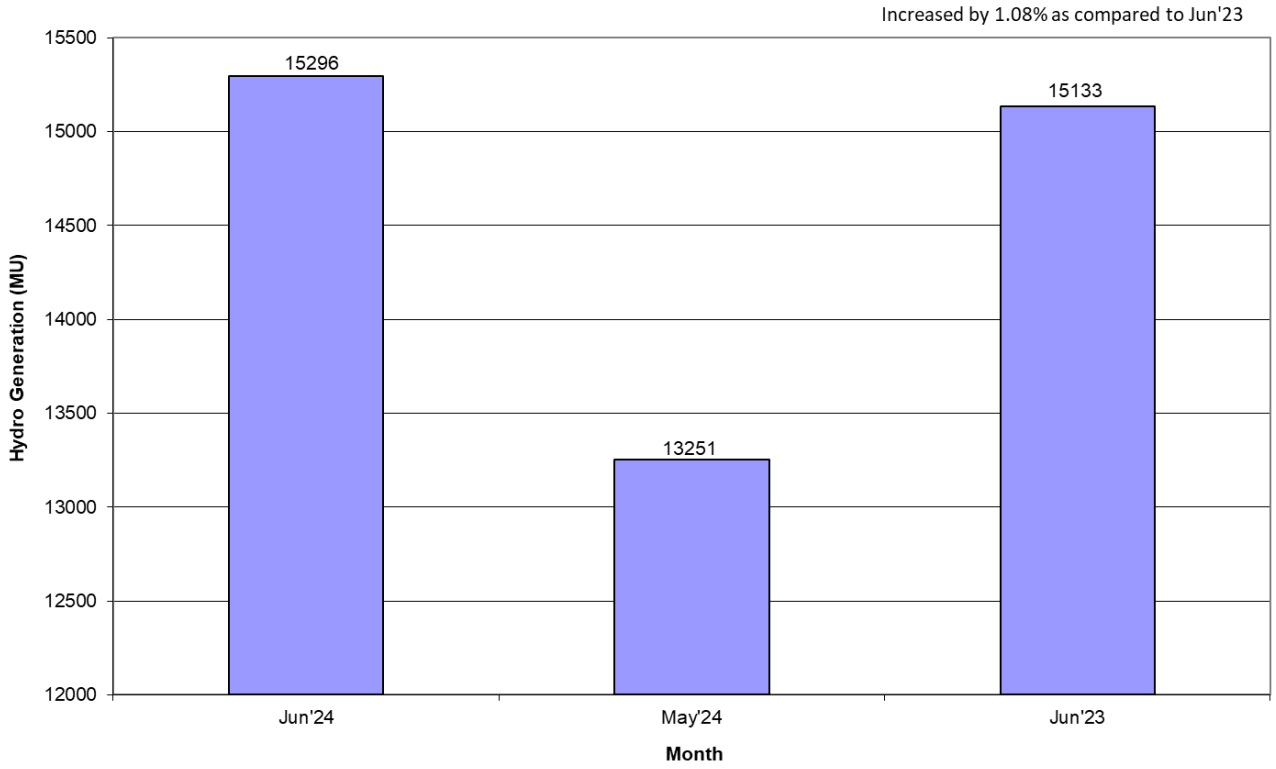
ENERGY MET AT NATIONAL LEVEL (MU)



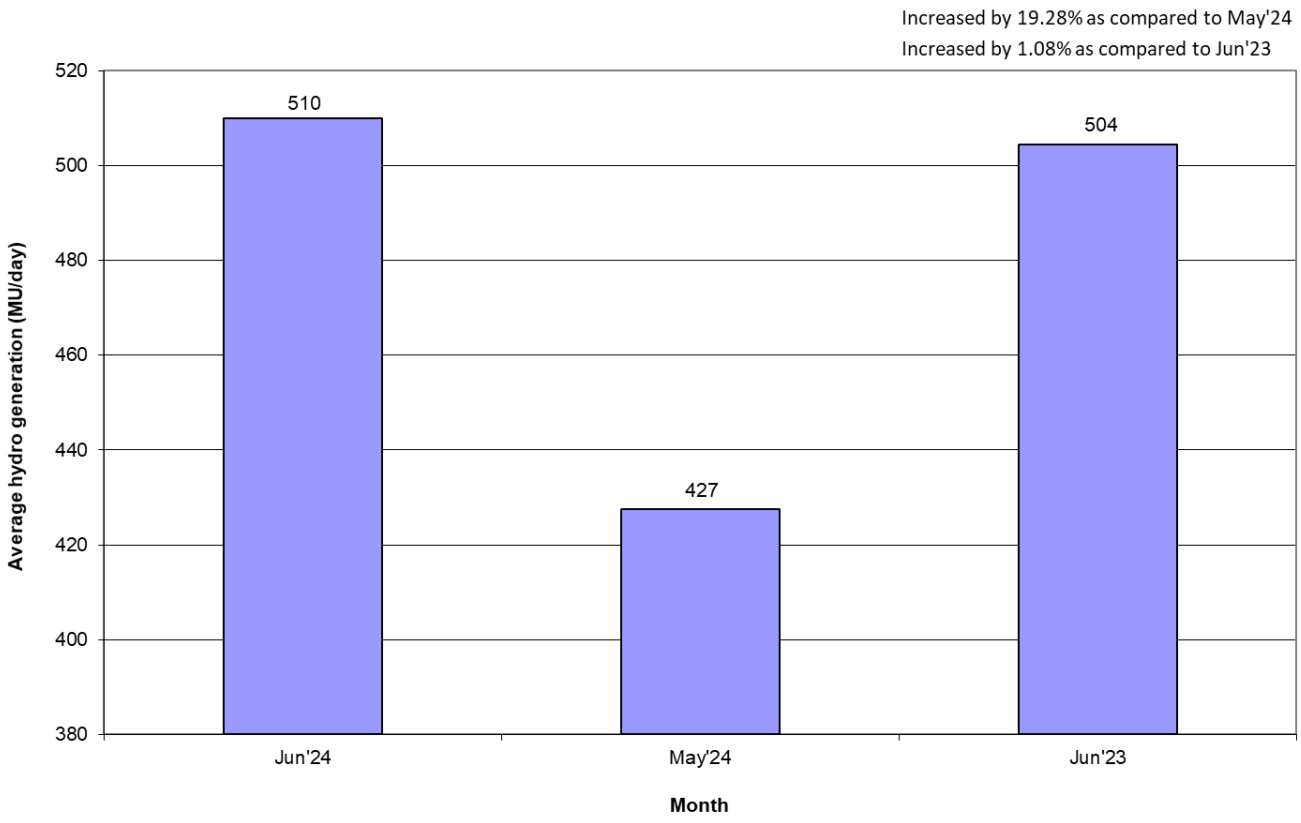
AVERAGE ENERGY MET AT NATIONAL LEVEL (MU/Day)



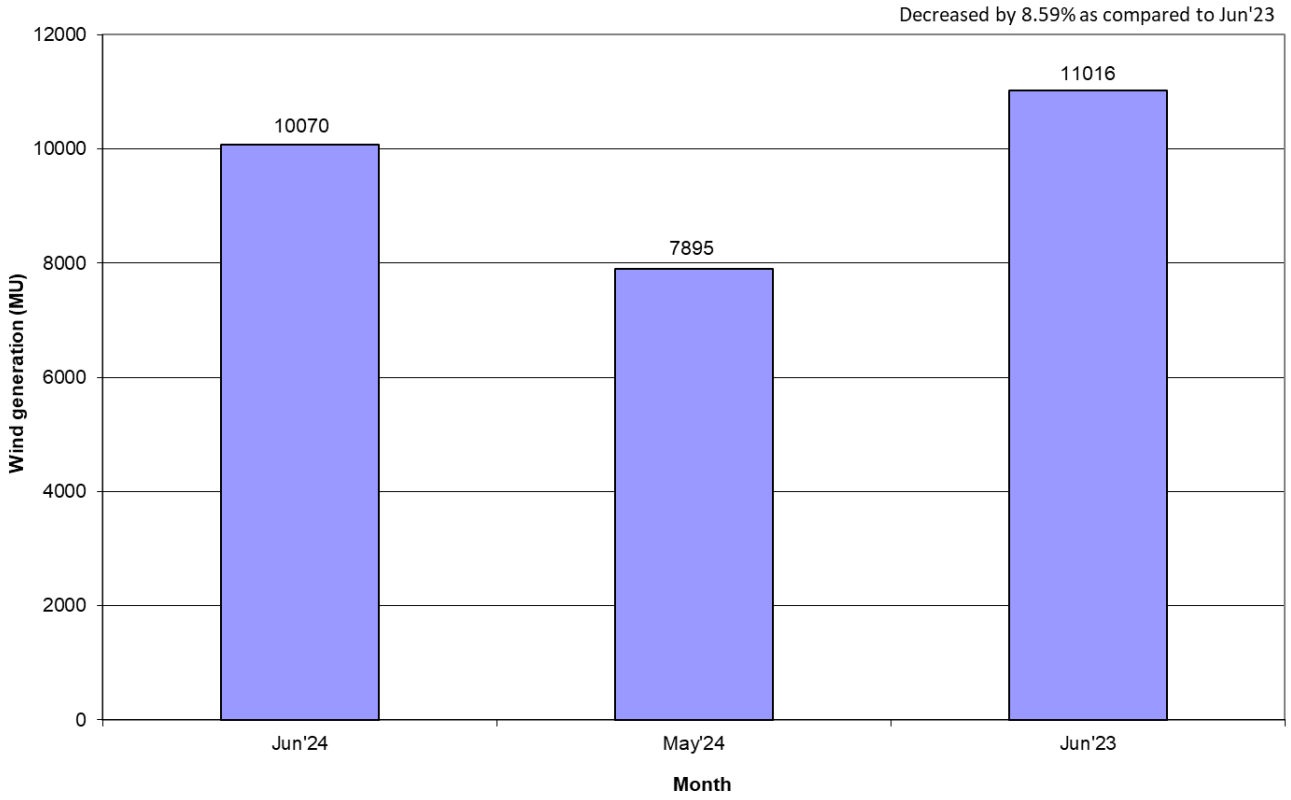
HYDRO GENERATION AT NATIONAL LEVEL (MU)



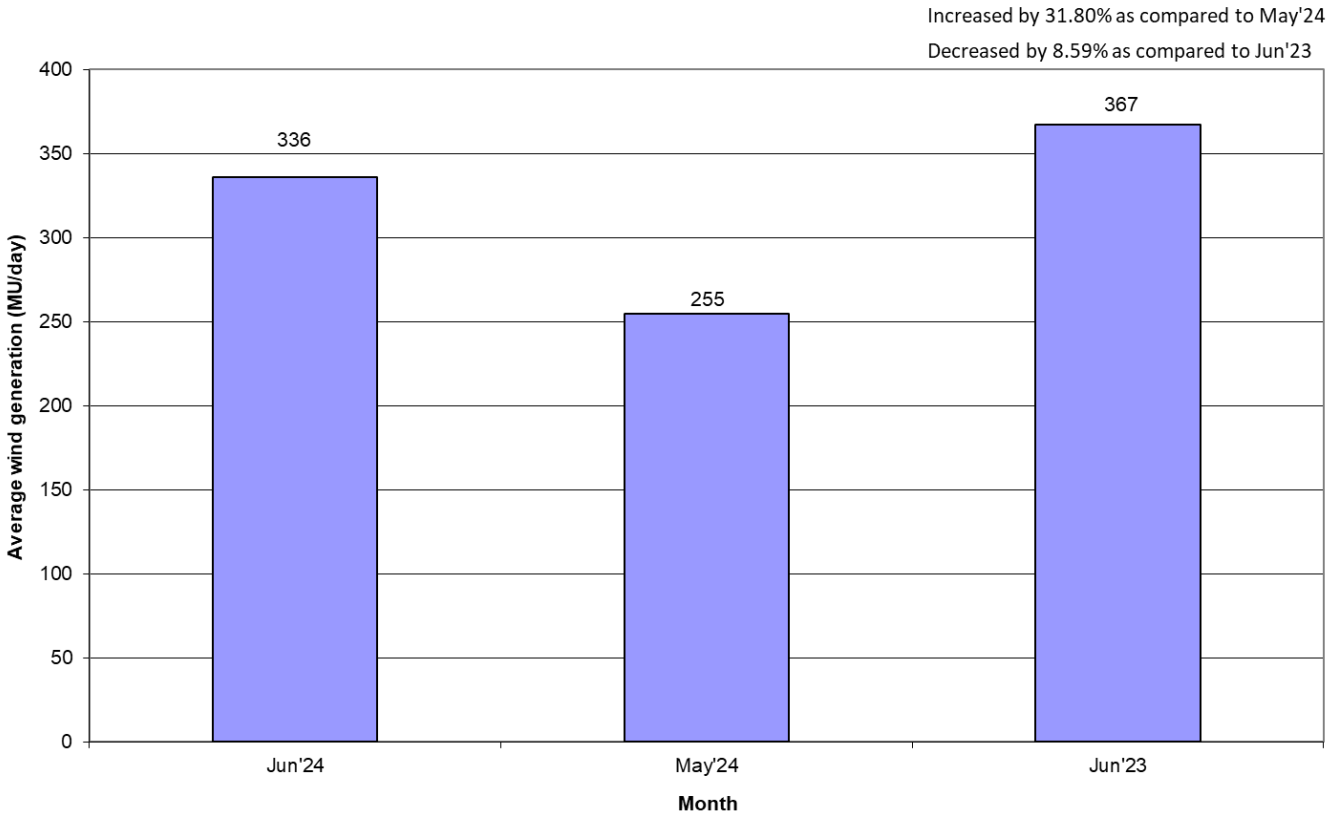
AVERAGE HYDRO GENERATION AT NATIONAL LEVEL (MU/Day)



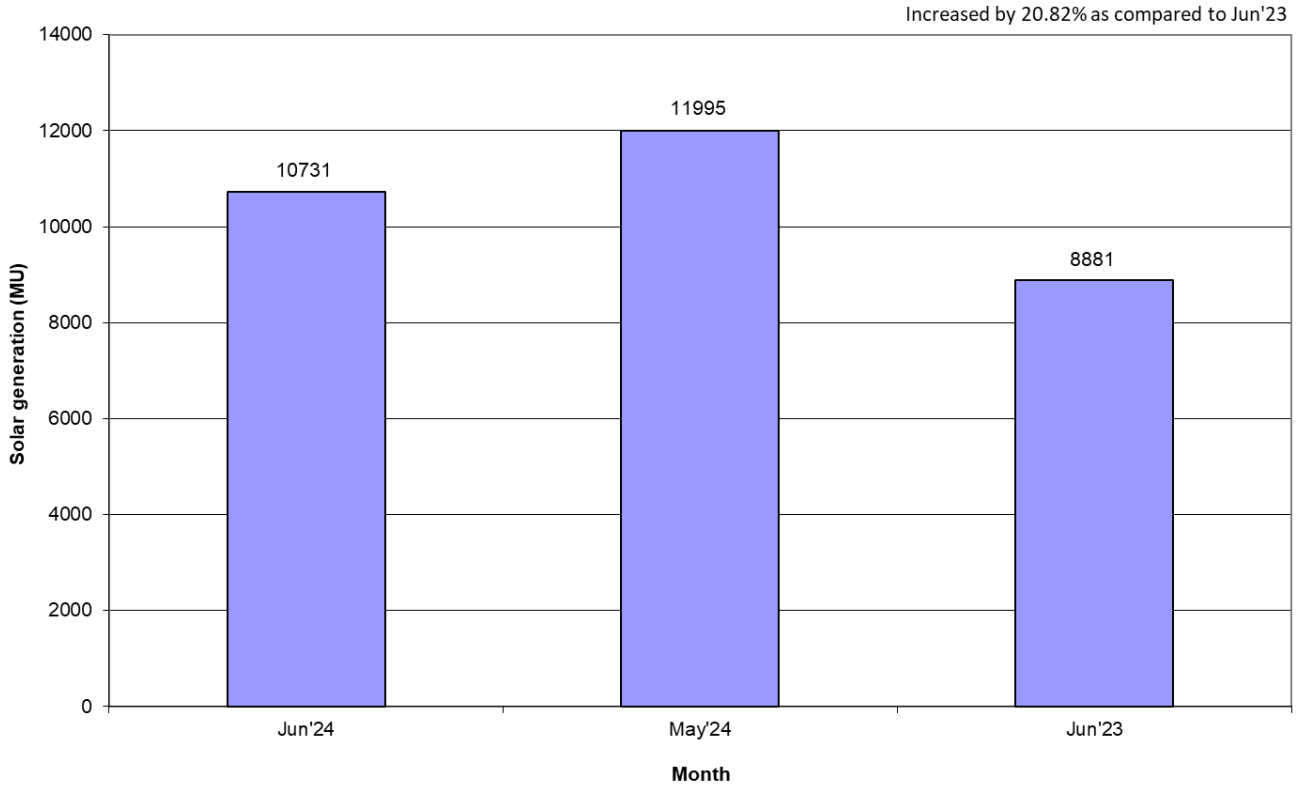
WIND GENERATION AT NATIONAL LEVEL (MU)



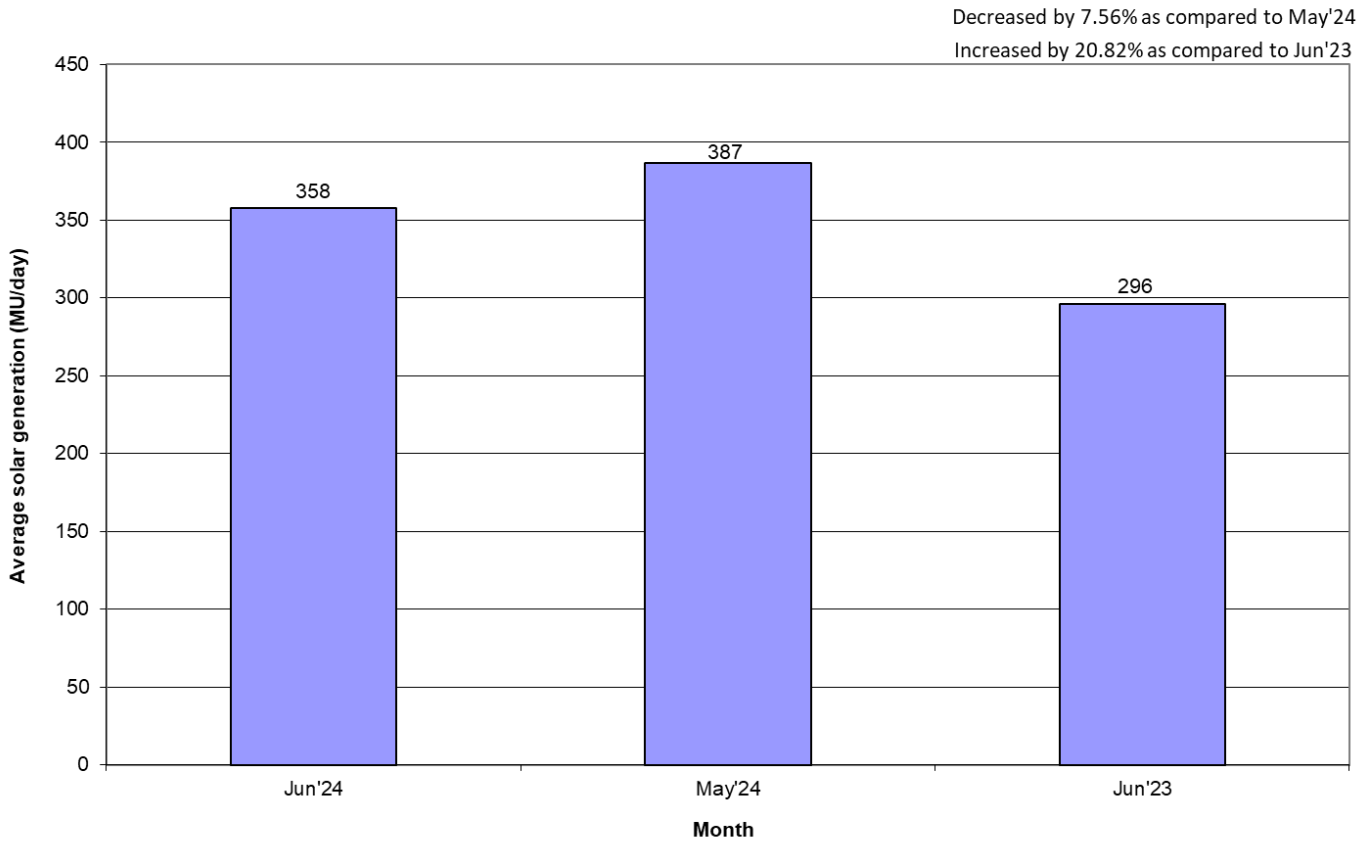
AVERAGE WIND GENERATION AT NATIONAL LEVEL (MU/Day)



SOLAR GENERATION AT NATIONAL LEVEL (MU)



AVERAGE SOLAR GENERATION AT NATIONAL LEVEL (MU/Day)



2. ALL INDIA REGIONWISE INSTALLED CAPACITY

As on 30.06.2024
(All figures are in MW)

S No	Region	THERMAL					NUCLEAR	HYDRO	RES @ MNRE	GRAND TOTAL
		COAL	LIGNITE	GAS	DIESEL	TOTAL				
1	NR	56658	1580	5995	0	64233	1620	20830	39941	126624
2	WR	74663	1400	10806	0	86869	3240	7563	50116	147788
3	SR	49717	3640	6272	434	60063	3320	11827	55363	130574
4	ER	28690	0	80	0	28770	0	4764	2026	35560
5	NER	1242	0	1665	36	2943	0	1944	598	5485
6	ISLANDS	0	0	0	120	120	0	0	40	160
	ALL INDIA	210970	6620	24818	589	242997	8180	46928	148085	446190

Remarks: Data from CEA website

3. राष्ट्रीय स्तर पर संध्याकालीन शिखर विद्युत मांग पूर्ति

EVENING PEAK HOUR DEMAND MET AT NATIONAL LEVEL

माह: जून 2024 MONTH:- JUN 2024

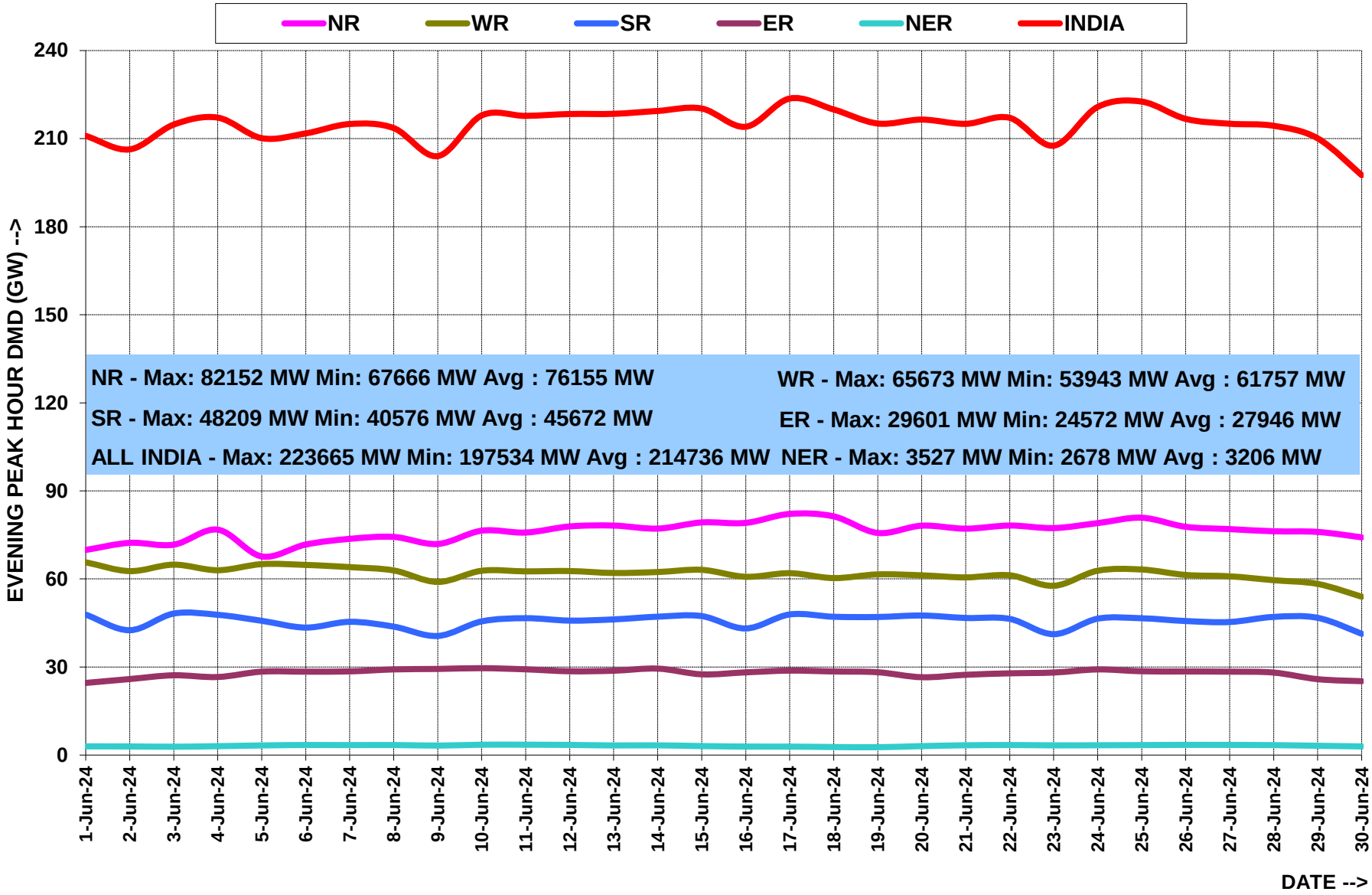
सभी आंकड़े मेगावाट मे All figures in MW

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
1-Jun-24	69873	65673	47847	24572	2968	210933
2-Jun-24	72296	62670	42531	25898	2935	206330
3-Jun-24	71670	64865	48209	27181	2848	214773
4-Jun-24	76790	62968	47782	26591	3025	217156
5-Jun-24	67666	65044	45740	28423	3277	210150
6-Jun-24	71717	64777	43436	28415	3432	211777
7-Jun-24	73657	64023	45395	28505	3397	214977
8-Jun-24	74318	62896	43756	29186	3425	213581
9-Jun-24	71907	58982	40576	29340	3232	204037
10-Jun-24	76444	62792	45544	29601	3525	217906
11-Jun-24	75806	62579	46632	29207	3527	217751
12-Jun-24	77893	62685	45804	28545	3459	218386
13-Jun-24	78180	62036	46220	28733	3275	218444
14-Jun-24	77157	62351	47145	29456	3316	219425
15-Jun-24	79252	63097	47341	27509	3065	220264
16-Jun-24	79099	60740	43127	28182	2894	214042
17-Jun-24	82152	61955	47902	28772	2884	223665
18-Jun-24	81306	60291	47113	28475	2737	219922
19-Jun-24	75665	61552	47040	28217	2678	215152
20-Jun-24	78143	61229	47541	26519	3049	216481
21-Jun-24	77127	60526	46704	27351	3331	215039
22-Jun-24	78192	61232	46393	27843	3422	217082
23-Jun-24	77328	57660	41181	28120	3307	207596
24-Jun-24	79015	62805	46462	29175	3342	220799
25-Jun-24	80850	63212	46601	28544	3405	222612
26-Jun-24	77756	61357	45659	28479	3476	216727
27-Jun-24	76974	60896	45327	28413	3460	215070
28-Jun-24	76236	59572	47077	28106	3380	214371
29-Jun-24	76015	58305	46753	25851	3164	210088
30-Jun-24	74167	53943	41312	25171	2941	197534
उच्चतम MAXIMUM	82152	65673	48209	29601	3527	223665
निम्नतम MINIMUM	67666	53943	40576	24572	2678	197534
औसत AVERAGE	76155	61757	45672	27946	3206	214736
अब तक का उच्चतम All Time Max.	**82152**	67482	54798	29695	3544	227354
दिनांक Date	**17.06.24**	30.05.24	29.04.24	29.05.24	24.05.24	29.05.24

नोट : सभी आंकड़े 2000 बजे की मांगपूर्ति पर आधारित है ।

Note: The above figures correspond to value at 2000 Hrs

EVENING PEAK HOUR DEMAND (2000hrs) MET DURING THE MONTH OF JUNE 2024



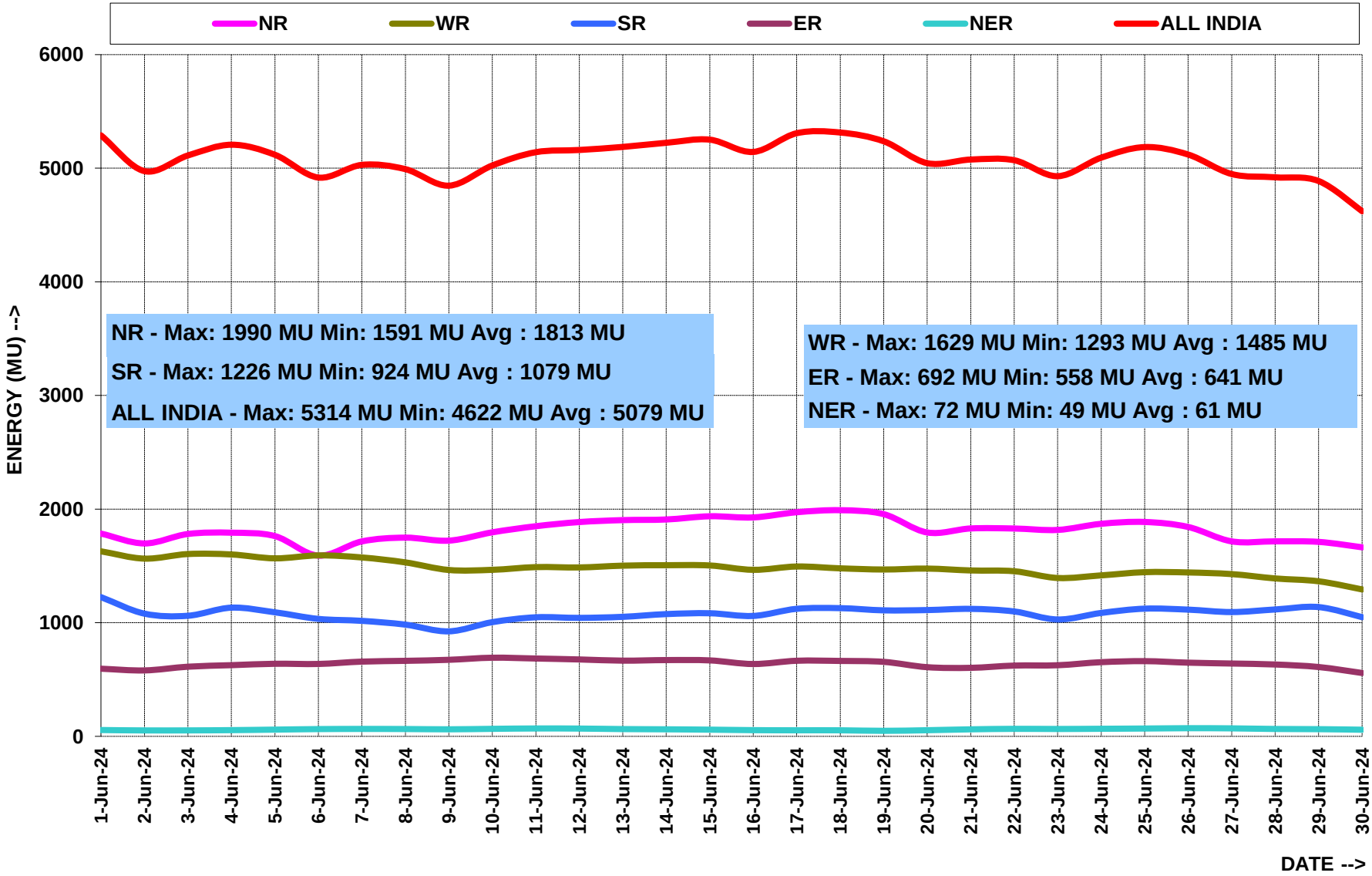
4. राष्ट्रीय स्तर पर विद्युत ऊर्जा आपूर्ति ENERGY MET AT NATIONAL LEVEL

माह: जून 2024 MONTH:- JUN 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jun-24	1785	1629	1226	595	56	5290
02-Jun-24	1698	1564	1080	580	52	4975
03-Jun-24	1781	1605	1061	613	52	5112
04-Jun-24	1792	1601	1133	626	55	5207
05-Jun-24	1763	1567	1092	639	59	5119
06-Jun-24	1591	1593	1034	637	64	4918
07-Jun-24	1716	1575	1016	658	65	5030
08-Jun-24	1749	1531	983	665	64	4992
09-Jun-24	1723	1464	924	674	62	4847
10-Jun-24	1796	1466	1005	692	66	5025
11-Jun-24	1849	1490	1048	685	69	5141
12-Jun-24	1886	1487	1043	677	69	5161
13-Jun-24	1903	1502	1052	666	64	5188
14-Jun-24	1909	1506	1076	671	62	5224
15-Jun-24	1937	1504	1083	668	59	5252
16-Jun-24	1927	1466	1060	637	55	5144
17-Jun-24	1973	1495	1123	665	53	5308
18-Jun-24	1990	1479	1128	664	53	5314
19-Jun-24	1955	1468	1109	656	49	5237
20-Jun-24	1794	1477	1111	609	54	5045
21-Jun-24	1830	1460	1122	603	62	5077
22-Jun-24	1829	1453	1099	623	66	5070
23-Jun-24	1817	1394	1029	626	65	4930
24-Jun-24	1871	1417	1086	653	67	5093
25-Jun-24	1887	1445	1124	662	69	5186
26-Jun-24	1843	1443	1115	648	72	5121
27-Jun-24	1716	1428	1094	641	70	4949
28-Jun-24	1716	1390	1117	632	65	4920
29-Jun-24	1712	1365	1138	610	63	4888
30-Jun-24	1664	1293	1049	558	58	4622
कुल TOTAL	54400	44553	32359	19233	1839	152384
उच्चतम MAXIMUM	1990	1629	1226	692	72	5314
निम्नतम MINIMUM	1591	1293	924	558	49	4622
औसत AVERAGE	1813	1485	1079	641	61	5079
संचयी 2024-25 Cumulative 2024-25	142534	140306	108824	56106	5203	452973
अब तक का उच्चतम All Time Max.	**1990**	1648	1416	**692**	72	5466
दिनांक Date	**18.06.24**	06.05.24	05.04.24	**10.06.24**	15.09.23	30.05.24

ENERGY MET DURING THE MONTH OF JUNE' 2024



5. वर्ष 2024-25 के लिए आवृति रूपरेखा FREQUENCY PROFILE FOR YEAR 2024-25

राष्ट्रीय ग्रिड NATIONAL GRID									
फ्रिक्वेंसी रूपरेखा (Hz)			<49.9	49.9-50.05	>50.05		उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी
% समय	Apr-24	All India Grid	5.25	78.56	16.19		50.43	49.55	50.00
	May-24	All India Grid	2.49	80.04	17.47		50.46	49.72	50.01
	Jun-24	All India Grid	4.50	79.18	16.32		50.67	49.63	50.00
	2024-25 (upto Jun)	All India Grid	4.06	79.27	16.67		50.67	49.55	50.00

5.1 जून 2024 के लिए आवृत्ति रूपरेखा FREQUENCY PROFILE FOR JUNE 2024

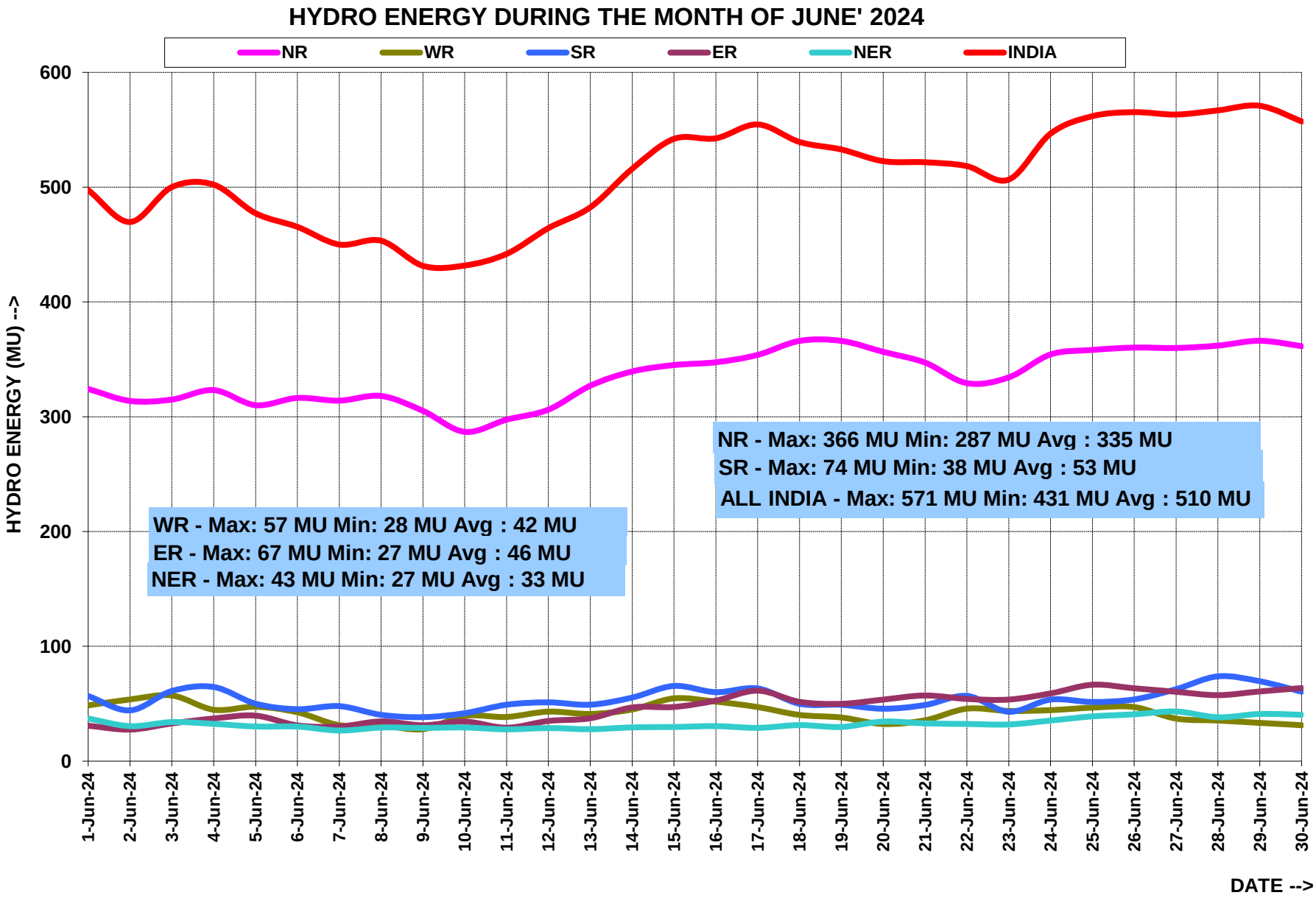
फ्रिक्वेंसी रूपरेखा (Hz)		<49.9	49.9-50.05	>50.05		उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी	एफ़.वी.आई.
% समय	01-Jun-24	4.05	85.20	10.75		50.15	49.82	49.99	0.03
	02-Jun-24	6.01	88.65	5.35		50.13	49.68	49.98	0.03
	03-Jun-24	4.14	80.01	15.84		50.18	49.76	50.01	0.03
	04-Jun-24	4.03	78.89	17.08		50.20	49.75	50.01	0.03
	05-Jun-24	3.53	69.59	26.88		50.23	49.80	50.02	0.04
	06-Jun-24	2.57	71.59	25.84		50.24	49.77	50.02	0.03
	07-Jun-24	2.09	81.94	15.96		50.20	49.75	50.01	0.02
	08-Jun-24	2.89	90.46	6.64		50.12	49.77	49.99	0.02
	09-Jun-24	7.06	80.20	12.74		50.18	49.71	49.99	0.03
	10-Jun-24	3.60	72.80	23.60		50.17	49.77	50.02	0.03
	11-Jun-24	0.47	73.41	26.11		50.27	49.86	50.03	0.03
	12-Jun-24	0.73	86.24	13.03		50.15	49.86	50.01	0.02
	13-Jun-24	0.39	82.73	16.88		50.15	49.88	50.01	0.02
	14-Jun-24	3.53	79.04	17.43		50.21	49.76	50.00	0.04
	15-Jun-24	16.39	77.95	5.66		50.14	49.71	49.97	0.05
	16-Jun-24	7.91	83.66	8.44		50.17	49.63	49.99	0.04
	17-Jun-24	3.72	83.61	12.67		50.67	49.77	50.00	0.04
	18-Jun-24	1.85	71.61	26.54		50.22	49.84	50.02	0.03
	19-Jun-24	0.10	66.55	33.34		50.21	49.67	50.03	0.03
	20-Jun-24	6.31	71.74	21.96		50.28	49.81	50.01	0.04
	21-Jun-24	4.22	79.70	16.08		50.26	49.80	50.01	0.03
	22-Jun-24	9.97	73.52	16.52		50.19	49.77	49.99	0.04
	23-Jun-24	7.96	76.83	15.21		50.16	49.71	49.99	0.04
	24-Jun-24	8.43	77.81	13.76		50.20	49.75	49.99	0.04
	25-Jun-24	5.81	82.23	11.96		50.19	49.77	49.99	0.03
	26-Jun-24	2.07	79.72	18.21		50.16	49.86	50.01	0.02
	27-Jun-24	2.93	79.02	18.06		50.16	49.85	50.00	0.02
	28-Jun-24	3.22	82.04	14.75		50.22	49.83	50.00	0.03
	29-Jun-24	3.15	84.41	12.44		50.16	49.77	50.00	0.02
	30-Jun-24	5.87	84.17	9.97		50.19	49.78	49.99	0.03

6.राष्ट्रीय स्तर पर जल विद्युत उत्पादन
HYDRO GENERATION AT NATIONAL LEVEL

माह: जून 2024 MONTH:- JUN 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jun-24	324	49	57	31	37	497
02-Jun-24	314	54	44	27	30	470
03-Jun-24	315	57	61	33	34	500
04-Jun-24	323	45	65	37	32	502
05-Jun-24	310	47	50	40	30	477
06-Jun-24	316	43	45	31	30	465
07-Jun-24	314	31	48	30	27	450
08-Jun-24	318	31	40	35	29	453
09-Jun-24	305	28	38	31	29	431
10-Jun-24	287	39	42	34	29	432
11-Jun-24	297	38	49	29	28	442
12-Jun-24	306	43	51	35	29	464
13-Jun-24	327	41	49	37	28	482
14-Jun-24	339	45	55	47	29	516
15-Jun-24	345	55	65	47	30	542
16-Jun-24	347	52	60	53	31	543
17-Jun-24	354	47	63	61	29	555
18-Jun-24	366	40	50	52	31	539
19-Jun-24	366	38	49	50	30	533
20-Jun-24	357	32	46	54	34	523
21-Jun-24	347	36	49	57	33	522
22-Jun-24	329	46	57	54	32	518
23-Jun-24	334	44	43	54	32	507
24-Jun-24	354	44	54	59	35	547
25-Jun-24	358	47	52	67	39	562
26-Jun-24	360	47	54	63	41	565
27-Jun-24	360	37	63	60	43	563
28-Jun-24	362	35	74	57	38	567
29-Jun-24	366	33	70	61	41	571
30-Jun-24	361	31	60	64	40	557
कुल TOTAL	10065	1256	1603	1390	982	15296
उच्चतम MAXIMUM	366	57	74	67	43	571
निम्नतम MINIMUM	287	28	38	27	27	431
औसत AVERAGE	335	42	53	46	33	510
संचयी 2024-25 Cumulative 2024-25	23759	4142	4413	2898	1900	37113
अब तक का उच्चतम All Time Max.	443	167	208	157	**43**	877
दिनांक Date	01.08.23	18.12.14	31.08.18	14.09.22	**27.06.24**	30.08.22


HYDRO ENERGY DURING THE MONTH OF JUNE' 2024
NR
WR
SR
ER
NER
INDIA
HYDRO ENERGY (MU) -->
600
500
400
300
200
100
0
NR - Max: 366 MU Min: 287 MU Avg : 335 MU
SR - Max: 74 MU Min: 38 MU Avg : 53 MU
ALL INDIA - Max: 571 MU Min: 431 MU Avg : 510 MU
WR - Max: 57 MU Min: 28 MU Avg : 42 MU
ER - Max: 67 MU Min: 27 MU Avg : 46 MU
NER - Max: 43 MU Min: 27 MU Avg : 33 MU
1-Jun-24
2-Jun-24
3-Jun-24
4-Jun-24
5-Jun-24
6-Jun-24
7-Jun-24
8-Jun-24
9-Jun-24
10-Jun-24
11-Jun-24
12-Jun-24
13-Jun-24
14-Jun-24
15-Jun-24
16-Jun-24
17-Jun-24
18-Jun-24
19-Jun-24
20-Jun-24
21-Jun-24
22-Jun-24
23-Jun-24
24-Jun-24
25-Jun-24
26-Jun-24
27-Jun-24
28-Jun-24
29-Jun-24
30-Jun-24
DATE -->

7. राष्ट्रीय स्तर पर पवन ऊर्जा उत्पादन
WIND GENERATION AT NATIONAL LEVEL

माह: जून 2024 MONTH:- JUN 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jun-24	43	202	126	----------	----------	371
02-Jun-24	32	107	77	----------	----------	217
03-Jun-24	43	121	47	----------	----------	211
04-Jun-24	66	144	69	----------	----------	280
05-Jun-24	71	156	78	----------	----------	305
06-Jun-24	34	134	55	----------	----------	223
07-Jun-24	24	110	58	----------	----------	192
08-Jun-24	25	96	115	----------	----------	236
09-Jun-24	15	120	175	----------	----------	310
10-Jun-24	16	136	210	----------	----------	362
11-Jun-24	24	168	223	----------	----------	415
12-Jun-24	42	177	192	----------	----------	411
13-Jun-24	56	175	170	----------	----------	401
14-Jun-24	47	183	183	----------	----------	414
15-Jun-24	29	143	179	----------	----------	352
16-Jun-24	29	95	155	----------	----------	278
17-Jun-24	37	101	121	----------	----------	259
18-Jun-24	64	178	165	----------	----------	408
19-Jun-24	72	229	176	----------	----------	477
20-Jun-24	58	250	201	----------	----------	508
21-Jun-24	20	210	209	----------	----------	439
22-Jun-24	17	102	198	----------	----------	317
23-Jun-24	7	84	216	----------	----------	307
24-Jun-24	10	78	217	----------	----------	305
25-Jun-24	16	70	256	----------	----------	343
26-Jun-24	15	64	268	----------	----------	347
27-Jun-24	10	74	266	----------	----------	351
28-Jun-24	7	111	223	----------	----------	341
29-Jun-24	14	136	197	----------	----------	347
30-Jun-24	25	147	170	----------	----------	342
कुल TOTAL	970	4103	4998	----------	----------	10070
उच्चतम MAXIMUM	72	250	268	----------	----------	508
निम्नतम MINIMUM	7	64	47	----------	----------	192
औसत AVERAGE	32	137	167	----------	----------	336
संचयी 2024-25 Cumulative 2024-25	2281	11155	9085	----------	----------	22521
अब तक का उच्चतम All Time Max.	86	310	298	----------	----------	619
दिनांक Date	07.08.23	28.05.24	21.07.23	----------	----------	28.05.24

*__Source:__ As reported by SLDCs. Limited visibility of embedded wind generator data.

8. राष्ट्रीय स्तर पर सौर ऊर्जा उत्पादन
SOLAR GENERATION AT NATIONAL LEVEL

माह: जून 2024 MONTH:- JUN 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jun-24	181	92	121	3.0	1.0	398
02-Jun-24	174	92	119	3.1	0.9	389
03-Jun-24	173	92	89	3.8	0.7	359
04-Jun-24	173	93	109	3.9	0.7	380
05-Jun-24	179	92	123	3.9	1.0	398
06-Jun-24	171	87	105	4.0	1.0	368
07-Jun-24	166	94	91	4.0	1.0	356
08-Jun-24	167	84	101	3.2	1.1	356
09-Jun-24	172	90	107	4.1	0.9	373
10-Jun-24	170	81	107	3.6	0.9	363
11-Jun-24	173	92	107	3.5	1.0	376
12-Jun-24	174	89	92	3.3	0.9	360
13-Jun-24	168	85	105	3.4	0.6	363
14-Jun-24	176	83	113	2.3	0.9	375
15-Jun-24	178	85	134	3.6	0.5	401
16-Jun-24	173	84	131	2.9	0.4	392
17-Jun-24	175	80	124	4.0	0.4	383
18-Jun-24	174	85	124	3.2	0.3	386
19-Jun-24	173	79	112	3.1	0.3	367
20-Jun-24	169	85	120	3.2	0.6	377
21-Jun-24	149	75	112	2.2	1.0	339
22-Jun-24	139	80	110	2.4	1.0	332
23-Jun-24	164	71	112	3.3	0.9	352
24-Jun-24	164	69	114	3.2	0.8	351
25-Jun-24	152	74	110	3.0	0.9	340
26-Jun-24	127	79	100	3.0	1.1	311
27-Jun-24	134	65	84	2.9	0.8	287
28-Jun-24	121	67	114	2.4	0.6	305
29-Jun-24	113	68	114	2.3	0.5	297
30-Jun-24	157	53	86	2.4	0.5	298
कुल TOTAL	4879	2443	3289	96	23	10731
उच्चतम MAXIMUM	181	94	134	4.1	1.1	401
निम्नतम MINIMUM	113	53	84	2.2	0.3	287
औसत AVERAGE	163	81	110	3.2	0.8	358
संचयी 2024-25 Cumulative 2024-25	14725	7960	10993	294	86.4	34060
अब तक का उच्चतम All Time Max.	186	103	149	6.7	2.4	426
दिनांक Date	01.05.24	30.04.24	31.03.24	23.03.23	22.06.22	30.04.24

*__Source__: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

9. राष्ट्रीय स्तर पर दैनिक अधिकतम विद्युत मांग पूर्ति

DAILY MAXIMUM DEMAND MET AT NATIONAL LEVEL

माह: जून 2024 MONTH:- JUN 2024

सभी आंकड़े मेगावाट मे All figures in MW

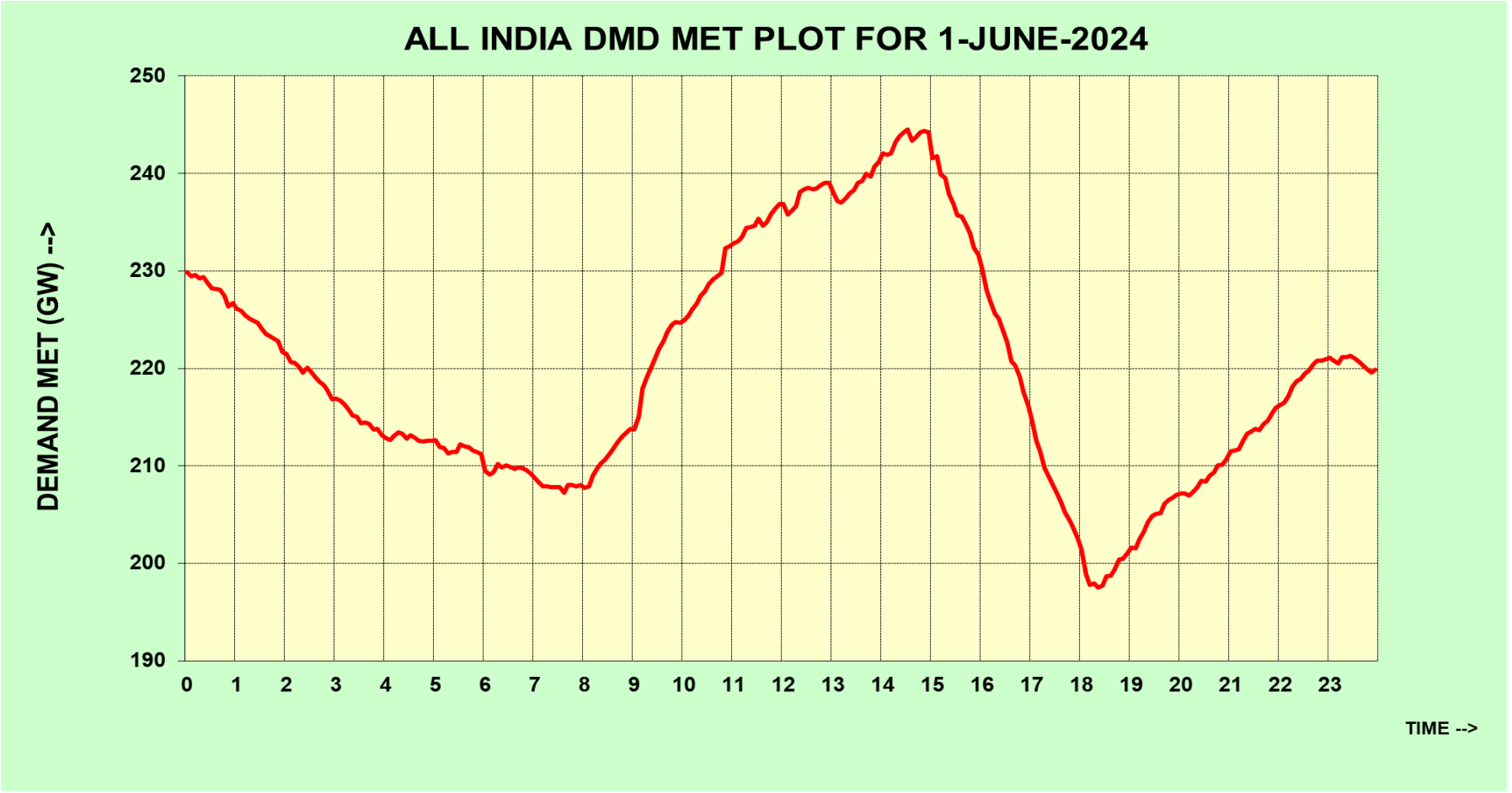
दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL		विभिन्नता फ़ैक्टर Diversity Factor*
01-Jun-24	83373	74974	59053	27646	3044	245419		1.011
02-Jun-24	77749	70047	50108	27268	2944	225900		1.010
03-Jun-24	84086	74791	51134	28984	2970	236379		1.024
04-Jun-24	84088	74934	52322	28967	3103	240890		1.010
05-Jun-24	84928	72274	51440	31224	3341	238588		1.019
06-Jun-24	78903	73055	47761	30924	3448	230073		1.017
07-Jun-24	81044	73405	46142	30448	3434	231951		1.011
08-Jun-24	79794	69932	44446	31579	3439	224763		1.020
09-Jun-24	77316	66803	41785	31226	3300	217242		1.015
10-Jun-24	81749	68035	46178	32531	3573	228617		1.015
11-Jun-24	84226	67570	47624	32467	3582	231017		1.019
12-Jun-24	85953	67594	46855	31561	3491	233391		1.009
13-Jun-24	86548	68160	48237	31557	3288	234856		1.012
14-Jun-24	87046	67102	48521	31440	3341	233615		1.016
15-Jun-24	89177	67411	49448	31170	3193	238957		1.006
16-Jun-24	85961	65691	48211	31162	2892	231161		1.012
17-Jun-24	89513	67425	53819	31226	2955	240135		1.020
18-Jun-24	89526	66493	52191	30893	2746	239329		1.011
19-Jun-24	91215	65608	49874	31296	2712	239060		1.007
20-Jun-24	85448	65174	50787	29839	3043	229780		1.020
21-Jun-24	83385	65789	51864	29346	3331	227661		1.027
22-Jun-24	82051	64269	50921	29643	3466	226637		1.016
23-Jun-24	81526	60899	46130	30106	3331	216260		1.027
24-Jun-24	84861	63199	50509	31346	3360	227001		1.028
25-Jun-24	85282	63435	52024	30928	3503	230598		1.020
26-Jun-24	81084	62580	50965	30577	3480	224164		1.020
27-Jun-24	78855	62190	50359	30370	3478	216122		1.042
28-Jun-24	78128	59868	51635	29806	3385	215592		1.034
29-Jun-24	77917	58876	52874	29167	3302	212167		1.047
30-Jun-24	76295	56878	49030	27415	3049	205936		1.033
उच्चतम MAXIMUM	91215	74974	59053	32531	3582	245419		1.047
निम्नतम MINIMUM	76295	56878	41785	27268	2712	205936		1.006
औसत AVERAGE	83234	66815	49742	30404	3251	229109		1.019
अब तक का उच्चतम All Time Max.	**91215**	**74974**	68735	**32531**	3712	250070		
दिनांक Date	**19.06.24**	**01.06.24**	13.03.24	**10.06.24**	04.09.23	30.05.24		

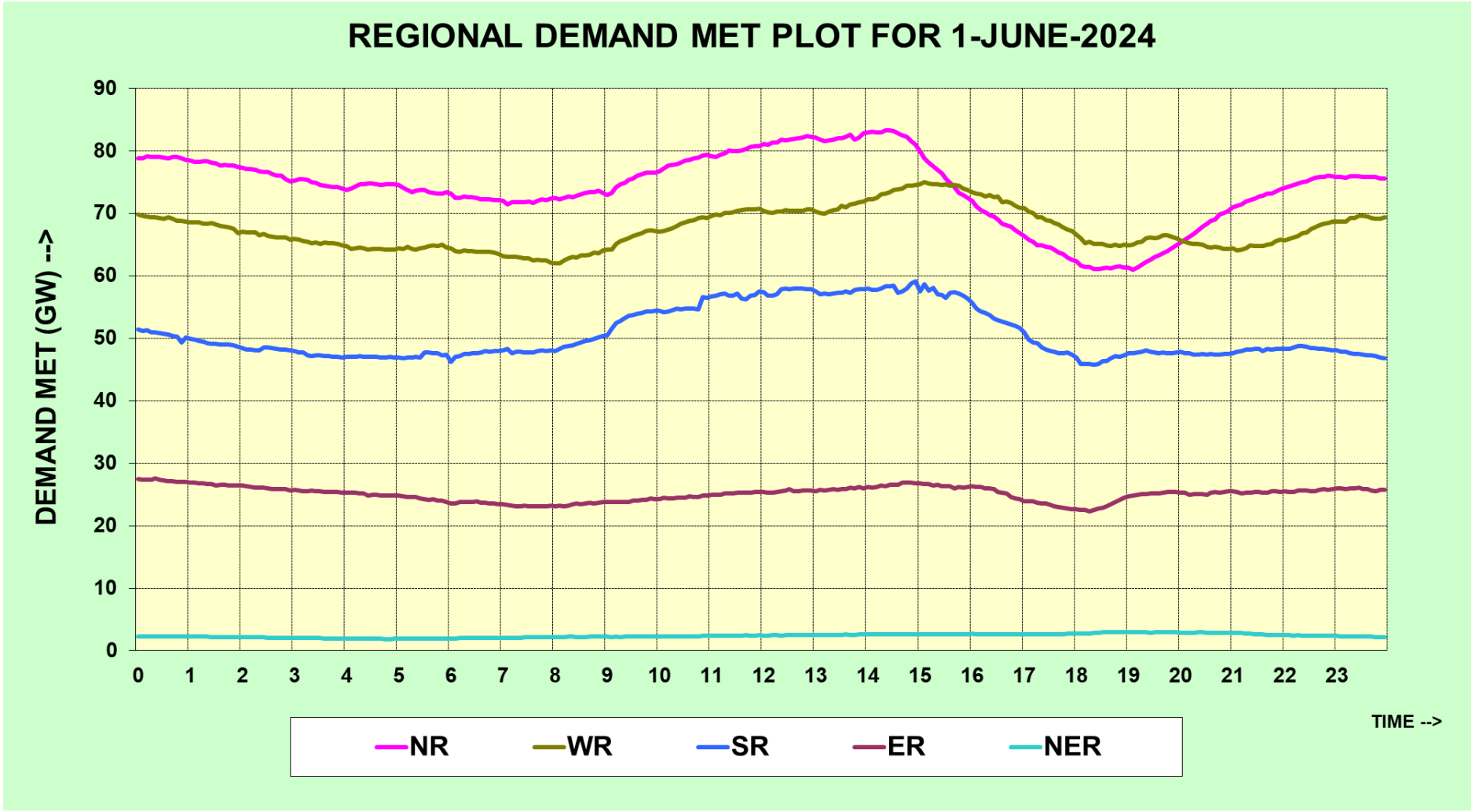
* Diversity factor = (Sum of regional max demands) / All India max demand

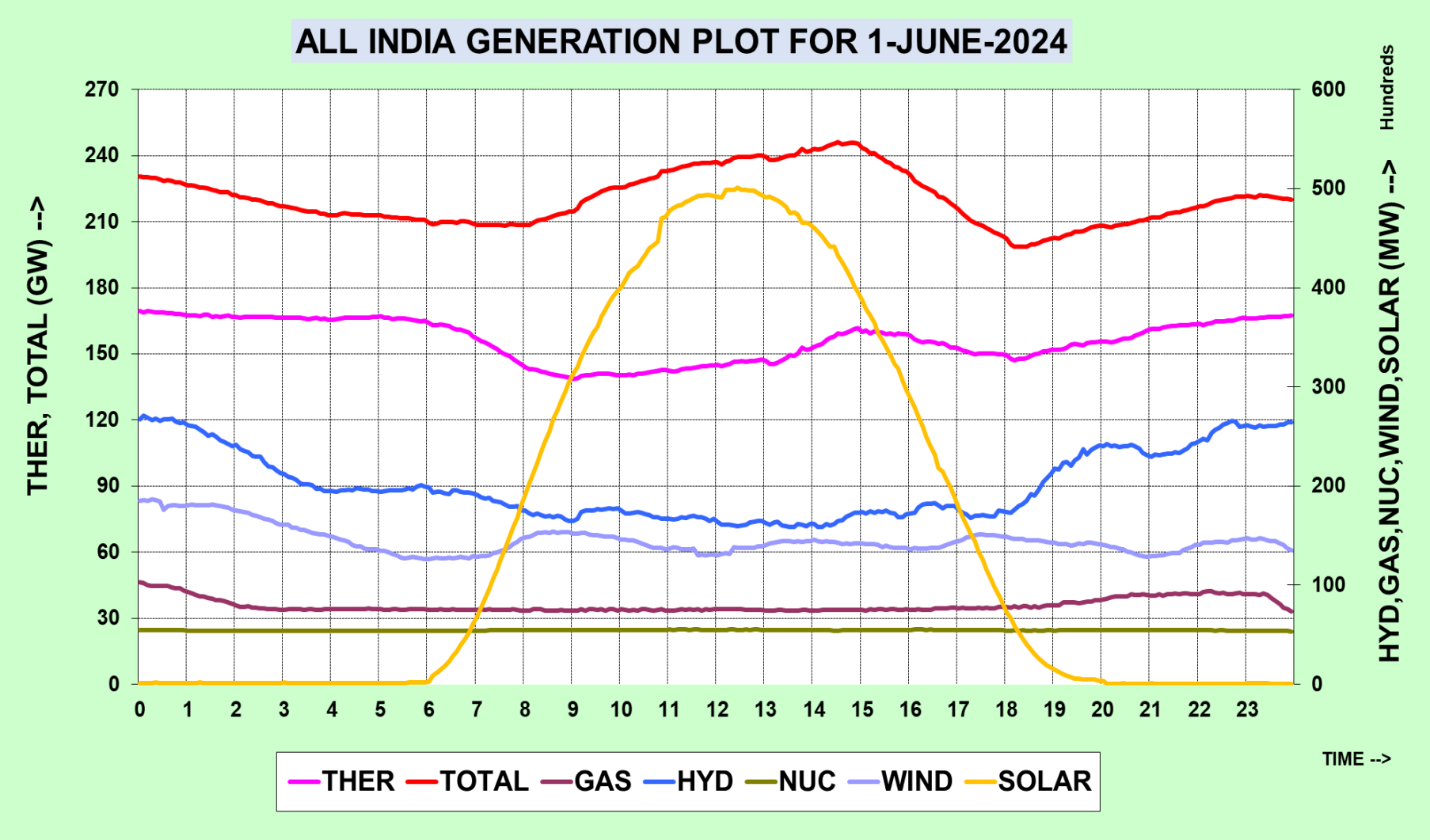
नोट : यह आंकड़े एनएलडीसी स्काडा प्रणाली में दर्ज दैनिक अधिकतम मांगपूर्ति दर्शाते हैं ।

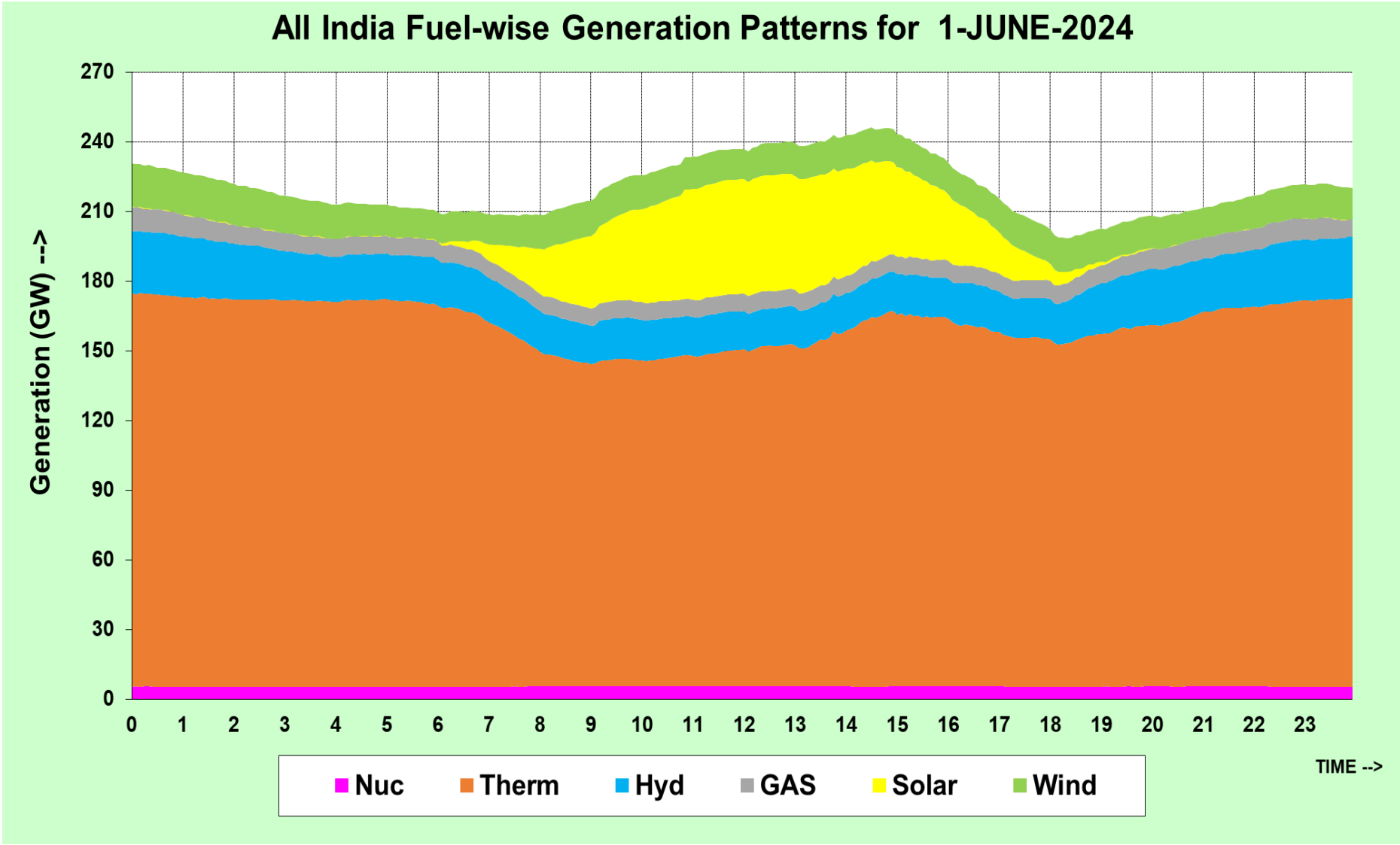
Note: The above figures denote daily maximum demand met recorded in NLDC SCADA.

10. DEMAND AND GENERATION PLOTS FOR 01 JUN (MAXIMUM DEMAND MET)




REGIONAL DEMAND MET PLOT FOR 1-JUNE-2024
DEMAND MET (GW) -->
90
80
70
60
50
40
30
20
10
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
TIME -->
NR
WR
SR
ER
NER


ALL INDIA GENERATION PLOT FOR 1-JUNE-2024
THER, TOTAL (GW) -->
HYD,GAS,NUC,WIND,SOLAR (MW) --> Hundreds
TIME -->
270
240
210
180
150
120
90
60
30
0
600
500
400
300
200
100
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
THER TOTAL GAS HYD NUC WIND SOLAR


All India Fuel-wise Generation Patterns for 1-JUNE-2024
Generation (GW) -->
270
240
210
180
150
120
90
60
30
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
TIME -->
Nuc
Therm
Hyd
GAS
Solar
Wind

01 जून 2024 (अधिकतम माँग) की आल इंडिया माँग आपूर्ति

ALL INDIA DEMAND MET FOR 01 JUNE 2024 (MAXIMUM DEMAND*)

Time (hrs)	All India Demand (MW)		Time (hrs)	All India Demand (MW)
00:05	229435		**12:05**	235767
00:20	229361		**12:20**	238085
00:35	228128		**12:35**	238390
00:50	226354		**12:50**	239039
01:05	225887		**13:05**	237143
01:20	224922		**13:20**	237960
01:35	223542		**13:35**	239290
01:50	222785		**13:50**	240666
02:05	220669		**14:05**	241891
02:20	219561		**14:20**	243820
02:35	219067		**14:35**	243374
02:50	217713		**14:50**	244372
03:05	216676		**15:05**	241808
03:20	215176		**15:20**	237897
03:35	214414		**15:35**	235599
03:50	213788		**15:50**	232430
04:05	212673		**16:05**	228016
04:20	213269		**16:20**	225096
04:35	212870		**16:35**	220757
04:50	212591		**16:50**	217525
05:05	211917		**17:05**	212570
05:20	211449		**17:20**	208981
05:35	211971		**17:35**	206258
05:50	211422		**17:50**	203616
06:05	209131		**18:05**	198878
06:20	209855		**18:20**	197519
06:35	209659		**18:35**	198745
06:50	209512		**18:50**	200494
07:05	208261		**19:05**	201559
07:20	207828		**19:20**	204229
07:35	207261		**19:35**	205141
07:50	207881		**19:50**	206769
08:05	207924		**20:05**	207173
08:20	210290		**20:20**	207784
08:35	211838		**20:35**	209002
08:50	213353		**20:50**	210155
09:05	215033		**21:05**	211543
09:20	219983		**21:20**	213306
09:35	222847		**21:35**	213626
09:50	224755		**21:50**	215408
10:05	225391		**22:05**	216471
10:20	227455		**22:20**	218702
10:35	229134		**22:35**	219814
10:50	232327		**22:50**	220802
11:05	233029		**23:05**	220802
11:20	234518		**23:20**	221165
11:35	234644		**23:35**	220751
11:50	236354		**23:50**	219567
Maximum Demand of 245419 MW met@14:57 hrs (from 1 min. interval SCADA DATA)				

* 15 minute interval SCADA DATA of instanteneous demand

11. ACTUAL POWER SUPPLY POSITION - JUNE 2024

REGION	STATE	Requirement/Availability in MU/DAY			Requirement/Availability in MU				Peak Demand/Peak Met in MW			
		Requirement	Energy met	Surplus(+)/ Deficit(-)	Requirement	Energy met	Surplus(+)/ Deficit(-)	% Shortage	Requirement	Peak Demand Met	Surplus(+)/ Deficit(-)	% Shortage
NR	Chandigarh	8	8	0.0	237	237	0	0.0	449	449	0	0.0
	Delhi	152	152	0.0	4554	4554	0	0.0	8656	8656	0	0.0
	Haryana	260	260	-0.4	7800	7789	-11	-0.1	14469	14469	0	0.0
	Himachal Pradesh	38	38	0.0	1149	1149	0	0.0	1919	1919	0	0.0
	J&K(UT) and Ladakh(UT)	55	55	-0.3	1648	1638	-9	-0.6	2902	2902	0	0.0
	Punjab	293	293	0.0	8796	8796	0	0.0	16089	16089	0	0.0
	Rajasthan	349	345	-4.1	10469	10345	-123	-1.2	17774	17774	0	0.0
	Uttar Pradesh	603	602	-1.5	18103	18057	-46	-0.3	30618	30618	0	0.0
	Uttarakhand	54	54	0.0	1615	1614	-1	-0.1	2863	2863	0	0.0
	Railway	4	4	0.0	127	127	0	0.0	236	236	0	0.0
	NFL	0	0	0.0	2	2	0	0.0	9	9	0	0.0
	TOTAL	1817	1810	-6.3	54501	54311	-190	-0.3	91592	90747	-845	-0.9
WR	Gujarat	477	477	0.0	14324	14324	0	0.0	25588	25588	0	0.0
	Madhya Pradesh	264	264	0.0	7916	7916	0	0.0	13953	13953	0	0.0
	Chhattisgarh	116	116	0.0	3470	3469	-1	0.0	5438	5438	0	0.0
	Maharashtra	554	554	-0.1	16628	16626	-2	0.0	28310	28310	0	0.0
	Goa	14	14	-0.2	434	428	-6	-1.4	799	799	0	0.0
	DNHDDPDCL	30	30	0.0	907	907	0	0.0	1353	1353	0	0.0
	AMNSIL	17	17	0.0	508	508	0	0.0	850	850	0	0.0
	BALCO	12	12	0.0	375	375	0	0.0	526	526	0	0.0
	TOTAL	1485	1485	-0.3	44562	44553	-9	0.0	74933	74933	0	0.0
SR	Andhra Pradesh	212	212	0.0	6349	6349	0	0.0	12578	12578	0	0.0
	Telangana	203	203	0.0	6085	6085	0	0.0	11264	11244	-20	-0.2
	Karnataka	213	213	0.0	6394	6394	0	0.0	14335	14335	0	0.0
	Kerala	81	81	0.0	2435	2435	0	0.0	4404	4400	-4	-0.1
	Tamil Nadu	359	359	0.0	10784	10784	0	0.0	18771	18771	0	0.0
	Puducherry	10	10	0.0	310	310	0	0.0	517	517	0	0.0
	TOTAL	1079	1079	0.0	32359	32359	0	0.0	59053	59053	0	0.0
ER	Bihar	152	151	-0.9	4568	4542	-27	-0.6	7588	7588	0	0.0
	Jharkhand	48	47	-0.4	1433	1419	-13	-0.9	2330	2330	0	0.0
	DVC	76	76	0.0	2279	2279	0	0.0	3673	3673	0	0.0
	Odisha	131	131	0.0	3938	3938	0	0.0	7046	7046	0	0.0
	West Bengal	238	238	0.0	7125	7125	0	0.0	12412	12412	0	0.0
	Sikkim	1	1	0.0	41	41	0	0.0	99	99	0	0.0
	Railways_ER ISTS	0	0	0.0	5	5	0	0.0	25	25	0	0.0
	TOTAL	646	645	-1.3	19390	19350	-40	-0.2	33033	32585	-448	-1.4
NER	Arunachal Pradesh	3	3	0.0	84	84	0	0.0	170	170	0	0.0
	Assam	40	40	0.0	1192	1192	0	0.0	2386	2386	0	0.0
	Manipur	3	3	0.0	94	94	0	0.0	225	225	0	0.0
	Meghalaya	6	6	0.0	165	165	0	0.0	330	330	0	0.0
	Mizoram	2	2	0.0	53	53	0	0.0	126	126	0	0.0
	Nagaland	3	3	0.0	86	86	0	0.0	184	184	0	0.0
	Tripura	5	5	0.0	163	163	0	0.0	353	353	0	0.0
	TOTAL	61	61	0.0	1837	1837	0	0.0	3579	3579	0	0.0
ALL INDIA		5088	5080	-8.0	152648	152409	-239	-0.2				

NOTE: BASED ON THE DATA FURNISHED BY ENTITIES ON DAILY BASIS

12. ENERGY COMPARISON OF JUNE 2024 vs JUNE 2023

REGION	STATE	Energy Requirement (MU)					Energy Met (MU)				
		Jun-23	Jun-24	Difference	% Change	Average MU/day for Jun-24	Jun-23	Jun-24	Difference	% Change	Average MU/day for Jun-24
NR	Chandigarh	182	237	55	30	8	182	237	55	30	8
	Delhi	3667	4554	887	24	152	3666	4554	888	24	152
	Haryana	6159	7800	1641	27	260	6137	7789	1652	27	260
	Himachal Pradesh	935	1149	214	23	38	933	1149	216	23	38
	J&K(UT) and Ladakh(UT)	1586	1648	62	4	55	1560	1638	78	5	55
	Punjab	7077	8796	1718	24	293	7077	8796	1718	24	293
	Rajasthan	8136	10469	2332	29	349	8133	10345	2212	27	345
	Uttar Pradesh	15123	18103	2980	20	603	15116	18057	2941	19	602
	Uttarakhand	1495	1615	120	8	54	1491	1614	123	8	54
	Railway/NFL	105	130	25	23	4	105	130	25	23	4
	TOTAL	44466	54501	10035	23	1817	44402	54311	9909	22	1810
WR	Gujarat	11759	14324	2565	22	477	11759	14324	2565	22	477
	Madhya Pradesh	7234	7916	683	9	264	7234	7916	683	9	264
	Chhattisgarh	3196	3470	274	9	116	3195	3469	275	9	116
	Maharashtra	17535	16628	-907	-5	554	17512	16626	-885	-5	554
	Goa	440	434	-7	-2	14	440	428	-12	-3	14
	DNHDDPDCL	879	907	27	3	30	879	907	27	3	30
	AMNSIL	537	508	-29	-5	17	537	508	-29	-5	17
	BALCO	372	375	3	1	12	372	375	3	1	12
	TOTAL	41952	44562	2610	6	1485	41927	44553	2626	6	1485
SR	Andhra Pradesh	7113	6349	-763	-11	212	7113	6349	-763	-11	212
	Telangana	6000	6085	86	1	203	6075	6085	11	0	203
	Karnataka	7309	6394	-914	-13	213	7383	6394	-989	-13	213
	Kerala	2415	2435	20	1	81	2439	2435	-4	0	81
	Tamil Nadu	10967	10784	-183	-2	359	11065	10784	-280	-3	359
	Puducherry	309	310	2	1	10	311	310	-1	0	10
	TOTAL	34112	32359	-1753	-5	1079	34386	32359	-2027	-6	1079
ER	Bihar	4112	4568	456	11	152	4047	4542	494	12	151
	Jharkhand	1111	1433	322	29	48	1096	1419	323	30	47
	DVC	2300	2279	-22	-1	76	2300	2279	-22	-1	76
	Odisha	3632	3938	306	8	131	3631	3938	307	8	131
	West Bengal	6402	7125	723	11	238	6402	7125	723	11	238
	Sikkim	42	41	-1	-2	1	42	41	-1	-2	1
	Railways_ER ISTS	3	5	2	46	0	3	5	2	46	0
	TOTAL	17603	19390	1787	10	646	17522	19350	1827	10	645
NER	Arunachal Pradesh	79	84	5	6	3	79	84	5	6	3
	Assam	1141	1192	51	4	40	1139	1192	54	5	40
	Manipur	71	94	23	32	3	71	94	23	32	3
	Meghalaya	177	165	-12	-7	6	143	165	22	16	6
	Mizoram	51	53	2	4	2	51	53	2	4	2
	Nagaland	81	86	5	6	3	81	86	5	6	3
	Tripura	150	163	12	8	5	150	163	12	8	5
	TOTAL	1751	1837	87	5	61	1714	1837	123	7	61
ALL INDIA		139883	152648	12765	9	5088	139951	152409	12458	9	5080

13. DEMAND COMPARISON OF JUNE 2024 vs JUNE 2023

REGION	STATE	Peak Requirement (MW)				Peak Met (MW)			
		Jun-23	Jun-24	Difference	% Change	Jun-23	Jun-24	Difference	% Change
NR	Chandigarh	377	449	72	19.1	377	449	72	19.1
	Delhi	7226	8656	1430	19.8	7226	8656	1430	19.8
	Haryana	11634	14469	2835	24.4	11634	14469	2835	24.4
	Himachal Pradesh	1753	1919	166	9.5	1753	1919	166	9.5
	J&K(UT) and Ladakh(UT)	2542	2902	360	14.2	2542	2902	360	14.2
	Punjab	15293	16089	796	5.2	15293	16089	796	5.2
	Rajasthan	15840	17774	1934	12.2	15840	17774	1934	12.2
	Uttar Pradesh	27611	30618	3007	10.9	27611	30618	3007	10.9
	Uttarakhand	2536	2863	327	12.9	2436	2863	427	17.5
	Railway/NFL	193	244	51	26.5	193	244	51	26.5
WR	Gujarat	22225	25588	3363	15	22225	25588	3363	15.1
	Madhya Pradesh	11837	13953	2116	18	11837	13953	2116	17.9
	Chhattisgarh	5289	5438	150	3	5289	5438	150	2.8
	Maharashtra	28546	28310	-236	-1	28546	28310	-236	-0.8
	Goa	776	799	23	3	776	799	23	2.9
	DNHDDPDCL	1311	1353	42	3.2	1311	1353	42	3.2
	AMNSIL	863	850	-13	-1.5	863	850	-13	-1.5
	BALCO	525	526	1	0.2	525	526	1	0.2
SR	Andhra Pradesh	12903	12578	-325	-2.5	12900	12578	-322	-2.5
	Telangana	11436	11264	-171	-1.5	11417	11244	-173	-1.5
	Karnataka	14198	14335	137	1.0	14198	14335	137	1.0
	Kerala	4644	4404	-240	-5.2	4616	4400	-216	-4.7
	Tamil Nadu	18527	18771	244	1.3	18522	18771	249	1.3
	Puducherry	505	517	12	2.3	505	517	12	2.4
ER	Bihar	7071	7588	517	7.3	7071	7588	517	7.3
	Jharkhand	1923	2330	407	21.2	1923	2330	407	21.2
	DVC	3594	3673	79	2.2	3594	3673	79	2.2
	Odisha	6854	7046	192	2.8	6854	7046	192	2.8
	West Bengal	11806	12412	606	5.1	11806	12412	606	5.1
	Sikkim	101	99	-2	-2.0	101	99	-2	-2.0
	Railways_ER ISTS	41	25	-16	-39.0	41	25	-16	-39.0
NER	Arunachal Pradesh	155	170	15	9.5	155	170	15	9.5
	Assam	2307	2386	79	3.4	2307	2386	79	3.4
	Manipur	185	225	40	21.9	185	225	40	21.9
	Meghalaya	333	330	-3	-0.9	333	330	-3	-0.9
	Mizoram	125	126	1	0.5	125	126	1	0.5
	Nagaland	167	184	17	10.3	167	184	17	10.3
	Tripura	362	353	-9	-2.5	362	353	-9	-2.5

14. SCHEDULE AND DRAWAL OF CONSTITUENTS - JUNE 2024

REGION	STATE	SCHEDULE (MU)	ACTUAL DRAWAL (MU)	Over drawal(+)/ Under Drawal(-) (MU)	% OD/UD	SCHEDULE (MU/DAY)	ACTUAL DRAWAL (MU/DAY)	Over drawal(+)/ Under Drawal(-) (MU/DAY)
NR	Chandigarh	239.85	240.94	1.09	0.45	8.00	8.03	0.04
	Delhi	3800.80	3798.58	-2.22	-0.06	126.69	126.62	-0.07
	Haryana	5740.15	5700.05	-40.10	-0.70	191.34	190.00	-1.34
	Himachal Pradesh	124.77	119.70	-5.07	-4.06	4.16	3.99	-0.17
	J&K(UT) and Ladakh(UT)	863.23	815.66	-47.57	-5.51	28.77	27.19	-1.59
	Punjab	5072.09	4963.61	-108.48	-2.14	169.07	165.45	-3.62
	Rajasthan	3409.05	3427.35	18.30	0.54	113.64	114.24	0.61
	Uttar Pradesh	9200.83	9173.79	-27.04	-0.29	306.69	305.79	-0.90
	Uttarakhand	902.68	927.58	24.90	2.76	30.09	30.92	0.83
	Railway	111.21	127.19	15.98	14.37	3.71	4.24	0.53
	NFL	2.81	2.47	-0.34	-12.23	0.09	0.08	-0.01
	Total	29467.48	29296.93	-170.55	-0.58	982.25	976.56	-5.69
WR	Gujarat	4674.58	4554.90	-119.68	-2.56	155.82	151.83	-3.99
	Madhya Pradesh	4322.32	4191.93	-130.39	-3.02	144.08	139.73	-4.35
	Chhattisgarh	1828.66	1790.56	-38.10	-2.08	60.96	59.69	-1.27
	Maharashtra	5532.00	5398.73	-133.27	-2.41	184.40	179.96	-4.44
	Goa	405.39	418.92	13.53	3.34	13.51	13.96	0.45
	DNHDDPDCL	908.31	906.84	-1.47	-0.16	30.28	30.23	-0.05
	AMNSIL	195.29	202.91	7.62	3.90	6.51	6.76	0.25
	BALCO*	374.22	374.81	0.59	0.16	12.47	12.49	0.02
	Total	18240.77	17839.61	-401.16	-2.20	608.03	594.65	-13.37
SR	Andhra Pradesh	1514.44	1475.15	-39.29	-2.59	50.48	49.17	-1.31
	Telangana	2908.82	2910.25	1.43	0.05	96.96	97.01	0.05
	Karnataka	1561.21	1504.05	-57.16	-3.66	52.04	50.14	-1.91
	Kerala	1611.29	1636.06	24.77	1.54	53.71	54.54	0.83
	Tamil Nadu	4955.87	4909.37	-46.50	-0.94	165.20	163.65	-1.55
	Puducherry	300.81	293.41	-7.40	-2.46	10.03	9.78	-0.25
	Goa (SR)	105.62	61.50	-44.12	-41.77	3.52	2.05	-1.47
	Total	12958.06	12789.79	-168.27	-1.30	431.94	426.33	-5.61
ER	Bihar	4207.68	4228.18	20.51	0.49	140.26	140.94	0.68
	Jharkhand	1070.53	1095.83	25.30	2.36	35.68	36.53	0.84
	DVC	-1201.44	-1197.58	3.86	-0.32	-40.05	-39.92	0.13
	Odisha	1635.12	1620.04	-15.08	-0.92	54.50	54.00	-0.50
	West Bengal	2918.77	2884.37	-34.41	-1.18	97.29	96.15	-1.15
	Sikkim	43.59	41.48	-2.11	-4.85	1.45	1.38	-0.07
	Railways_ER ISTS	5.16	4.96	-0.20	-3.86	0.17	0.17	-0.01
	Total	8679.41	8677.27	-2.14	-0.02	289.31	289.24	-0.07
NER	Arunachal Pradesh	84.17	81.27	-2.90	-3.45	2.81	2.71	-0.10
	Assam	907.38	910.82	3.44	0.38	30.25	30.36	0.11
	Manipur	88.65	85.58	-3.07	-3.46	2.96	2.85	-0.10
	Meghalaya	77.46	73.42	-4.04	-5.22	2.58	2.45	-0.13
	Mizoram	53.96	45.97	-7.99	-14.81	1.80	1.53	-0.27
	Nagaland	82.42	80.10	-2.32	-2.81	2.75	2.67	-0.08
	Tripura	153.73	157.34	3.61	2.35	5.12	5.24	0.12
	Total	1447.77	1434.50	-13.27	-0.92	48.26	47.82	-0.44
All India Total		70793.49	70038.09	-755.40	-1.07	2359.78	2334.60	-25.18

15. INTER REGIONAL EXCHANGES 2024-25

(All figures in MU)

	Apr'24	May'24	June'24	Fin. Year 2024-25
Name of Line				
Import of NR from WR (WR-NR)				
WR - NR 765 kV Gwalior - Agra 2xS/C	529.65	1235.15	1669.21	**3434.01**
WR-NR 765 kV Gwalior - Phagi 2xS/C	157.44	396.56	594.39	**1148.39**
WR-NR 765 kV Jabalpur-Orai D/C	304.53	976.67	1388.19	**2669.39**
WR-NR 765 kV Satna-Orai	535.85	669.33	731.25	**1936.43**
WR-NR 765 kV Gwalior-Orai	0.00	0.00	0.00	**0.00**
WR - NR HVDC Champa - Kurukshetra	1525.63	2816.73	3112.34	**7454.70**
WR - NR HVDC Vindhyachal	0.00	100.25	234.97	**335.22**
WR - NR 400 kV Zerda Kankroli	0.00	35.28	83.46	**118.74**
WR - NR 400 kV Zerda Bhinmal	0.00	1.38	0.00	**1.38**
WR - NR 400 kV RAPP C - Shujalpur D/C	0.00	60.74	171.23	**231.97**
WR - NR 220 kV Bhanpura - Ranpur	55.64	76.03	72.42	**204.09**
WR - NR 220 kV Bhanpura - Modak	69.87	76.33	55.61	**201.81**
WR - NR 220 kV Auraiya - Malanpur/Mehgaon	0.00	3.05	7.54	**10.59**
WR - NR 400 kV Vindhyachal-Rihand D/C	0.00	0.00	0.00	**0.00**
WR - NR HVDC Mundra - Mohindergarh	717.97	1144.70	1159.61	**3022.28**
WR-NR 765 kV Banaskata-Chittorgarh D/C	0.67	262.23	420.74	**683.64**
WR-NR 765 kV Varanasi-Vindhyachal	1373.47	2029.86	2265.31	**5668.64**
WR -NR 765 kV Neemach - Chittorgarh D/C	0.00	35.36	144.70	**180.06**
Total WR - NR	**5270.72**	**9919.65**	**12110.97**	**27301.34**
Export of NR to WR (NR-WR)				
NR - WR 765 kV Gwalior - Agra 2xS/C	0.00	0.00	0.00	**0.00**
NR - WR 765 kV Gwalior - Phagi 2xS/C	10.49	1.42	0.00	**11.91**
NR - WR 765 kV Jabalpur-Orai D/C	0.00	0.00	0.00	**0.00**
NR - WR 765 kV Satna-Orai	0.00	0.00	0.00	**0.00**
NR - WR 765 kV Gwalior-Orai	245.97	283.52	291.19	**820.68**
NR - WR HVDC Champa- Kurukshetra	0.00	0.00	0.00	**0.00**
NR - WR HVDC Vindhyachal	259.46	128.55	20.24	**408.25**
NR - WR 400 kV Zerda Kankroli	240.09	83.35	1.32	**324.76**
NR - WR 400 kV Zerda Bhinmal	8.05	0.00	1.50	**9.55**

	Apr'24	May'24	June'24	Fin. Year 2024-25
Name of Line				
NR - WR 400 kV RAPP C - Shujalpur D/C	159.07	31.59	0.21	**190.87**
NR - WR 220 kV Bhanpura - Ranpur	0.00	0.00	0.00	**0.00**
NR - WR 220 kV Bhanpura - Modak	0.00	0.00	0.00	**0.00**
NR - WR 220 kV Auraiya - Malanpur/Mehgaon	59.69	31.06	9.61	**100.36**
NR-WR 400 kV Rihand-Vindhyachal D/C	609.68	680.87	661.37	**1951.92**
NR - WR HVDC Mundra - Mohindergarh	0.00	0.00	0.00	**0.00**
NR - WR 765 kV Banaskata-Chittorgarh D/C	463.90	132.00	0.70	**596.60**
NR-WR 765 kV Varanasi-Vindhyachal	0.00	0.00	0.00	**0.00**
NR -WR 765 kV Chittorgarh-Neemach D/C	139.03	35.99	2.22	**177.24**
Total NR - WR	**2195.43**	**1408.35**	**988.36**	**4592.14**

	Apr'24	May'24	June'24	Fin. Year 2024-25
Name of Line				
Import of NR from ER (ER-NR)				
ER - NR 765 kV Sasaram - Fatehpur	111.08	133.89	160.45	**405.42**
ER - NR 765 kV Gaya - Varanasi 2*S/C	50.67	114.34	92.48	**257.49**
ER - NR 765 kV Gaya - Balia	213.79	293.69	320.79	**828.27**
ER - NR HVDC Alipurduar - Agra	0.00	0.00	0.00	**0.00**
ER-NR 400 kV Patna - Balia D/C	192.05	288.47	317.66	**798.18**
ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	3.14	135.70	339.63	**478.47**
ER - NR 400 kV Biharshariff - Balia D/C	0.00	10.98	26.88	**37.86**
ER - NR 400 kV Motihari - Gorakhpur D/C	63.12	164.75	249.69	**477.56**
ER - NR 400 kV Biharshariff - Varanasi D/C	3.31	1.28	6.08	**10.67**
ER - NR 400 kV Sasaram -Varanasi	41.13	16.74	11.73	**69.60**
ER - NR 400 kV Sasaram -Allahabad	0.13	1.97	22.79	**24.89**
ER - NR 220 kV Sahupuri-Karamnasa	26.02	26.80	20.84	**73.66**
ER - NR 132 kV Sahupuri - Karamnasa	2.88	2.88	1.01	**6.77**
ER - NR 132 kV 132 kV Rihand - Nagar Untari	0.00	0.66	0.00	**0.66**
ER - NR 132 kV Rihand - Garhwa	0.00	0.72	0.00	**0.72**
ER-NR 400 kV Naubatpur - Balia D/C	44.01	81.16	88.27	**213.44**
ER-NR 400 kV Biharshriff - Sahupuri D/C	2.89	29.05	73.37	**105.31**
Total ER-NR	**754.22**	**1303.08**	**1731.67**	**3788.97**
Import of NR from NER (NER-NR)				
NER-NR HVDC Biswanath Chariali - Agra	0.00	0.00	0.00	**0.00**
Total NER - NR	**0.00**	**0.00**	**0.00**	**0.00**

	Apr'24	May'24	June'24	Fin. Year 2024-25
Name of Line				
Export of NR to ER (NR-ER)				
NR - ER 765 kV Sasaram - Fatehpur	0.00	0.46	0.00	**0.46**
NR - ER 765 kV Gaya - Varanasi 2*S/C	65.96	56.17	23.53	**145.66**
NR - ER 765 kV Gaya - Balia	0.00	0.00	0.00	**0.00**
NR - ER HVDC Alipurduar - Agra	0.00	0.00	0.00	**0.00**
NR - ER 400 kV Patna - Balia 2xD/C	0.00	0.00	0.00	**0.00**
NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	87.39	32.41	0.00	**119.80**
NR - ER 400 kV Biharshariff - Balia D/C	148.68	86.56	34.72	**269.96**
ER - NR 400 kV Motihari - Gorakhpur D/C	0.29	1.01	0.00	**1.30**
NR - ER 400 kV Biharshariff - Varanasi D/C	19.06	0.00	0.00	**19.06**
NR - ER 400 kV Sasaram -Varanasi	0.00	2.63	4.50	**7.13**
NR - ER 400 kV Sasaram -Allahabad	10.41	12.82	11.40	**34.63**
NR - ER 220 kV Sahupuri-Karamnasa	0.00	0.00	0.00	**0.00**
NR - ER 132 kV Sahupuri - Karamnasa	1.44	1.23	0.00	**2.67**
NR - ER 132 kV 132 kV Rihand - Nagar Untari	0.00	0.00	0.92	**0.92**
ER - NR 132 kV Rihand - Garhwa	18.99	19.86	16.50	**55.35**
NR-ER 400 kV Balia- Naubatpur D/C	0.00	1.31	0.00	**1.31**
NR-ER 400 kV Sahupuri- Biharshariff D/C	0.00	1.60	2.33	**3.93**
Total NR - ER	**352.22**	**216.06**	**93.90**	**662.18**
Export of NR to NER (NER-NR)				
NR-NER HVDC Biswanath Chariali - Agra	433.33	466.77	450.41	**1350.51**
Total NR - NER	**433.33**	**466.77**	**450.41**	**1350.51**

	Apr'24	May'24	June'24	Fin. Year 2024-25
Name of Line				
Export of WR to ER (WR-ER)				
WR - ER 765 kV Ranchi - Dharamjaygarh 2xS/C	388.72	751.76	1168.90	**2309.38**
WR - ER 765 kV Dharamjaygarh - Jharsuguda 2xD/C	721.67	524.74	158.97	**1405.37**
WR - ER 765 kV Durg - Jharsuguda D/C	0.05	1.25	9.28	**10.58**
WR - ER 400 kV Sipat - Ranchi D/C	51.26	216.17	285.29	**552.71**
WR - ER 400 kV Raigarh - Jharsuguda 2xD/C	3.12	46.71	321.05	**370.88**
WR - ER 220 kV Korba - Budhipadar D/C	8.86	26.39	12.71	**47.97**
WR - ER 220 kV Raigarh - Budhipadar	7.90	2.55	1.47	**11.91**
Total WR - ER	**1181.58**	**1569.56**	**1957.67**	**4708.81**
Import of WR from ER (ER - WR)				
ER - WR 765 kV Ranchi - Dharamjaygarh 2xS/C	48.43	15.64	0.11	**64.18**
ER - WR 765 kV Dharamjaygarh - Jharsuguda 2XD/C	2.75	169.79	403.73	**576.28**
ER - WR 765 kV Durg - Jharsuguda D/C	376.32	377.94	175.66	**929.92**
ER - WR 400 kV Sipat - Ranchi D/C	22.01	4.57	0.14	**26.72**
ER - WR 400 kV Raigarh - Jharsuguda 2xD/C	195.37	120.88	235.08	**551.33**
ER - WR 220 kV Korba - Budhipadar D/C	1.27	8.92	26.67	**36.86**
ER - WR 220 kV Raigarh - Budhipadar	15.69	25.32	47.74	**88.75**
Total ER - WR	**661.85**	**723.06**	**889.13**	**2274.05**
Export of ER to NER (ER - NER)				
ER - NER 400 kV Binaguri - Bongaigaon D/C	20.06	23.09	0.13	**43.28**
ER - NER 400 kV Alipurduar - Bongaigaon D/C	16.74	47.75	0.06	**64.56**
ER - NER 220 kV Birpara - Salakati D/C	2.74	6.73	0.18	**9.65**
Total ER - NER	**39.54**	**77.57**	**0.38**	**117.48**
Import of ER from NER (NER - ER)				
NER - ER 400 kV Binaguri - Bongaigaon D/C	79.80	122.93	220.43	**423.16**
NER - ER 400 kV Alipurduar - Bongaigaon D/C	164.15	208.72	302.14	**675.01**
NER - ER 220 kV Birpara - Salakati D/C	18.58	38.63	53.31	**110.52**
Total NER - ER	**262.52**	**370.28**	**575.88**	**1208.69**

	Apr'24	May'24	June'24	Fin. Year 2024-25
Name of Line				
Export of ER to SR (ER - SR)				
ER - SR HVDC Gazuwaka	413.69	249.41	22.79	**685.88**
ER - SR HVDC Talchar - Kolar	1069.04	1104.35	1220.43	**3393.82**
ER - SR 765 kV Angul- Srikakulam D/C	1640.93	1313.30	1089.30	**4043.52**
Total ER - SR	**3123.66**	**2667.05**	**2332.52**	**8123.23**
Import of ER from SR (SR - ER)				
SR - ER HVDC Gazuwaka	0.00	16.28	175.68	**191.96**
SR - ER HVDC Talchar - Kolar	0.00	0.00	0.00	**0.00**
SR - ER 765 kV Angul- Srikakulam D/C	0.00	0.00	0.00	**0.00**
Total SR - ER	**0.00**	**16.28**	**175.68**	**191.96**
Export of WR to SR (WR-SR)				
WR - SR HVDC Bhadrawati	601.92	221.20	27.24	**850.36**
WR - SR 765 kV Sholapur - Raichur 2xS/C	802.05	292.07	41.83	**1135.96**
WR - SR 765 kV Wardha - Nizamabad D/C	1407.32	763.52	504.48	**2675.33**
WR - SR 400KV Kolhapur-Kudgi D/C	0.00	0.02	0.00	**0.02**
WR-SR HVDC Raigarh-Pugalur	2668.37	1464.59	692.13	**4825.09**
WR-SR 220kV Xeldem - Ambewadi S/C	73.01	75.81	61.15	**209.98**
WR-SR 220kV Ponda - Ambewadi S/C	43.06	54.26	44.46	**141.78**
WR-SR 765 kV Warora - Warangal D/C	1603.13	954.21	557.06	**3114.40**
Total WR - SR	**7198.86**	**3825.69**	**1928.36**	**12952.91**
Import of WR from SR (SR - WR)				
SR - WR HVDC Bhadrawati	2.36	182.40	422.51	**607.27**
SR - WR 765 kV Raichur - Sholapur 2xS/C	0.55	262.88	727.33	**990.76**
SR - WR 765 kV Wardha - Nizamabad D/C	0.00	6.86	32.86	**39.72**
SR - WR 400KV Kolhapur-Kudgi D/C	429.02	646.95	994.86	**2070.83**
SR-WR HVDC Pugalur-Raigarh	0.00	0.00	22.86	**22.86**
WR-SR 220kV Ambewadi - Xeldem S/C	0.00	0.00	0.00	**0.00**
WR-SR 220kV Ambewadi-Ponda S/C	0.00	0.00	0.00	**0.00**
SR-WR 765 kV Warangal - Warora D/C	0.00	6.90	27.23	**34.13**
Total SR - WR	**431.93**	**1105.99**	**2227.65**	**3765.57**
TOTAL ALL INDIA	**21906**	**23669**	**25463**	**71038**

***In case of mutiple ckt / DC pole year corres. to the commissioning of final element**

15.1 Import-Export of NR with WR during June 2024

Import of NR from WR (WR-NR)

Date	WR - NR 765 kV Gwalior - Agra 2xS/C	WR-NR 765 kV Gwalior-Phagi 2xS/C	WR-NR 765 kV Jabalpur- Orai D/C	WR-NR 765 kV Satna- Orai	WR-NR 765 kV Gwalior-Orai	WR - NR HVDC Champa-Kurukshetra	WR - NR HVDC Vindhyachal	WR - NR 400 kV Zerda-Kankroli	WR - NR 400 kV Zerda-Bhinmal	WR-NR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	WR - NR 220 kV Auraiya - Malanpur / Mehgaon	WR-NR 400 KV Rihand Vindhyachal D/C	WR - NR HVDC Mundra - M'garh	WR -NR 765 kV Banaskata - Chittorgarh D/C	WR-NR 765 kV Varanasi-Vindhyachal	WR -NR 765 kV Neemach - Chittorgarh D/C	Total WR - NR
1-Jun-24	47.24	10.85	37.51	22.22	0.00	94.28	11.99	5.45		2.36	2.42	2.04	0.00	0.00	37.05	23.56	73.86	0.42	**371.25**
2-Jun-24	41.05	19.44	38.44	22.60	0.00	88.10	11.99	0.00		3.47	2.38	1.96	0.00	0.00	34.20	1.36	64.62	2.41	**332.02**
3-Jun-24	46.45	16.91	39.95	23.36	0.00	111.99	12.07	0.00		3.33	2.54	1.95	0.00	0.00	36.31	5.74	65.06	2.10	**367.76**
4-Jun-24	46.13	12.31	35.10	22.73	0.00	110.92	5.01	1.45		1.60	2.66	2.03	0.00	0.00	36.31	14.48	70.96	0.40	**362.09**
5-Jun-24	44.24	3.10	34.19	21.78	0.00	116.25	2.70	1.95		0.00	2.68	1.78	0.00	0.00	37.39	17.27	72.89	0.00	**356.22**
6-Jun-24	41.34	5.54	31.45	20.76	0.00	84.12	0.00	2.15		0.06	2.84	1.86	0.00	0.00	25.49	10.47	64.89	0.00	**290.97**
7-Jun-24	49.79	10.66	42.08	23.27	0.00	102.05	0.00	2.54		2.99	2.69	2.41	0.14	0.00	31.28	10.15	74.56	1.79	**356.40**
8-Jun-24	50.36	13.37	41.43	22.89	0.00	119.12	3.63	1.75		2.54	2.66	2.77	0.00	0.00	42.34	8.62	75.74	2.13	**389.35**
9-Jun-24	55.25	17.59	47.46	23.13	0.00	109.35	10.85	1.83		5.09	2.63	2.62	0.66	0.00	42.38	10.20	80.75	4.05	**413.84**
10-Jun-24	60.10	19.63	51.17	24.68	0.00	120.05	7.25	3.60		6.49	2.60	2.54	0.49	0.00	42.38	15.53	85.92	5.10	**447.53**
11-Jun-24	61.02	14.12	50.64	24.04	0.00	138.92	11.96	6.21		5.89	2.52	2.46	0.00	0.00	42.38	25.80	88.99	3.81	**478.76**
12-Jun-24	58.93	13.21	48.59	23.48	0.00	134.67	12.01	4.70		4.69	2.58	2.43	0.00	0.00	42.37	23.92	87.18	3.16	**461.92**
13-Jun-24	58.95	12.51	44.92	23.57	0.00	127.79	9.46	4.41	0.00	5.11	2.53	1.91	0.00	0.00	42.37	24.40	79.56	4.55	**442.04**
14-Jun-24	56.48	13.21	45.69	23.35	0.00	128.02	10.78	3.55		5.09	2.59	1.98	0.00	0.00	39.78	21.99	78.06	4.36	**434.93**
15-Jun-24	58.38	20.11	47.96	24.35	0.00	142.26	9.84	2.64		6.77	2.55	2.30	0.00	0.00	34.26	15.25	80.96	6.27	**453.90**
16-Jun-24	59.86	23.95	49.98	25.35	0.00	119.56	9.84	2.39		7.82	2.64	2.00	0.00	0.00	26.98	12.29	67.96	7.56	**418.18**
17-Jun-24	61.10	24.08	52.13	25.95	0.00	114.91	12.00	0.45		6.62	2.80	2.00	0.00	0.00	41.63	7.80	71.71	6.31	**429.49**
18-Jun-24	61.76	17.10	47.88	26.10	0.00	113.70	10.28	2.73		6.19	2.76	2.07	0.41	0.00	42.38	18.96	75.54	5.37	**433.23**
19-Jun-24	61.68	18.45	47.51	24.66	0.00	100.50	6.88	5.35		6.00	2.62	2.32	0.00	0.00	42.37	28.08	70.80	4.22	**421.44**
20-Jun-24	58.13	15.09	44.64	22.75		45.89	6.02	9.15		6.11	2.74	1.25	0.87	0.00	28.69	38.52	70.92	4.12	**354.89**
21-Jun-24	55.99	21.88	42.99	24.04	0.00	82.44	3.49	5.33		6.91	2.04	1.01	0.64	0.00	30.14	25.43	75.37	4.85	**382.55**
22-Jun-24	63.68	30.39	54.46	27.42	0.00	83.98	4.99	2.16		9.39	2.17	1.23	0.50	0.00	30.25	9.21	80.82	8.91	**409.56**
23-Jun-24	59.66	28.46	51.35	26.24	0.00	123.86	5.02	3.58		9.27	2.50	1.27	0.95	0.00	15.95	12.95	80.51	8.20	**429.77**
24-Jun-24	57.30	28.50	52.31	25.69	0.00	129.50	4.49	3.36		8.64	0.63	1.20	0.00	0.00	46.75	3.16	79.09	7.92	**448.54**
25-Jun-24	60.70	29.80	51.10	26.53	0.00	129.30	0.84	0.21		8.24	2.38	1.11	0.00	0.00	57.05	2.52	81.05	8.07	**458.90**
26-Jun-24	66.34	35.46	59.13	27.99	0.00	101.58	6.25	1.70		10.25	2.03	1.14	0.71	0.00	53.22	5.01	76.70	9.91	**457.42**
27-Jun-24	59.24	31.18	50.59	25.40	0.00	82.96	11.82	0.00		7.65	1.89	1.64	0.65	0.00	47.72	0.00	75.93	8.06	**404.73**
28-Jun-24	59.86	31.56	53.21	25.99	0.00	55.04	9.50	2.55		8.10	2.11	1.90	0.00	0.00	46.95	10.32	79.57	7.87	**394.53**
29-Jun-24	60.20	32.93	54.71	26.63	0.00	48.23	12.02	2.12		9.36	2.10	1.26	1.52	0.00	44.45	12.62	72.04	8.78	**388.97**
30-Jun-24	48.00	23.00	39.62	24.30	0.00	53.00	11.99	0.15	0.00	5.20	2.14	1.17	0.00	0.00	38.79	5.13	63.30	4.00	**319.79**
Total	**1669.21**	**594.39**	**1388.19**	**731.25**	**0.00**	**3112.34**	**234.97**	**83.46**	**0.00**	**171.23**	**72.42**	**55.61**	**7.54**	**0.00**	**1159.61**	**420.74**	**2265.31**	**144.70**	**12110.97**

Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day

15.1 Import-Export of NR with WR during June 2024

Export of NR to WR (NR- WR)

Date	NR - WR 765 kV Gwalior - Agra 2xS/C	NR-WR 765 kV Gwalior-Phagi 2xS/C	NR-WR 765 kV Jabalpur- Orai D/C	NR-WR 765 kV Satna- Orai	NR-WR 765 kV Gwalior-Orai	NR - WR HVDC Champa-Kurukshetra	NR - WR HVDC Vindhyachal	NR - WR 400 kV Zerda-Kankroli	NR - WR 400 kV Zerda-Bhinmal	NR-WR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	NR - WR 220 kV Auraiya - Malanpur / Mehgaon	NR-WR 400 kV Rihand-Vindhyachal D/C	NR - WR HVDC Mundra - M'garh	NR - WR 765 kV Banaskata - Chittorgarh D/C	NR-WR 765 kV Varanasi-Vindhyachal	WR -NR 765 kV Chittorgarh-Neemach D/C	Total NR - WR
1-Jun-24	0.00	0.00	0.00	0.00	8.67	0.00	0.00	0.00		0.00	0.00	0.00	0.50	22.35	0.00	0.00	0.00	0.00	**31.52**
2-Jun-24	0.00	0.00	0.00	0.00	11.33	0.00	0.00	0.48		0.00	0.00	0.00	0.02	22.44	0.00	0.00	0.00	0.00	**34.27**
3-Jun-24	0.00	0.00	0.00	0.00	11.51	0.00	0.00	0.36		0.00	0.00	0.00	0.21	22.30	0.00	0.00	0.00	0.00	**34.38**
4-Jun-24	0.00	0.00	0.00	0.00	8.48	0.00	0.00	0.00		0.00	0.00	0.00	0.79	22.16	0.00	0.00	0.00	0.00	**31.43**
5-Jun-24	0.00	0.00	0.00	0.00	7.03	0.00	2.58	0.00		0.21	0.00	0.00	0.79	21.76	0.00	0.00	0.00	1.70	**34.07**
6-Jun-24	0.00	0.00	0.00	0.00	8.18	0.00	5.72	0.00		0.00	0.00	0.00	0.17	21.25	0.00	0.00	0.00	0.52	**35.84**
7-Jun-24	0.00	0.00	0.00	0.00	9.15	0.00	4.97	0.00		0.00	0.00	0.00	0.00	22.12	0.00	0.00	0.00	0.00	**36.24**
8-Jun-24	0.00	0.00	0.00	0.00	9.78	0.00	0.00	0.00		0.00	0.00	0.00	0.65	22.47	0.00	0.00	0.00	0.00	**32.90**
9-Jun-24	0.00	0.00	0.00	0.00	10.79	0.00	0.00	0.00		0.00	0.00	0.00	0.00	22.47	0.00	0.00	0.00	0.00	**33.26**
10-Jun-24	0.00	0.00	0.00	0.00	10.42	0.00	0.00	0.00		0.00	0.00	0.00	0.00	22.01	0.00	0.00	0.00	0.00	**32.43**
11-Jun-24	0.00	0.00	0.00	0.00	9.22	0.00	0.00	0.00		0.00	0.00	0.00	0.06	21.79	0.00	0.00	0.00	0.00	**31.07**
12-Jun-24	0.00	0.00	0.00	0.00	8.91	0.00	0.00	0.00		0.00	0.00	0.00	0.29	22.17	0.00	0.00	0.00	0.00	**31.37**
13-Jun-24	0.00	0.00	0.00	0.00	8.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	22.27	0.00	0.00	0.00	0.00	**30.95**
14-Jun-24	0.00	0.00	0.00	0.00	9.36	0.00	0.00	0.00		0.00	0.00	0.00	0.25	22.30	0.00	0.00	0.00	0.00	**31.91**
15-Jun-24	0.00	0.00	0.00	0.00	10.86	0.00	0.00	0.00		0.00	0.00	0.00	0.49	22.52	0.00	0.00	0.00	0.00	**33.87**
16-Jun-24	0.00	0.00	0.00	0.00	11.30	0.00	0.00	0.00		0.00	0.00	0.00	0.59	22.43	0.00	0.00	0.00	0.00	**34.32**
17-Jun-24	0.00	0.00	0.00	0.00	11.49	0.00	0.00	0.00		0.00	0.00	0.00	0.76	22.53	0.00	0.00	0.00	0.00	**34.78**
18-Jun-24	0.00	0.00	0.00	0.00	8.39	0.00	0.00	0.00		0.00	0.00	0.00	0.00	21.74	0.00	0.00	0.00	0.00	**30.13**
19-Jun-24	0.00	0.00	0.00	0.00	6.67	0.00	0.00	0.00		0.00	0.00	0.00	0.40	21.74	0.00	0.00	0.00	0.00	**28.81**
20-Jun-24	0.00	0.00	0.00	0.00		0.00	0.00	0.00		0.00	0.00	0.00	0.00	20.58	0.00	0.00	0.00	0.00	**20.58**
21-Jun-24	0.00	0.00	0.00	0.00	2.38	0.00	0.00	0.00		0.00	0.00	0.00	0.00	22.20	0.00	0.00	0.00	0.00	**24.58**
22-Jun-24	0.00	0.00	0.00	0.00	11.34	0.00	0.87	0.00		0.00	0.00	0.00	0.00	22.40	0.00	0.00	0.00	0.00	**34.61**
23-Jun-24	0.00	0.00	0.00	0.00	11.10	0.00	0.00	0.00		0.00	0.00	0.00	0.00	22.24	0.00	0.00	0.00	0.00	**33.34**
24-Jun-24	0.00	0.00	0.00	0.00	11.77	0.00	0.51	0.00		0.00	0.00	0.00	0.63	22.46	0.00	0.00	0.00	0.00	**35.37**
25-Jun-24	0.00	0.00	0.00	0.00	12.65	0.00	5.59	0.00		0.00	0.00	0.00	0.11	22.39	0.00	0.00	0.00	0.00	**40.74**
26-Jun-24	0.00	0.00	0.00	0.00	13.93	0.00	0.00	0.00		0.00	0.00	0.00	0.00	22.15	0.00	0.00	0.00	0.00	**36.08**
27-Jun-24	0.00	0.00	0.00	0.00	12.79	0.00	0.00	0.48		0.00	0.00	0.00	0.00	20.72	0.00	0.70	0.00	0.00	**34.69**
28-Jun-24	0.00	0.00	0.00	0.00	12.26	0.00	0.00	0.00		0.00	0.00	0.00	1.81	22.06	0.00	0.00	0.00	0.00	**36.13**
29-Jun-24	0.00	0.00	0.00	0.00	12.84	0.00	0.00	0.00		0.00	0.00	0.00	0.00	22.14	0.00	0.00	0.00	0.00	**34.98**
30-Jun-24	0.00	0.00	0.00	0.00	9.93	0.00	0.00	0.00	1.50	0.00	0.00	0.00	1.07	21.21	0.00	0.00	0.00	0.00	**33.71**
Total	**0.00**	**0.00**	**0.00**	**0.00**	**291.19**	**0.00**	**20.24**	**1.32**	**1.50**	**0.21**	**0.00**	**0.00**	**9.61**	**661.37**	**0.00**	**0.70**	**0.00**	**2.22**	**988.36**

Disclaimer:- Blank entry if the line under outage/shutdown on corrosponding day

15.2 Import-Export of NR with ER & NER during June 2024

Date	Import of NR from ER (ER-NR)																		Import of NR from NER (NER-NR)	
	ER - NR 765 kV Sasaram - Fatehpur	ER - NR 765 kV Gaya - Varanasi 2*S/C	ER - NR 765 kV Gaya Balia	ER - NR HVDC Alipurduar - Agra	ER-NR 400 kV Patna - Balia D/C	ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Balia D/C	ER - NR 400 kV Motihari - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Varanasi D/C	ER-NR 400 kV Sasaram - Varanasi	ER-NR 400 kV Sasaram - Allahabad	ER - NR 220 kV Sahupuri-Karamnasa	ER - NR 132 kV Sahupuri - Karamnasa	ER - NR 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	ER-NR 400 kV Naubatpur - Balia D/C	ER-NR 400 kV Biharshriff - Sahupuri D/C	Total ER -NR	NER-NR HVDC Biswanath Chariali - Agra	Total NER-NR
1-Jun-24	4.95	0.98	11.22		13.50	10.22	1.65	8.97		0.22	0.00	0.80	0.00		0.00	3.96	0.00	56.47	0.00	0.00
2-Jun-24	6.87	5.00	11.11		12.85	10.40	2.35	9.65		0.51	0.10	0.78	0.00		0.00	5.25	2.87	67.74	0.00	0.00
3-Jun-24	6.50	5.00	11.20		15.90	8.45	2.60	9.68		0.51	0.01	0.94		0.00		5.09	3.40	69.28	0.00	0.00
4-Jun-24	4.90	0.00	9.86		12.80	8.61	1.00	8.61	1.40	0.87	0.00	0.93			0.00	4.30	1.41	54.69	0.00	0.00
5-Jun-24	3.68	0.00	10.69		12.89	7.39	0.10	6.87	0.00	0.10	0.00	0.71			0.00	3.69	1.78	47.90	0.00	0.00
6-Jun-24	3.33	0.00	8.45		10.01	3.87	0.00	4.54		0.08	0.00	0.81			0.00	2.72	1.23	35.04	0.00	0.00
7-Jun-24	4.97	0.00	12.10		12.45	3.47	0.00	4.65		0.58	0.00	0.90			0.00	3.45	1.26	43.83	0.00	0.00
8-Jun-24	3.67	0.00	11.42		9.76	4.61	0.00	5.67		0.00	0.00	0.65			0.00	2.68	1.43	39.89	0.00	0.00
9-Jun-24	3.48	0.00	9.47		4.34	4.43	0.00	4.60		0.00	0.00	0.63	0.00	0.00	0.00	0.42	1.53	28.90	0.00	0.00
10-Jun-24	3.98	0.00	10.22		5.07	4.27	0.00	5.25		0.00	1.05	0.64	0.00		0.00	0.71	1.45	32.64	0.00	0.00
11-Jun-24	2.32	0.00	10.00		3.23	2.20	0.00	3.82		0.00	0.00	0.42			0.00	0.56	0.00	22.55	0.00	0.00
12-Jun-24	3.12	0.00	10.19		5.01	5.41	0.00	6.17		0.00	0.00	0.55	0.48		0.00	1.57	0.00	32.50	0.00	0.00
13-Jun-24	3.92	0.00	9.97		7.89	8.18	0.00	7.46		0.00	0.00	0.52	0.00		0.00	1.51	1.12	40.57	0.00	0.00
14-Jun-24	4.96	0.44	10.73		8.01	10.71	0.00	7.77		0.03	0.00	0.87	0.00		0.00	1.79	2.01	47.32	0.00	0.00
15-Jun-24	5.06	0.70	11.01		8.27	11.75	0.00	7.78	2.02	0.03	0.00	0.80	0.00	0.00	0.00	2.20	2.01	51.63	0.00	0.00
16-Jun-24	7.81	10.25	13.06		16.96	17.21	0.00	11.88		0.96	1.05	0.92		0.00	0.00	4.56	4.64	89.30	0.00	0.00
17-Jun-24	7.65	8.68	12.23		16.00	16.25	1.39	11.60		0.68	0.95	0.95	0.05	0.00	0.00	4.50	4.35	85.28	0.00	0.00
18-Jun-24	6.50	3.45	10.55		9.75	14.65	0.10	9.85		0.45	0.35	0.80			0.00	3.25	2.55	62.25	0.00	0.00
19-Jun-24	6.72	2.20	11.00		9.53	15.17	0.42	8.50	2.66	1.15	1.11	0.54			0.00	2.90	1.50	63.40	0.00	0.00
20-Jun-24	5.30	1.24	7.90		6.92	14.57	0.75	8.00		0.62	1.54	0.42			0.00	1.68	2.35	51.29	0.00	0.00
21-Jun-24	6.97	6.59	11.15		10.59	15.78	2.25	9.54	0.00	0.42	1.62	0.72	0.00		0.00	2.88	4.18	72.69	0.00	0.00
22-Jun-24	7.77	6.51	12.14		11.39	15.35	1.88	9.15		0.27	1.88	0.79			0.00	2.89	4.42	74.44	0.00	0.00
23-Jun-24	5.32	2.72	11.36		9.85	13.37	0.34	8.04		0.62	1.69	0.47	0.00		0.00	2.11	2.78	58.67	0.00	0.00
24-Jun-24	3.92	2.25	10.46		10.62	12.67	1.10	9.23		0.77	1.35	0.42	0.48		0.00	2.86	2.35	58.48	0.00	0.00
25-Jun-24	5.03	2.77	10.77		12.43	16.10	0.92	10.45		1.00	1.21	1.01	0.00		0.00	3.24	3.23	68.16	0.00	0.00
26-Jun-24	7.01	8.01	9.89		12.70	17.91	2.80	11.31		0.55	1.60	0.29	0.00		0.00	3.46	4.21	79.74	0.00	0.00
27-Jun-24	5.82	4.39	11.00		9.42	15.00	1.21	8.96		0.24	2.00	0.65			0.00	2.72	3.36	64.77	0.00	0.00
28-Jun-24	5.90	6.68	12.50		13.20	15.55	1.35	11.39		0.18	2.00	0.62	0.00	0.00	0.00	3.86	3.75	76.98	0.00	0.00
29-Jun-24	6.92	8.62	10.14		13.52	16.88	1.62	9.80		0.14	1.72	0.64	0.00		0.00	4.06	4.20	78.26	0.00	0.00
30-Jun-24	6.10	6.00	9.00		12.80	19.20	3.05	10.50		0.75	1.56	0.65			0.00	3.40	4.00	77.01	0.00	0.00
Total	**160.45**	**92.48**	**320.79**	**0.00**	**317.66**	**339.63**	**26.88**	**249.69**	**6.08**	**11.73**	**22.79**	**20.84**	**1.01**	**0.00**	**0.00**	**88.27**	**73.37**	**1731.67**	**0.00**	**0.00**

Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day

15.2 Import-Export of NR with ER & NER during June 2024

Date	Export of NR to ER (NR- ER)																		Export of NR to NER (NR-NER)	
	NR - ER 765 kV Sasaram - Fatehpur	NR - ER 765 kV Gaya - Varanasi 2*S/C	NR - ER 765 kV Gaya Balia	NR - ER HVDC Alipurduar - Agra	NR-ER 400 kV Patna - Balia D/C	NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	NR - ER 400 kV Biharshariff - Balia D/C	NR - ER 400 kV Motihari - Gorakhpur D/C	NR - ER 400 kV Biharshariff - Varanasi D/C	NR-ER 400 kV Sasaram - Varanasi	NR-ER 400 kV Sasaram - Allahabad	NR - ER 220 kV Sahupuri-Karamnasa	NR - ER 132 kV Sahupuri - Karamnasa	NR - ER 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	NR-ER 400 kV Balia-Naubatpur D/C	NR-ER 400 kV Sahupuri-Biharshariff D/C	Total NR -ER	NR-NER HVDC Biswanath Chariali - Agra	Total NR-NER
1-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.25	0.00	0.00		0.60	0.00	1.25	**2.10**	16.92	**16.92**
2-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.56	0.00	0.00	**0.56**	16.26	**16.26**
3-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00		0.64		0.00	0.00	**0.64**	17.20	**17.20**
4-Jun-24	0.00	1.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.60	0.00			0.44	0.00	0.00	**2.04**	16.98	**16.98**
5-Jun-24	0.00	2.31	0.00		0.00	0.00	0.00	0.00	0.00	0.00	1.28	0.00			0.65	0.00	0.00	**4.24**	17.25	**17.25**
6-Jun-24	0.00	1.59	0.00		0.00	0.00	2.24	0.00		0.00	0.72	0.00			0.59	0.00	0.00	**5.14**	15.98	**15.98**
7-Jun-24	0.00	0.28	0.00		0.00	0.00	2.05	0.00		0.00	0.50	0.00			0.60	0.00	0.00	**3.43**	16.86	**16.86**
8-Jun-24	0.00	1.24	0.00		0.00	0.00	1.82	0.00		0.37	1.30	0.00			0.61	0.00	0.00	**5.34**	17.13	**17.13**
9-Jun-24	0.00	4.41	0.00		0.00	0.00	4.52	0.00		0.87	1.51	0.00	0.00	0.08	0.62	0.00	0.00	**12.01**	23.14	**23.14**
10-Jun-24	0.00	2.45	0.00		0.00	0.00	4.60	0.00		0.58	0.00	0.00	0.00		0.65	0.00	0.00	**8.28**	11.93	**11.93**
11-Jun-24	0.00	6.09	0.00		0.00	0.00	6.36	0.00		1.38	2.00	0.00			0.58	0.00	0.73	**17.14**	13.77	**13.77**
12-Jun-24	0.00	3.68	0.00		0.00	0.00	5.05	0.00		1.02	1.57	0.00	0.00		0.59	0.00	0.35	**12.26**	24.09	**24.09**
13-Jun-24	0.00	0.48	0.00		0.00	0.00	3.01	0.00		0.28	0.91	0.00	0.00		0.54	0.00	0.00	**5.22**	20.32	**20.32**
14-Jun-24	0.00	0.00	0.00		0.00	0.00	1.64	0.00		0.00	0.48	0.00	0.00		0.61	0.00	0.00	**2.73**	15.87	**15.87**
15-Jun-24	0.00	0.00	0.00		0.00	0.00	1.45	0.00	0.00	0.00	0.28	0.00	0.00	0.00	0.46	0.00	0.00	**2.19**	15.30	**15.30**
16-Jun-24	0.00	0.00	0.00		0.00	0.00	1.98	0.00		0.00	0.00	0.00		0.10	0.51	0.00	0.00	**2.59**	16.01	**16.01**
17-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.10	0.52	0.00	0.00	**0.62**	14.10	**14.10**
18-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00			0.50	0.00	0.00	**0.50**	11.89	**11.89**
19-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.45	0.00	0.00	**0.45**	17.34	**17.34**
20-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00			0.44	0.00	0.00	**0.44**	14.50	**14.50**
21-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.67	0.00	0.00	**0.67**	9.24	**9.24**
22-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00			0.61	0.00	0.00	**0.61**	8.48	**8.48**
23-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.60	0.00	0.00	**0.60**	8.34	**8.34**
24-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.62	0.00	0.00	**0.62**	8.31	**8.31**
25-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.66	0.00	0.00	**0.66**	16.86	**16.86**
26-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.61	0.00	0.00	**0.61**	13.22	**13.22**
27-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00			0.48	0.00	0.00	**0.48**	13.86	**13.86**
28-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.64	0.00	0.00	**0.64**	13.20	**13.20**
29-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.58	0.00	0.00	**0.58**	12.98	**12.98**
30-Jun-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00			0.51	0.00	0.00	**0.51**	13.08	**13.08**
Total	**0.00**	**23.53**	**0.00**	**0.00**	**0.00**	**0.00**	**34.72**	**0.00**	**0.00**	**4.50**	**11.40**	**0.00**	**0.00**	**0.92**	**16.50**	**0.00**	**2.33**	**93.90**	**450.41**	**450.41**
Disclaimer:-	Blank entry if the line under outage/shutdown on corrosponding day																			

15.3 Import and Export of ER with WR & NER during June 2024

Date	Export of WR to ER (WR-ER)								Import of WR from ER (ER - WR)								Export of ER to NER (ER-NER)				Import of ER from NER (NER-ER)			
	WR - ER 765 kV Dharamjaygarh - Ranchi 2xS/C	WR - ER 765 kV Dharamjaygarh - Jharsuguda Q/C	WR - ER 765 kV Durg - Jharsuguda D/C	WR - ER 400 kV Sipat - Ranchi D/C	WR - ER 400 kV Raigarh - Jharsuguda- 2xD/C	WR - ER 220 kV Korba - Budhipadar D/C	WR - ER 220 kV Raigarh - Budhipadar	Total WR - ER	ER - WR 765 kV Dharamjaygarh - Ranchi 2xS/C	ER-WR 765 kV Dharamjaygarh - Jharsuguda D/C	ER-WR 765 kV Durg - Jharsuguda D/C	ER - WR 400 kV Sipat - Ranchi D/C	ER - WR 400 kV Raigarh - Jharsuguda- 2xD/C	ER - WR 220 kV Korba - Budhipadar D/C	ER - WR 220 kV Raigarh - Budhipadar	Total ER - WR	ER - NER 400 kV Binaguri - Bongaigaon D/C	ER - NER 400 kV Alipurduar - Bongaigaon D/C	ER - NER 220 kV Birpara - Salakati D/C	Total ER - NER	NER - ER 400 kV Binaguri - Bongaigaon D/C	NER - ER 400 kV Alipurduar - Bongaigaon 2xD/C	NER - ER 220 kV Birpara - Salakati D/C	Total NER - ER
1-Jun-24	28.8	14.6	0.0	8.2	2.2	0.4	0.0	**54.17**	0.0	4.1	8.4	0.0	4.3	0.8	1.5	**19.15**	0.00	0.00	0.00	**0.00**	9.77	15.35	2.94	**28.06**
2-Jun-24	24.1	5.2	0.0	6.2	1.9	0.3	0.0	**37.62**	0.0	6.6	9.4	0.0	3.6	0.9	1.6	**22.11**	0.00	0.00	0.00	**0.00**	8.71	16.76	3.18	**28.65**
3-Jun-24	27.8	6.5	0.0	6.6	1.7	0.4	0.1	**43.14**	0.0	8.3	10.1	0.0	3.2	0.8	1.3	**23.73**	0.00	0.00	0.00	**0.00**	8.74	20.25	3.87	**32.86**
4-Jun-24	31.5	7.7	0.0	7.5	3.2	0.7	0.0	**50.52**	0.0	2.6	7.8	0.0	3.0	0.5	1.5	**15.32**	0.00	0.00	0.00	**0.00**	8.82	14.19	2.80	**25.81**
5-Jun-24	30.9	12.2	0.0	8.0	4.7	1.6	1.2	**58.75**	0.0	2.4	5.1	0.0	2.1	0.1	0.4	**10.22**	0.00	0.00	0.00	**0.00**	7.31	10.85	2.20	**20.36**
6-Jun-24	37.0	12.3	0.0	9.6	4.7	1.4	0.0	**64.90**	0.0	6.2	3.5	0.0	2.8	0.0	0.7	**13.09**	0.00	0.00	0.00	**0.00**	4.59	7.26	1.47	**13.32**
7-Jun-24	38.5	5.3	0.0	9.4	2.9	0.5	0.0	**56.56**	0.0	11.9	4.1	0.0	3.8	0.2	1.3	**21.32**	0.00	0.00	0.00	**0.00**	3.23	5.74	0.69	**9.66**
8-Jun-24	41.6	0.7	0.1	9.8	2.9	0.6	0.0	**55.85**	0.0	14.1	3.9	0.0	4.2	0.3	1.2	**23.61**	0.00	0.00	0.01	**0.01**	4.96	8.17	1.07	**14.20**
9-Jun-24	49.9	3.5	1.2	12.0	3.6	0.6	0.0	**70.74**	0.0	17.3	1.7	0.0	4.4	0.3	0.9	**24.56**	0.00	0.00	0.00	**0.00**	7.78	12.45	1.83	**22.06**
10-Jun-24	51.8	2.5	1.9	12.2	4.8	0.9	0.0	**74.03**	0.0	13.0	1.0	0.0	3.3	0.1	1.1	**18.45**	0.00	0.00	0.07	**0.07**	3.37	5.82	0.51	**9.70**
11-Jun-24	53.1	3.0	3.1	13.3	6.2	1.2	0.0	**80.02**	0.0	15.1	0.8	0.0	2.2	0.0	0.8	**18.91**	0.13	0.06	0.10	**0.29**	1.58	2.90	0.31	**4.78**
12-Jun-24	46.6	3.7	1.9	11.5	9.8	1.4	0.0	**74.81**	0.0	7.9	1.5	0.0	3.2	0.0	1.3	**13.85**	0.00	0.00	0.00	**0.00**	5.51	7.96	0.96	**14.42**
13-Jun-24	43.6	1.9	0.4	10.2	16.4	0.7	0.0	**73.28**	0.0	12.0	2.2	0.0	9.4	0.1	1.5	**25.16**	0.00	0.00	0.00	**0.00**	6.65	8.05	1.10	**15.81**
14-Jun-24	43.2	1.5	0.0	10.3	16.8	0.0	0.0	**71.87**	0.0	16.7	4.2	0.0	11.5	0.5	0.3	**33.19**	0.00	0.00	0.00	**0.00**	7.23	7.28	1.19	**15.70**
15-Jun-24	44.5	0.3	0.1	11.2	16.8	0.7	0.0	**73.54**	0.0	20.7	3.6	0.0	9.3	0.4	2.2	**36.07**	0.00	0.00	0.00	**0.00**	8.38	11.29	1.87	**21.54**
16-Jun-24	32.2	2.0	0.0	8.3	12.3	0.5	0.0	**55.29**	0.0	13.6	6.6	0.0	7.3	0.8	2.5	**30.83**	0.00	0.00	0.00	**0.00**	10.37	14.05	2.37	**26.79**
17-Jun-24	36.1	2.9	0.0	8.2	13.6	0.0	0.0	**60.82**	0.0	12.6	6.2	0.0	9.3	2.0	2.5	**32.61**	0.00	0.00	0.00	**0.00**	9.34	12.49	2.16	**23.99**
18-Jun-24	41.0	4.3	0.0	9.6	16.2	0.0	0.0	**71.08**	0.0	12.1	6.1	0.0	11.0	1.6	2.9	**33.74**	0.00	0.00	0.00	**0.00**	9.56	12.94	2.26	**24.76**
19-Jun-24	39.9	8.6	0.1	9.4	15.8	0.2	0.0	**73.97**	0.0	7.6	4.9	0.0	10.7	1.0	2.6	**26.87**	0.00	0.00	0.00	**0.00**	11.27	14.61	2.74	**28.62**
20-Jun-24	43.1	13.9	0.3	10.1	16.8	0.0	0.0	**84.22**	0.0	10.9	4.4	0.0	12.3	2.1	1.9	**31.62**	0.00	0.00	0.00	**0.00**	10.29	12.78	2.62	**25.70**
21-Jun-24	30.6	6.6	0.0	8.0	13.6	0.0	0.0	**58.87**	0.0	15.6	7.1	0.0	10.7	1.6	0.0	**35.01**	0.00	0.00	0.00	**0.00**	6.26	6.10	1.33	**13.68**
22-Jun-24	38.1	6.2	0.0	10.0	16.0	0.0	0.0	**70.26**	0.0	20.8	8.6	0.0	10.8	1.3	2.5	**44.14**	0.00	0.00	0.00	**0.00**	4.63	5.23	0.95	**10.82**
23-Jun-24	40.1	1.1	0.0	11.1	17.7	0.3	0.0	**70.33**	0.0	24.4	7.6	0.0	10.5	0.8	1.6	**44.93**	0.00	0.00	0.00	**0.00**	4.03	4.31	0.82	**9.15**
24-Jun-24	42.7	1.5	0.0	11.1	16.6	0.2	0.0	**72.08**	0.0	21.5	8.3	0.0	10.2	0.9	1.6	**42.46**	0.00	0.01	0.00	**0.01**	5.59	5.57	1.06	**12.22**
25-Jun-24	42.8	1.4	0.0	11.0	16.0	0.1	0.0	**71.40**	0.0	22.2	8.4	0.0	11.4	0.9	1.5	**44.38**	0.00	0.00	0.00	**0.00**	9.06	10.49	1.75	**21.30**
26-Jun-24	42.4	2.4	0.0	9.8	15.1	0.0	0.0	**69.63**	0.0	23.6	7.1	0.0	12.7	1.4	1.9	**46.81**	0.00	0.00	0.00	**0.00**	7.46	8.17	1.36	**16.99**
27-Jun-24	43.2	2.2	0.0	9.4	15.0	0.0	0.0	**69.79**	0.0	21.3	7.4	0.0	12.6	1.8	2.1	**45.18**	0.00	0.00	0.00	**0.00**	7.64	8.57	1.45	**17.66**
28-Jun-24	40.9	6.4	0.0	9.2	14.9	0.0	0.0	**71.42**	0.0	17.2	7.6	0.0	13.4	1.9	2.2	**42.28**	0.00	0.00	0.00	**0.00**	7.77	8.69	1.90	**18.36**
29-Jun-24	34.7	8.4	0.0	7.5	12.7	0.0	0.0	**63.32**	0.0	12.7	8.8	0.0	10.8	1.9	2.3	**36.53**	0.00	0.00	0.00	**0.00**	9.68	11.07	2.12	**22.87**
30-Jun-24	28.4	10.3	0.0	6.2	10.5	0.0	0.0	**55.40**	0.1	8.6	9.2	0.1	11.3	1.5	2.1	**32.97**	0.00	0.00	0.00	**0.00**	10.85	12.76	2.43	**26.04**
Total	**1168.90**	**158.97**	**9.28**	**285.29**	**321.05**	**12.71**	**1.47**	**1957.67**	**0.11**	**403.73**	**175.66**	**0.14**	**235.08**	**26.67**	**47.74**	**889.13**	**0.13**	**0.06**	**0.18**	**0.38**	**220.43**	**302.14**	**53.31**	**575.88**

15.4 Import and Export of SR with ER & WR during June 2024

Date	Export of ER to SR (ER - SR)				Import of ER from SR (SR - ER)				Export of WR to SR (WR-SR)									Import of WR from SR (SR - WR)								
	ER - SR HVDC Gazuwaka	ER - SR HVDC Talchar -Kolar	ER-SR 765 kV Angul-Srikakulam D/C	Total ER - SR	SR - ER HVDC Gazuwaka	SR - ER HVDC Talchar -	SR-ER 765 kV Angul-Srikakulam D/C	Total SR - ER	WR - SR HVDC Bhadrawati	WR - SR 765 kV Sholapur - Raichur 2xS/C	WR-SR 765 kV Wardha - Nizamabad D/C	WR-SR 400KV Kolhapur-Kudgi D/C	WR-SR HVDC Raigarh-Puglur	WR-SR 220kV Xeldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	WR-SR 765 kV Warora - Warangal D/C	Total WR - SR	SR - WR HVDC Bhadrawati	SR - WR 765 kV Raichur - Sholapur 2xS/C	SR-WR 765 kV Wardha - Nizamabad D/C	SR-WR 400KV Kolhapur-Kudgi D/C	SR-WR HVDC Raigarh-Puglur	WR-SR 220kV Xeldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	SR-WR 765 kV Warangal - Warora D/C	Total SR - WR
1-Jun-24	2.50	40.99	43.56	**87.05**	0.00	0.00	0.00	**0.00**	0.00	4.69	29.54	0.00	39.40	2.72	1.78	33.60	**111.74**	7.20	3.83	0.00	25.84	0.00	0.00	0.00	0.00	**36.86**
2-Jun-24	3.71	40.99	34.68	**79.38**	0.00	0.00	0.00	**0.00**	0.00	0.00	13.61	0.00	29.57	2.43	2.12	19.11	**66.85**	7.20	16.15	0.28	31.27	0.00	0.00	0.00	0.10	**55.00**
3-Jun-24	3.70	40.97	33.52	**78.19**	0.00	0.00	0.00	**0.00**	0.00	4.76	15.43	0.00	26.65	2.60	2.08	20.47	**71.98**	7.19	10.16	2.48	25.51	0.00	0.00	0.00	1.23	**46.57**
4-Jun-24	3.69	39.65	40.80	**84.15**	0.00	0.00	0.00	**0.00**	0.00	5.81	25.88	0.00	36.23	2.47	1.59	31.09	**103.06**	7.19	3.98	0.00	25.66	0.00	0.00	0.00	0.00	**36.82**
5-Jun-24	3.69	40.98	37.83	**82.50**	0.00	0.00	0.00	**0.00**	1.75	3.84	22.33	0.00	42.35	2.48	1.55	26.44	**100.74**	5.37	7.42	0.00	27.59	0.00	0.00	0.00	0.00	**40.38**
6-Jun-24	2.84	40.98	32.44	**76.26**	0.52	0.00	0.00	**0.52**	7.27	2.35	14.01	0.00	38.58	2.45	1.71	19.89	**86.25**	0.00	12.35	0.20	27.49	0.00	0.00	0.00	0.03	**40.08**
7-Jun-24	0.00	34.55	29.29	**63.84**	3.32	0.00	0.00	**3.32**	7.28	1.04	7.91	0.00	36.21	2.27	1.68	12.74	**69.13**	0.00	19.26	1.88	28.75	0.00	0.00	0.00	0.84	**50.74**
8-Jun-24	0.00	40.97	23.41	**64.38**	3.44	0.00	0.00	**3.44**	7.30	0.00	2.59	0.00	29.69	2.07	1.53	5.33	**48.51**	0.00	36.51	5.76	37.72	0.00	0.00	0.00	3.35	**83.34**
9-Jun-24	0.00	40.96	18.25	**59.21**	6.20	0.00	0.00	**6.20**	3.64	0.00	2.25	0.00	16.70	1.87	1.40	3.58	**29.44**	11.44	46.88	10.34	37.06	1.12	0.00	0.00	9.03	**115.88**
10-Jun-24	0.00	40.99	26.50	**67.48**	8.05	0.00	0.00	**8.05**	0.00	0.00	7.58	0.00	5.35	2.07	1.42	8.87	**25.29**	23.86	41.79	3.36	13.41	16.83	0.00	0.00	2.76	**102.01**
11-Jun-24	0.00	40.98	27.70	**68.69**	10.02	0.00	0.00	**10.02**	0.00	0.00	8.43	0.00	19.30	2.13	1.97	9.29	**41.12**	21.53	33.85	2.79	27.56	2.30	0.00	0.00	2.29	**90.30**
12-Jun-24	0.00	40.97	34.70	**75.67**	9.60	0.00	0.00	**9.60**	0.00	0.17	13.44	0.00	15.74	2.14	1.70	16.85	**50.03**	15.21	26.45	1.70	29.81	0.00	0.00	0.00	1.11	**74.28**
13-Jun-24	0.49	40.98	33.33	**74.80**	8.41	0.00	0.00	**8.41**	0.00	0.26	10.45	0.00	21.14	2.12	1.60	14.18	**49.74**	7.20	25.33	0.37	33.50	0.00	0.00	0.00	0.21	**66.61**
14-Jun-24	2.17	40.98	34.28	**77.43**	4.59	0.00	0.00	**4.59**	0.00	1.04	14.78	0.00	18.59	2.05	1.68	18.40	**56.54**	13.72	23.06	0.04	35.47	0.00	0.00	0.00	0.02	**72.31**
15-Jun-24	0.00	40.96	35.01	**75.97**	10.33	0.00	0.00	**10.33**	0.00	0.00	10.77	0.00	20.58	2.12	1.53	13.40	**48.40**	23.65	33.62	0.29	41.76	0.00	0.00	0.00	0.12	**99.43**
16-Jun-24	0.00	40.99	40.34	**81.32**	11.00	0.00	0.00	**11.00**	0.00	0.00	15.78	0.00	18.71	2.18	1.50	18.22	**56.39**	23.71	29.46	0.00	38.93	0.00	0.00	0.00	0.00	**92.11**
17-Jun-24	0.00	40.97	41.29	**82.26**	7.22	0.00	0.00	**7.22**	0.00	0.00	19.02	0.00	20.55	2.24	1.59	22.37	**65.77**	23.85	20.89	0.00	36.69	0.00	0.00	0.00	0.00	**81.43**
18-Jun-24	0.00	40.97	46.56	**87.53**	6.81	0.00	0.00	**6.81**	0.00	0.95	28.55	0.00	17.88	2.01	1.66	29.06	**80.11**	20.36	13.40	0.00	30.46	0.00	0.00	0.00	0.00	**64.22**
19-Jun-24	0.00	40.97	49.77	**90.74**	4.97	0.00	0.00	**4.97**	0.00	4.26	35.13	0.00	16.68	1.91	1.57	35.16	**94.71**	11.99	9.03	0.00	28.10	0.00	0.00	0.00	0.00	**49.13**
20-Jun-24	0.00	40.95	47.32	**88.26**	4.96	0.00	0.00	**4.96**	0.00	3.37	31.91	0.00	17.80	1.98	1.78	30.61	**87.44**	11.70	12.58	0.00	30.37	0.00	0.00	0.00	0.00	**54.66**
21-Jun-24	0.00	40.96	44.24	**85.21**	4.95	0.00	0.00	**4.95**	0.00	2.74	25.21	0.00	25.04	1.64	1.18	25.09	**80.90**	7.20	21.10	0.01	33.78	0.00	0.00	0.00	0.14	**62.23**
22-Jun-24	0.00	40.97	38.28	**79.24**	4.95	0.00	0.00	**4.95**	0.00	0.00	17.10	0.00	16.10	2.01	0.93	16.17	**52.32**	13.89	29.73	0.01	38.96	0.00	0.00	0.00	0.13	**82.73**
23-Jun-24	0.00	40.94	27.48	**68.42**	6.71	0.00	0.00	**6.71**	0.00	0.00	6.20	0.00	17.40	1.93	1.08	5.62	**32.22**	23.84	44.18	1.24	41.66	0.00	0.00	0.00	1.97	**112.89**
24-Jun-24	0.00	40.96	33.64	**74.60**	9.63	0.00	0.00	**9.63**	0.00	0.00	10.39	0.00	24.24	1.99	1.60	11.17	**49.40**	23.85	35.41	0.51	41.34	0.00	0.00	0.00	0.93	**102.04**
25-Jun-24	0.00	40.98	33.58	**74.56**	9.62	0.00	0.00	**9.62**	0.00	0.00	9.74	0.00	18.78	2.08	1.63	9.17	**41.40**	23.86	36.50	0.09	40.64	0.00	0.00	0.00	0.23	**101.32**
26-Jun-24	0.00	40.98	35.39	**76.36**	11.30	0.00	0.00	**11.30**	0.00	0.00	11.75	0.00	16.45	2.22	1.56	10.71	**42.68**	23.85	38.39	0.66	43.54	0.00	0.00	0.00	1.23	**107.68**
27-Jun-24	0.00	39.92	36.02	**75.94**	10.48	0.00	0.00	**10.48**	0.00	0.00	13.97	0.00	17.50	2.06	1.15	12.90	**47.57**	23.82	37.08	0.78	41.03	2.61	0.00	0.00	1.20	**106.52**
28-Jun-24	0.00	40.99	39.09	**80.08**	8.94	0.00	0.00	**8.94**	0.00	0.67	18.39	0.00	22.32	2.11	1.38	16.61	**61.48**	15.83	31.53	0.06	38.32	0.00	0.00	0.00	0.30	**86.05**
29-Jun-24	0.00	40.99	43.36	**84.35**	4.81	0.00	0.00	**4.81**	0.00	1.76	26.90	0.00	18.63	0.82	0.52	25.42	**74.05**	11.99	17.45	0.00	34.00	0.00	0.00	0.00	0.00	**63.45**
30-Jun-24	0.00	40.99	47.64	**88.63**	4.82	0.00	0.00	**4.82**	0.00	4.13	35.44	0.00	27.98	0.00	0.00	35.55	**103.09**	11.98	9.93	0.00	28.64	0.00	0.00	0.00	0.00	**50.56**
TOTAL	**22.79**	**1220.43**	**1089.30**	**2332.52**	**175.68**	**0.00**	**0.00**	**175.68**	**27.24**	**41.83**	**504.48**	**0.00**	**692.13**	**61.15**	**44.46**	**557.06**	**1928.36**	**422.51**	**727.33**	**32.86**	**994.86**	**22.86**	**0.00**	**0.00**	**27.23**	**2227.65**

16. भूटान , नेपाल, बांग्लादेश एवं म्यान्मार के साथ अंतरराष्ट्रीय विद्युत विनिमय

अप्रैल 2024 से मार्च 2025 April 2024 to March 2025

अंतरराष्ट्रीय विद्युत विनिमय [भारत 'से निर्यात'/ 'को आयात']

Transnational Exchange ('Export from'/'Import to' India)

माह MONTH	भूटान BHUTAN		नेपाल NEPAL		बांग्लादेश BANGLADESH		म्यान्मार MYANMAR	
	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)
अप्रैल 2024 APR'24	86.47	18.58	346.08	0.00	694.34	0.00	0.76	0.00
मई 2024 MAY'24	51.37	68.17	271.84	3.92	655.55	0.00	0.74	0.00
जून 2024 JUN'24	0.00	520.23	20.51	152.32	709.14	0.00	0.87	0.00
कुल Total	**137.84**	**606.98**	**638.42**	**156.24**	**2059.03**	**0.00**	**2.37**	**0.00**

* Based on daily operational data

16.1 Import from neighbouring countries during Jun 2024

(All figures in MU)

Date	Import from Bhutan								Import from Bangladesh		Import from Nepal			Import from Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurdwar D/C*	400 kV Jigmeling-Alipurdwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Jun-24	1.63	0.00	0.00	0.00	2.12	3.18	0.56	0.00	0.00	0.00	0.00	2.38	0.00	0.00
2-Jun-24	1.64	0.00	0.00	0.00	1.66	2.49	0.59	0.00	0.00	0.00	0.00	2.06	0.00	0.00
3-Jun-24	1.41	0.00	0.00	0.00	1.79	2.69	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4-Jun-24	0.14	0.00	0.00	0.00	2.38	3.58	0.00	0.00	0.00	0.00	0.00	2.98	0.00	0.00
5-Jun-24	2.48	0.18	0.00	0.00	2.77	4.17	1.09	0.00	0.00	0.00	0.00	1.87	0.00	0.00
6-Jun-24	0.57	0.00	0.00	0.00	2.29	3.45	1.13	0.00	0.00	0.00	0.00	2.64	0.00	0.00
7-Jun-24	0.20	0.00	0.00	0.00	1.65	2.48	0.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00
8-Jun-24	0.00	0.00	0.00	0.00	1.97	2.96	1.04	0.00	0.00	0.00	0.00	2.05	0.00	0.00
9-Jun-24	0.35	0.00	0.00	0.00	2.00	3.01	0.82	0.00	0.00	0.00	0.00	0.53	0.00	0.00
10-Jun-24	0.09	0.00	0.00	0.00	1.55	2.32	1.05	0.00	0.00	0.00	0.00	2.00	0.00	0.00
11-Jun-24	0.45	0.00	0.00	0.00	1.57	2.36	0.94	0.00	0.00	0.00	0.00	2.74	0.00	0.00
12-Jun-24	0.94	0.00	0.00	0.00	3.15	4.74	1.07	0.01	0.00	0.00	0.00	3.68	0.00	0.00
13-Jun-24	3.29	0.30	0.00	0.00	4.94	7.43	1.18	0.00	0.00	0.00	0.00	4.50	0.00	0.00
14-Jun-24	5.62	1.66	0.00	0.00	6.02	9.04	1.12	0.43	0.00	0.00	0.00	6.12	0.00	0.00
15-Jun-24	5.92	2.03	0.00	0.00	3.98	5.99	0.80	0.00	0.00	0.00	0.00	7.32	0.00	0.00
16-Jun-24	8.25	3.14	0.21	0.00	4.22	6.34	0.79	0.06	0.00	0.00	0.00	6.20	0.02	0.00
17-Jun-24	9.65	3.07	0.07	0.00	4.05	6.10	0.82	0.08	0.00	0.00	0.00	6.20	0.03	0.00
18-Jun-24	8.75	3.25	0.00	0.00	3.85	5.79	0.78	0.01	0.00	0.00	0.00	8.69	0.03	0.00
19-Jun-24	10.02	3.23	0.00	0.00	3.94	5.91	0.73	0.00	0.00	0.00	0.00	11.63	0.03	0.00
20-Jun-24	14.41	3.67	0.23	0.00	5.58	8.40	0.92	0.19	0.00	0.00	0.00	12.26	0.23	0.00
21-Jun-24	10.92	3.08	0.35	0.00	6.17	9.28	1.23	0.41	0.00	0.00	0.00	11.61	0.21	0.00
22-Jun-24	8.63	1.75	0.01	0.00	4.40	6.61	1.26	0.00	0.00	0.00	0.15	11.15	0.04	0.00
23-Jun-24	9.38	2.00	0.16	0.00	4.34	6.52	1.25	0.25	0.00	0.00	0.00	10.03	0.03	0.00
24-Jun-24	8.63	2.29	0.41	0.00	5.69	8.56	0.98	0.42	0.00	0.00	0.00	7.67	0.01	0.00
25-Jun-24	15.45	1.48	0.49	0.00	6.60	9.24	1.07	0.34	0.00	0.00	0.00	10.86	0.01	0.00
26-Jun-24	13.64	3.37	0.43	0.00	6.25	9.39	0.90	0.40	0.00	0.00	0.00	10.84	0.15	0.00
27-Jun-24	12.49	3.96	0.41	0.00	5.99	9.01	1.30	0.27	0.00	0.00	0.11	10.62	0.06	0.00
28-Jun-24	10.03	2.59	0.00	0.00	6.32	9.49	1.24	0.32	0.00	0.00	0.09	11.39	0.03	0.00
29-Jun-24	13.43	3.41	0.96	0.00	6.67	10.00	1.14	0.40	0.00	0.00	0.56	11.97	0.05	0.00
30-Jun-24	13.88	3.59	0.78	0.00	6.77	10.16	1.02	0.36	0.00	0.00	0.82	11.69	0.01	0.00
Total	**192.31**	**48.05**	**4.53**	**0.00**	**120.67**	**180.69**	**27.86**	**3.95**	**0.00**	**0.00**	**1.73**	**193.69**	**0.94**	**0.00**

Based on SEM/Energy meter data for links where available (*Mangdechu generation reciept at APD through a bypassed arrangement at 400kV Punatsanchu station)

16.2 Export to neighbouring countries during Jun 2024

(All figures in MU)

Date	Export to Bhutan								Export to Bangladesh		Export to Nepal			Export to Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurdwar D/C*	400 kV Jigmeling-Alipurdwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Jun-24	0.00	0.18	2.15	1.40	0.00	0.00	0.00	0.31	21.16	2.04	0.99	0.00	1.56	0.03
2-Jun-24	0.00	0.20	2.39	1.55	0.00	0.00	0.00	0.39	21.14	2.11	0.92	0.00	1.50	0.03
3-Jun-24	0.00	0.31	2.29	1.60	0.00	0.00	0.00	0.47	21.18	1.91	1.04	0.14	2.04	0.03
4-Jun-24	0.00	0.82	1.90	1.52	0.00	0.00	0.00	0.23	21.48	1.99	1.06	0.00	1.61	0.03
5-Jun-24	0.00	0.00	1.89	1.31	0.00	0.00	0.00	0.19	21.40	2.10	1.11	0.00	1.89	0.03
6-Jun-24	0.00	0.87	2.70	1.27	0.00	0.00	0.00	0.19	21.55	2.01	1.04	0.00	1.27	0.03
7-Jun-24	0.00	1.06	2.02	0.35	0.00	0.00	0.00	0.11	21.55	1.69	0.95	0.06	1.77	0.03
8-Jun-24	0.52	1.42	2.49	0.00	0.00	0.00	0.00	0.10	21.48	1.85	1.05	0.00	1.75	0.03
9-Jun-24	0.00	1.00	2.79	0.00	0.00	0.00	0.00	0.22	21.48	2.10	1.07	0.00	1.78	0.03
10-Jun-24	0.00	1.06	2.33	0.00	0.00	0.00	0.00	0.09	21.40	1.85	1.06	0.00	1.99	0.03
11-Jun-24	0.00	0.80	2.86	0.00	0.00	0.00	0.00	0.05	21.33	1.97	1.08	0.00	1.77	0.03
12-Jun-24	0.00	1.37	2.36	0.00	0.00	0.00	0.00	0.00	21.49	2.08	1.06	0.00	1.99	0.01
13-Jun-24	0.00	0.00	1.08	0.00	0.00	0.00	0.00	0.24	21.56	1.65	1.02	0.00	2.20	0.01
14-Jun-24	0.00	0.00	0.68	0.04	0.00	0.00	0.00	0.00	16.82	1.50	0.92	0.00	2.24	0.02
15-Jun-24	0.00	0.00	0.27	0.41	0.00	0.00	0.00	0.08	21.35	1.64	1.00	0.00	1.91	0.02
16-Jun-24	0.00	0.00	0.00	0.29	0.00	0.00	0.00	0.00	21.20	1.68	0.96	0.00	0.00	0.03
17-Jun-24	0.00	0.00	0.00	0.46	0.00	0.00	0.00	0.00	22.17	1.76	0.85	0.00	0.00	0.09
18-Jun-24	0.00	0.00	0.19	0.82	0.00	0.00	0.00	0.00	22.49	1.73	1.01	0.00	0.00	0.03
19-Jun-24	0.00	0.00	0.11	0.84	0.00	0.00	0.00	0.00	22.55	1.34	0.86	0.00	0.00	0.03
20-Jun-24	0.00	0.00	0.00	0.45	0.00	0.00	0.00	0.00	22.54	1.39	0.06	0.00	0.00	0.03
21-Jun-24	0.00	0.00	0.00	0.26	0.00	0.00	0.00	0.00	22.43	1.59	0.45	0.00	0.00	0.01
22-Jun-24	0.00	0.00	0.00	0.52	0.00	0.00	0.00	0.19	22.43	1.85	0.00	0.00	0.00	0.02
23-Jun-24	0.00	0.00	0.00	0.36	0.00	0.00	0.00	0.00	22.32	2.00	0.66	0.00	0.00	0.03
24-Jun-24	0.00	0.00	0.00	0.38	0.00	0.00	0.00	0.00	22.31	2.13	0.80	0.00	0.00	0.03
25-Jun-24	0.00	0.00	0.00	0.57	0.00	0.00	0.00	0.00	22.43	2.14	0.89	0.00	0.00	0.04
26-Jun-24	0.00	0.00	0.00	0.11	0.00	0.00	0.00	0.00	22.40	1.96	0.58	0.00	0.00	0.03
27-Jun-24	0.00	0.00	0.00	0.09	0.00	0.00	0.00	0.00	22.03	2.04	0.00	0.00	0.00	0.03
28-Jun-24	0.00	0.00	1.35	0.60	0.00	0.00	0.00	0.00	22.13	1.75	0.00	0.00	0.00	0.03
29-Jun-24	0.00	0.00	0.00	0.36	0.00	0.00	0.00	0.00	22.40	1.79	0.00	0.00	0.00	0.03
30-Jun-24	0.00	0.00	0.00	0.48	0.00	0.00	0.00	0.00	22.57	1.67	0.00	0.00	0.00	0.03
Total	**0.52**	**9.08**	**31.84**	**16.02**	**0.00**	**0.00**	**0.00**	**2.86**	**650.77**	**55.34**	**22.48**	**0.20**	**27.26**	**0.87**

Based on SEM/Energy meter data for links where available (*Mangdechu generation reciept at APD through a bypassed arrangement at 400kV Punatsanchu station)

17. एसटीओए (द्विपक्षीय एवं सामूहिक) एवं डी एस एम बिलिंग का ब्योरा – जून 2024

STOA (BILATERAL AND COLLECTIVE) & DSM BILLING DETAILS - JUNE 2024

द्विपक्षीय एसटीओए BILATERAL SHORT TERM OPEN ACCESS

जून 2024 June 2024			Apr'24- Jun'24	
नोडल क्षे.भा.प्रे.के. का नाम Name of Nodal RLDC	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved(MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved (MU)
ऊ. क्षे. NR	1112	4831	2370	7316
प.क्षे. WR	326	576	1121	2149
द.क्षे. SR	310	462	1202	6599
पू.क्षे. ER	475	2114	1131	5042
पूर्वोत्तर क्षे. NER	106	110	328	148
कुल TOTAL	2329	8093	6152	21254

एसटीओए SHORT TERM OPEN ACCESS

	सामूहिक एसटीओए Collective STOA		द्विपक्षीय एसटीओए Bilateral STOA	
माह MONTH	क्रेताओं / विक्रेताओं की स. NO. of Buyers/Sellers	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)
अप्रैल 2024 Apr'24	9860	7128	1565	5921
मई 2024 May'24	497	8268	2258	7240
जून 2024 Jun'24	10971	8732	2329	8093
कुल TOTAL	21328	24128	6152	21254

मासिक डी एस एम बिलिंग का ब्योरा* 2024-25
MONTHLY DSM BILLING DETAILS* 2024-25

अनंतिम आँकड़े
Provisional data
subject to change

करोड़ रु. मे (RS. IN CRORES)

क्षेत्र REGION ⟹ / सप्ताह WEEK ⇓	उत्तरी क्षेत्र NORTH	पश्चिमी क्षेत्र WEST	दक्षिणी क्षेत्र SOUTH	पूर्वी क्षेत्र EAST	पूर्वोत्तर क्षेत्र NORTH EAST
03.06.24 to 09.06.24	285.87	307.00	345.51	228.26	66.41
10.06.24 to 16.06.24	342.18	324.70	371.05	288.08	68.36
17.06.24 to 23.06.24	237.72	299.36	408.31	247.00	61.37
24.06.24 to 30.06.24	182.85	278.89	349.39	203.69	57.18

* Amount shown is Payble to DSM pool ^Provisional Data

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
POWER MARKET INFORMATION (Source IEX & PXIL)
पावर एक्स्चेंज के माध्यम से विनिमय – माह:- जून 2024
EXCHANGES THROUGH POWER EXCHANGES -JUNE 2024

क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
1	AD HYDRO POWER LIMITED	उत्तरी क्षेत्र NR	90.71	0.00
2	ADANI SOLAR ENERGY JAISALMER TWO PRIVATE LIMITED(Project-2)		34.52	0.00
3	ADANI SOLAR ENERGY JODHPUR TWO LIMITED		11.84	0.00
4	ADANI SOLAR ENERGY RJ TWO PRIVATE LIMITED		47.03	0.00
5	ADANI SOLAR ENERGY RJ TWO PRIVATE LIMITED (infirm)		4.26	0.00
6	ADANI SOLAR ENERGY RJ TWO PRIVATE LIMITED(Phalodi)		34.65	0.00
7	Adept Renewable Technologies Private Limited		31.31	0.00
8	Amp Energy Green Six Private Limited		28.51	0.00
9	Amplus Ages Private Limited		16.36	0.00
10	Ayana Renewable Power Three Private Limited		24.16	0.00
11	Ayana Renewable Power Three Private Limited Infirm		11.34	0.00
12	Budhil HEP (Greenko Budhil Hydro Power Pvt. Ltd.)		8.29	0.00
13	Chandigarh (UT)		3.90	25.79
14	Delhi		361.89	98.56
15	Grian Energy Private Limited		26.03	0.00
16	Haryana		90.07	279.63
17	Himachal Pradesh		416.85	21.94
18	IGSTPS JHAJJAR		6.82	0.00
19	Jammu Kashmir		63.31	7.28
20	KARCHAM WANGTOO HYDRO ELECTRIC PLANT.		69.21	0.00
21	NEA Nepal Upper Chameliya Hydropower Project		2.24	0.00
22	NEA Nepal Upper Kalangagad Hydropower Project new		2.02	0.00
23	NEPAL ELECTRICITY AUTHORITY TANAKPUR		0.00	14.63
24	NTPC Dadri Stage I		4.29	0.00
25	NTPC Dadri Stage II		7.06	0.00
26	NTPC Rihand stage I		7.98	0.00
27	NTPC Rihand stage II		7.19	0.00
28	NTPC Rihand stage III		7.71	0.00
29	NTPC Singrauli		8.61	0.00
30	NTPC Tanda Stage II		4.12	0.00
31	NTPC Unchahar Stage I		1.39	0.00
32	NTPC Unchahar Stage II		3.28	0.00
33	NTPC Unchahar Stage III		1.44	0.00
34	NTPC Unchahar Stage IV		2.95	0.00
35	North Central Railway Prayagraj		0.00	40.87
36	Onevolt Energy Private Limited		25.73	0.00
37	Punjab		49.49	1396.55
38	RENEW SURYA PRATAP PRIVATE LIMITED		47.19	0.00
39	RENEW SURYA ROSHNI PRIVATE LIMITED-Fatehgarh-III PS		15.82	0.00
40	RENEW SURYA ROSHNI PRIVATE LIMITED-Fatehgarh-III PS_Infirm		13.50	0.00
41	Rajasthan		61.42	1103.30
42	ReNew Surya Ravi Private Limited		56.29	0.00
43	SHREE CEMENT LIMITED TPS		42.26	0.00
44	Sainj HEP		35.49	0.70

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
POWER MARKET INFORMATION (Source IEX & PXIL)
पावर एक्स्चेंज के माध्यम से विनिमय – माह:- जून 2024
EXCHANGES THROUGH POWER EXCHANGES -JUNE 2024

क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
45	Serentica Renewables India 4 Private Limited		8.47	0.00
46	Serentica Renewables India 4 Private Limited- Infirm		3.22	0.00
47	Singoli Bhatwari HEP		25.00	0.00
48	Sorang HEP (Himachal Sorang Power Pvt. Ltd.))		25.92	0.00
49	TPSL 200MW TPTCL Banderwala		24.82	0.00
50	TPSL Banderwala SECI 100_INF		21.60	0.00
51	Transition Energy Services Private Limited		17.60	0.00
52	Transition Green Energy Private Limited		25.63	0.00
53	Transition Sustainable Energy Services Private Limited		2.73	0.00
54	Uttar Pradesh		111.40	791.71
55	Uttarakhand		48.44	244.99

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
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			(DAM+HP DAM+RTM)	
56	ACB (INDIA) LIMITED	पश्चिमी क्षेत्र WR	43.05	0.10
57	ADANI GREEN ENERGY TWENTY SIX A LIMITED_infirm		0.47	0.00
58	ADANI RENEWABLE ENERGY FIFTY FIVE LIMITED_Infirm		20.00	0.00
59	ADANI RENEWABLE ENERGY HOLDING FOUR LIMITED		180.23	0.00
60	ADANI WIND ENERGY KUTCHH FOUR Ltd. Nakhatrana		84.05	0.00
61	ADANI WIND ENERGY KUTCHH FOUR Ltd. Nakhatrana (infirm)		9.57	0.00
62	Adani Green Energy Twenty Five A Limited_Infirm		4.52	0.00
63	Adani Green Energy Twenty Five B Limited_infirm		71.08	0.00
64	Adani Green Energy Twenty Six B Limited_infirm		11.85	0.00
65	Adani Power Limited - Raigarh TPP		156.44	0.86
66	Adani Power Limited-Raipur TPP		231.03	2.75
67	ArcelorMittal Nippon Steel India Limited		0.00	61.69
68	BHARAT ALUMINIUM COMPANY LTD		20.42	0.00
69	BHARAT ALUMINIUM COMPANY LTD (Bulk Consumer)		0.00	20.76
70	CONTINUUM POWER TRADING (TN) PRIVATE LIMITED		10.60	0.00
71	Chhattisgarh		495.26	4.71
72	D B Power Limited		99.20	0.00
73	DGEN MEGA POWER PROJECT		31.86	0.00
74	DHARIWAL STU OTH		5.33	0.00
75	Daman and Diu - Dadra and Nagar Haveli		0.42	167.40
76	Dhariwal ISTS		1.44	0.04
77	GMR Warora Energy Limited		7.34	0.00
78	Goa WR		29.20	5.03
79	Gujarat		104.53	830.60
80	Jaypee Nigrie Super Thermal Power Plant		218.46	0.00
81	Jhabua Power Limited		43.74	22.61
82	Jindal Power Limited, Stage-1		134.02	5.98
83	Jindal Power Limited, Stage-2		258.30	8.74
84	Jindal Steel & Power Ltd , DCPP		55.25	2.86
85	MAHAN ENERGEN LIMITED U#1		165.66	0.60
86	MAHAN ENERGEN LIMITED U#2		97.97	3.24
87	MB POWER (MADHYA PRADESH) LIMITED		8.46	4.67
88	Madhya Pradesh		475.82	64.22
89	Maharashtra		174.45	419.60
90	Maruti Clean Coal and Power Limited		8.21	0.46
91	NREL-Dayapar Wind		11.85	0.00
92	NTPC Gadarwara		11.97	0.00
93	NTPC Jhanor Gandhar GPS		0.00	0.47
94	NTPC Kawas GPS		0.00	0.42
95	NTPC Korba Stage I &II		1.98	0.00
96	NTPC Korba Stage III		0.13	0.00
97	NTPC Lara Stage I		2.01	0.00
98	NTPC Mouda Stage I		2.79	0.00
99	NTPC Mouda Stage II		4.12	0.00

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			(DAM+HP DAM+RTM)	
100	NTPC SAIL POWER COMPANY LIMITED		1.27	0.00
101	NTPC Sipat Stage I		2.23	0.00
102	NTPC Sipat Stage II		1.16	0.00
103	NTPC Solapur		1.74	0.00
104	NTPC Vindhyachal Stage I		2.20	0.00
105	NTPC Vindhyachal Stage II		1.43	0.00
106	NTPC Vindhyachal Stage III		1.47	0.00
107	NTPC Vindhyachal Stage IV		1.49	0.00
108	NTPC Vindhyachal Stage V		1.02	0.00
109	NTPC khargone		10.93	0.00
110	Nani Virani Wind Energy Private Limited		4.89	0.00
111	R.K.M POWERGEN PRIVATE LIMITED		53.90	0.00
112	Ratnagiri Gas & Power Private Limited		0.00	0.66
113	SKS Power Generation Chhattisgarh Limited		21.15	7.07
114	Sasan Power Limited		5.41	0.00
115	TRN ENERGY PRIVATE LIMITED		1.14	0.00
116	The Tata Power Co Ltd (MTPS)		19.32	0.00
117	WIND FIVE RENERGY LIMITED		13.89	0.00

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			(DAM+HP DAM+RTM)	
118	Andhra Pradesh	दक्षिणी क्षेत्र SR	258.06	231.56
119	COASTAL ENERGEN PRIVATE LIMITED		27.30	0.00
120	Goa SR		0.05	9.67
121	IL&FS TAMIL NADU POWER COMPANY LIMITED		6.86	0.00
122	JINDAL POWER LIMITED SIMHAPURI		18.20	9.76
123	JSW RENEW ENERGY LIMITED INFIRM		0.58	0.00
124	JSW RENEW ENERGY TWO LIMITED INFIRM		4.58	0.00
125	Karnataka		831.72	54.71
126	Kerala		55.34	54.92
127	MEENAKSHI ENERGY LIMITED		16.34	0.00
128	NLC INDIA LIMITED NEYVELI NEW THERMAL POWER STATION		10.36	0.00
129	NLC INDIA LIMITED THERMAL POWER STATION I EXPANSION		4.97	0.00
130	NLC INDIA LIMITED THERMAL POWER STATION II EXPANSION		0.34	0.00
131	NLC INDIA LIMITED THERMAL POWER STATION II STAGE I		0.49	0.00
132	NLC INDIA LIMITED THERMAL POWER STATION II STAGE II		0.91	0.00
133	NLC Tamilnadu Power Limited		3.84	0.00
134	NTECL VALLUR		3.17	0.00
135	NTPC KUDGI		8.69	0.00
136	NTPC Ramagundam Stage I &II		9.56	0.00
137	NTPC Ramagundam Stage III		1.84	0.00
138	NTPC Simhadri Stage I		2.31	0.00
139	NTPC Simhadri Stage II		2.73	0.00
140	NTPC Telangana		6.66	0.00
141	OSTRO KANNADA POWER PRIVATE LIMITED		27.82	0.00
142	Pondicherry UT		28.27	0.37
143	RENEW SURYA OJAS PRIVATE LIMITED-Wind		91.67	0.00
144	RENEW SURYA OJAS PRIVATE LIMITED_INFIRM		12.12	0.00
145	RENEW SURYA ROSHNI PRIVATE LIMITED Koppal PS		4.18	0.00
146	Ramagundam Floating solar		8.46	0.00
147	SEIL ENERGY INDIA LIMITED		67.08	0.00
148	SEIL Energy India Limited Project II		13.63	0.00
149	Serentica Renewables India 1 Private Limited- Koppal (Infirm)		19.58	0.00
150	Simhadri FSP 15 MW		2.29	0.00
151	Simhadri Floating solar (10 MW)		1.39	0.00
152	Tamil Nadu		199.91	339.65
153	Telangana		95.73	1155.28

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			(DAM+HP DAM+RTM)	
154	ADHUNIK POWER & NATURAL RESOURCES LIMITED	पूर्वी क्षेत्र ER	15.29	0.00
155	Basochhu Hydropower Plant Bhutan		14.58	0.00
156	Bihar		162.55	139.30
157	Chilime Hydro Power Project NEPAL ELECTRICITY AUTHORITY		4.71	0.00
158	Chuzachen HEP		0.17	0.00
159	DEVIGHAT NEPAL ELECTRICITY AUTHORITY		3.00	0.00
160	Damodar Valley Corporation		37.54	91.58
161	GMR KAMALANGA ENERGY LTD-CTU		2.40	0.00
162	IND BARATH ENERGY UTKAL LIMITED		9.01	0.00
163	JORETHANG LOOP HEP, DANS ENERGY PRIVATE LIMITED		0.59	0.00
164	Jharkhand		14.47	153.70
165	Jindal India Thermal Power Limited		86.42	0.00
166	KALI GANDAKI NEPAL ELECTRICITY AUTHORITY		41.16	0.00
167	KANTI BIJLEE UTPADAN NIGAM LIMITED		12.55	0.00
168	Kabeli B-1 Hydro Power Project NEPAL ELECTRICITY AUTHORITY		1.47	0.00
169	LIKHU-iV NEA		15.42	0.00
170	Lower Modi Hydro Power Project NEPAL ELECTRICITY AUTHORITY		1.05	0.00
171	MAITHON POWER LIMITED		2.93	0.00
172	MARSYANGDI NEPAL ELECTRICITY AUTHORITY		22.06	0.00
173	NABINAGAR POWER GENERATING COMPANY LIMITED		25.71	0.00
174	NEA - Bihar STU		0.00	27.78
175	NEA Nepal Upper Dordi A Hydropower Project		1.72	0.00
176	NEPAL ELECTRICITY AUTHORITY MUZAFFARPUR		0.00	17.86
177	NEPAL ELECTRICITY AUTHORITY-MIDDLE MARSYANGDI		21.51	0.00
178	NTPC BARH Stage I		8.20	0.00
179	NTPC BARH Stage II		33.19	0.00
180	NTPC Darlipali		0.54	0.00
181	NTPC Farakka stage I		20.30	0.00
182	NTPC Farakka stage III		7.71	0.00
183	NTPC Kahalgaon stage I		7.33	0.00
184	NTPC Kahalgaon stage II		10.26	0.00
185	NTPC North Karanpura STPS		17.97	0.00
186	NTPC Talcher Stage I		0.19	0.00
187	NTPP BRBCL		10.95	0.00
188	Nikachhu Hydropower Project		4.92	0.00
189	Odisha		121.33	282.90
190	Sikkim		18.57	7.06
191	Solu Hydropower Project NEPAL ELECTRICITY AUTHORITY		5.46	0.00
192	TRISHULI NEPAL ELECTRICITY AUTHORITY		5.89	0.00
193	Tashiding HEP, Shiga Energy Private Limited		0.33	0.00
194	West Bengal		152.33	416.44

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			(DAM+HP DAM+RTM)	
195	AGARTALA GAS BASED POWER STATION	पूर्वोत्तर क्षेत्र NER	1.46	0.00
196	ASSAM GAS BASED POWER STATION		3.27	0.00
197	Arunachal Pradesh		39.30	0.02
198	Assam		253.09	35.17
199	Bongaigaon Thermal Power Station NTPC		11.39	0.00
200	KAMENG HYDRO POWER STATION		19.64	0.00
201	Kopili 2 IN		2.06	0.00
202	Manipur		6.08	13.04
203	Meghalaya		53.26	9.44
204	Mizoram		11.55	0.00
205	Nagaland		3.73	7.50
206	Palatana Plant		6.63	0.00
207	Tripura		10.99	13.14
Total			8732	8732

19. INFORMATION ABOUT RENEWABLE ENERGY CERTIFICATE MECHANISM

MONTH : JUNE 2024

RE Source & Unit wise break up (01.06.2024-30.06.2024)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	12	1	1	1
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	9	4	4	1
5	Small Hydro	0	0	0	0
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	0	0	0	0
9	Bio-fuel cogeneration	0	0	0	0
	Total	**21**	**5**	**5**	**2**

RECs Issued (01.06.2024-30.06.2024)

Sr.No.	Non Solar	Solar	Total
1	193801	58079	251880

Redemption of REC (01.06.2024-30.06.2024)

Sr.No.	Non Solar	Solar	Total
1	670985	21840	692825

19. INFORMATION ABOUT RENEWABLE ENERGY CERTIFICATE MECHANISM

MONTH : JUNE 2024

RE Source & Unit wise break up (Apr'24-Jun'24)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	13	2	2	2
2	Urban or Municipal Waste	0	0	12	1
3	Solar Thermal	0	0	0	0
4	Solar PV	29	9	8	4
5	Small Hydro	24	1	24	1
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	0	0	0	0
9	Bio-fuel cogeneration	0	0	0	0
	Total	**66**	**12**	**46**	**8**

RECs Issued (Apr'24-Jun'24)

Sr.No.	Non Solar	Solar	Total
1	8008752	201492	8210244

Redemption of REC (Apr'24-Jun'24)

Sr.No.	Non Solar	Solar	Total
1	3400147	74288	3474435

19. INFORMATION ABOUT RENEWABLE ENERGY CERTIFICATE MECHANISM

MONTH : JUNE 2024

RE Source & Unit wise break up Since Inception to Jun'24

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	3146	532	2979	533
2	Urban or Municipal Waste	12	1	12	1
3	Solar Thermal	0	0	0	0
4	Solar PV	1664	583	1480	484
5	Small Hydro	869	43	881	45
6	Others	4	2	3	1
7	Geothermal	0	0	0	0
8	Biomass	402	37	378	35
9	Bio-fuel cogeneration	823	91	383	55
	Total	**6920**	**1289**	**6116**	**1154**

RECs Issued since Inception to Jun'24

Sr.No.	Non Solar	Solar	Total
1	123113790	13741870	136855660

Redemption of REC since Inception to Jun'24

Sr.No.	Non Solar	Solar	Total
1	81904476	12523545	94428021

REC Closing balance as on 30.06.2024

Sr.No.	Non Solar	Solar	Total
1	41209314	1218325	42427639

20.Details of Grid Events during the Month of June 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Delhi	01-06-2024 15:40	01-06-2024 18:05	02:25	0	98	0.000	0.131	64982	74546	i)220/66kV DSIDC(DV) has double main Bus arrangement at 220kV side. ii)During antecedent condition, incoming power at DSIDC(DV) through 220 kV DSIDC-Bawana (DV) ckt-1 & ckt-2 was approx. 44MW & 58MW respectively (as per SCADA). 220 kV DSIDC-Bawana (DV) ckt-1, 220/66kV 160MVA ICT-1 & 100MVA ICT-2 at DSIDC(DV) were on 220kV Bus-1 at DSIDC(DV) and 220 kV DSIDC-Bawana (DV) ckt-2, 220/66kV 100MVA ICT-3 & 160MVA ICT-4 at DSIDC(DV) were on 220kV Bus-2 at DSIDC(DV). Bus coupler of 220kV Bus-1 & Bus-2 was in open condition. 220kV DSIDC-Narela(DV) ckt-1 & ckt-2 were not in service. iii)As reported, at 15:40 hrs, 220 kV DSIDC-Bawana (DV) ckt-1 & ckt-2 tripped on R-Y phase to phase fault and 3-phase fault respectively. For both the faults, fault distance was 3.9 km from Bawana(DV) and zone-1 distance protection operated at Bawana(DV). iv)But, as per PMU and SCADA SOE, at 15:40 hrs, 220 kV DSIDC-Bawana (DV) ckt-2 tripped on Y-B phase to phase fault. Further at 15:41 hrs, 220 kV DSIDC-Bawana (DV) ckt-1 tripped on R-Y phase to phase fault. v)As 220kV DSIDC-Narela(DV) ckt-1 & ckt-2 were not in service and 220 kV DSIDC-Bawana (DV) ckt-1 & ckt-2 also tripped, complete blackout occurred at 220/66kV DSIDC(DV). vi)As per PMU at Abdullapur(PG), Y-B followed by R-Y phase to phase fault is observed with fault clearing time of 160ms & 120msec respectively. vii)As per SCADA, change in demand of approx. 308MW is observed in Delhi control area. But as reported, by SLDC-Delhi, load loss of approx. 98 MW occurred. viii)As reported, at 18:05hrs, power supply restored at 220/66kV DSIDC(DV) using 66kV Bawana-I sector ckt-1 & 2 from TPDDL.	1) 220 kV DSIDC-Bawana (DV) ckt-1 2) 220 kV DSIDC-Bawana (DV) ckt-2
2	GI-2	Haryana, Punjab, Uttar Pradesh and Rajasthan	01-06-2024 13:26	01-06-2024 13:52	00:26	1835	545	2.627	0.664	69862	82023	i)As reported, at 13:26 hrs, 765 kV Meerut-Bhiwani (PG) Ckt-1 tripped from Bhiwani(PG) end only on R-N phase to earth fault with fault current of 4.679 kA and fault distance of 148.715 km from Bhiwani(PG) end. Line was successfully auto-reclosed from Meerut(PG) end. ii)As per PMU at Bhiwani(PG), R-N phase to earth fault with fault clearing time of 80ms is observed. Voltage dipped upto 0.779 p.u. at Bhiwani(PG). iii)As per SCADA, change in NR total solar generation of approx. 1695 MW (ISTS Solar: ~1695 MW) was observed which almost revived within 05 minutes. Change in Rajasthan wind generation of approx. 140 MW is also observed. iv)Due to a significant dip in RE generation, frequency dropped by 0.121 Hz (from 49.976 Hz to 49.855 Hz). v)As per SCADA, total change in Northern region demand of approx. 545 MW (Punjab: ~445 MW, UP: ~100 MW) is observed. Demand dipped in Punjab area due to df/dt operation as informed by Punjab SLDC. As reported by SLDC Punjab, load loss of ~427 MW occurred due to df/dt operation in Punjab.	1) 765 KV Meerut-Bhiwani (PG) Ckt-1
3	GI-2	Delhi, Haryana, Punjab, Uttar Pradesh and Rajasthan	01-06-2024 13:43	01-06-2024 14:35	00:52	3180	1300	4.567	1.570	69630	82782	i)As reported, at 13:43 hrs, 400 KV Bawana-Mundka (DV) Ckt-1 & 2 tripped from Mundka(DV) end only on R-B phase to phase fault (phase sequence issue observed) (exact reason, nature and location of fault yet to be shared); no tripping reported at Bawana(DV) end. Fault was sensed in zone-4 at Bawana(DV), but zone-4 got reset as fault got cleared before zone-4 time delay. ii)As per PMU at Abdullapur(PG), R-Y phase to phase fault with fault clearing time of 80ms is observed. Voltage dipped upto 0.859 p.u. at Abdullapur(PG). iii)As per SCADA, change in NR total solar generation of approx. 3120 MW (ISTS Solar: ~2710 MW, Rajasthan Solar: ~410 MW) was observed which almost revived within 05 minutes. Change in Rajasthan wind generation of approx. 60 MW is also observed. iv)Due to a significant dip in RE generation, frequency dropped by 0.171Hz (from 49.948 Hz to 49.777 Hz). v)As per SCADA, total change in Northern region demand of approx. 1300 MW (Delhi: ~270 MW, Haryana: ~225 MW, Punjab: ~585 MW, UP: ~220 MW) is observed. Demand dipped in UP and Punjab area due to df/dt operation as informed by UP and Punjab SLDC. As reported by SLDC Punjab, load loss of ~533 MW occurred due to df/dt operation in Punjab.	1) 400 KV Bawana-Mundka (DV) Ckt-1 2) 400 KV Bawana-Mundka (DV) Ckt-2
4	GD-1	Rajasthan	02-06-2024 01:04	02-06-2024 02:18	01:14	0	247	0.000	0.335	53813	73790	i)220/132kV Kushkhera(RS) has double main Bus arrangement at 220kV side. ii)During antecedent condition, incoming power at Kushkhera(RS) S/s through 220 KV Neemrana(PG)-Kushkhera(RS) (RS) Ckt & 220 KV Bhiwadi(PG)-Kushkhera(RS) (RS) Ckt was approx. 91MW & 142MW respectively (as per SCADA). 220kV lines from Kushkhera to Alwar(RS), Kishangarh(RS) were not in service. Two buses are connected only through isolator. iii)As reported, at 01:04hrs, Y-phase CT of 220 KV Neemrana(PG)-Kushkhera(RS) (RS) Ckt at Kushkhera(RS) S/s blasted which resulted in Bus bar fault and bus bar protection operated. iv)As the two buses are connected only through isolator, all the 220kV lines from Kushkhera(RS) and 220/132kV 160MVA ICT-1 & 2 at Kushkhera(RS) tripped which led to total blackout at 220/132kV Kushkhera(RS) S/s. v)As further reported, 220 KV Neemrana(PG)-Kushkhera(RS) (RS) Ckt tripped from both ends, but 220 KV Bhiwadi(PG)-Kushkhera(RS) (RS) Ckt tripped only from Kushkhera(RS) end. vi)As per PMU at Bhiwadi(PG), at 01:04 hrs, Y-N followed by B-N phase to earth fault is observed with fault clearing time of 120ms. vii)As per SCADA, change in demand of approx. 247MW in Rajasthan control area is observed. viii)As reported, at 02:18 hrs, 220kV Kushkhera-Alwar(RS) line charged and at the same time 220/132kV 160MVA ICT-1 at Kushkhera(RS) also charged.	1) 220 KV Neemrana(PG)-Kushkhera(RS) (RS) Ckt 2) 220 KV Bhiwadi(PG)-Kushkhera(RS) (RS) Ckt 3) 220/132kV 160MVA ICT-1 at Kushkhera(RS) 4) 220/132kV 160MVA ICT-2 at Kushkhera(RS)
5	GD-1	Jammu and Kashmir	03-06-2024 17:33	03-06-2024 20:20	02:47	0	120	0.000	0.163	57513	73807	i)As reported, at 17:33hrs, 220 kV Barn(JK)-Kishenpur(PG) Ckt-1 tripped on R-N phase to earth fault with fault current of 5.094kA from Kishepur(PG) end (as per DR). As per DR, zone-1 distance protection operated at Kishepur(PG) end (exact reason and location of fault yet to be shared). ii)During the same time, 220 kV Barn(JK)-Kishenpur(PG) Ckt-2 also tripped from Barn(JK) end only on R-N phase to earth fault with fault current of 3.9kA from Kishepur(PG) end (as per DR). As per DR, fault sensed in zone-2 at Kishenpur(PG) and line remained charged from Kishenpur (PG) end. iii)Due to tripping of both 220 kV Barn(JK)-Kishenpur(PG) Ckt-1 & 2, complete blackout occurred at 220/132kV Barn(J&K) S/s. iv)As per PMU at Kishenpur(PG), R-N phase to earth fault with fault clearing time of 120ms is observed. v)As per SCADA, load loss of approx. 120MW occurred in J&K control area.	1)220 kV Barn(JK)-Kishenpur(PG) Ckt-1 2)220 kV Barn(JK)-Kishenpur(PG) Ckt-2

20.Details of Grid Events during the Month of June 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
6	GI-1	Haryana	03-06-2024 00:38	03-06-2024 01:26	00:48	0	565	0.000	0.763	53147	74075	i)As reported, at 00:38 hrs, bursting of B-ph CT of 220kV bus coupler-2 at Panipat(BB) end occurred which created B-N phase to earth fault in busbar differential zone. The reason of bursting of the B-ph CT was observed to be some internal fault in Heptacare make CT installed on the bay on 29th November 2018. ii)The Numerical low Impedance type MiCom P741 Bus-Bar Differential Protection Scheme (ALSTOM make) sensed the fault and operated tripping all the elements on either side of bus coupler i.e. 220kV Bus-1 & Bus-2 at Panipat(BB). iii)As per PMU at Panipat(BBMB), Y-N phase to earth fault is observed with fault clearing time of 120ms. (phase sequence issue observed) iv)As per SCADA, load loss of approx. 565 MW (~445 MW in Haryana and ~120 MW in Delhi control area) is observed. v)As reported by BBMB, 220kV Bus-1 at Panipat(BB) was charged by closing A-17 Breaker of 220 KV Panipat-Dhulkote (BB) Ckt-1 at 01:26 hrs and 220kV Bus-2 at Panipat(BB) was charged by closing A-18 Breaker of 220 KV Panipat-Dhulkote (BB) Ckt-2 at 01:36 hrs. vi)As remedial action taken, on 03rd June 2024 an old and used Rade Koncar make CT of same ratio i.e. 1200/1-1-1-1-1A was tested thoroughly and installed in place of bursted CT and bus coupler-2 was charged at 17:38 hrs on 03rd June 2024.	1)220 KV PanipatTH(HV)-Panipat(BB) (HVPNL) Ckt-1 2)220 KV PanipatTH(HV)-Panipat(BB) (HVPNL) Ckt-2 3)220 KV PanipatTH(HV)-Panipat(BB) (HVPNL) Ckt-3 4)220 KV PanipatTH(HV)-Panipat(BB) (HVPNL) Ckt-4 5)220 KV Panipat(BB)-Narela(DV) (BBMB) Ckt-1 6)220 KV Panipat(BB)-Narela(DV) (BBMB) Ckt-2 7)220 KV Panipat(BB)-Narela(DV) (BBMB) Ckt-3 8)220 KV Panipat(BB)-Chajpur(HV) (HVPNL) Ckt-1 9)220 KV Panipat(BB)-Chajpur(HV) (HVPNL) Ckt-2 10)220 KV Panipat-Dhulkote (BB) Ckt-1 11)220 KV Panipat-Dhulkote (BB) Ckt-2 12)220 KV Panipat-Charkhi Dadri (BB) Ckt 13)220 KV Panipat(BB)-Pipli Ckt 14)400/220kV 450 MVA ICT-1 at Panipat(BB) 15)400/220kV 500 MVA ICT-2 at Panipat(BB) 16)220/132kV 100 MVA ICT-1 at Panipat(BB) 17)220/132kV 100 MVA ICT-2 at Panipat(BB) 18)220/33kV 60 MVA ICT-1 at Panipat(BB) 19)220/33kV 60 MVA ICT-2 at Panipat(BB)
7	GD-1	Himachal Pradesh	03-06-2024 22:47	04-06-2024 00:14	01:27	238	0	0.430	0.000	55406	71913	i)During antecedent condition, 66MW Unit-1, 2, 3 & 6 at Pong HEP were running and generating approx. 58MW, 60MW, 60MW and 60MW respectively (as per SCADA). 66MW Unit-4 & 5 at Pong HEP were not in service. ii)As reported, at 22:47 hrs, while stopping 66MW Unit-1 at Pong(BB) B-ph limb failed to open due to failure of operating mechanism of CB. Generator Circuit Breaker did not open. Abnormal sound was observed from TG Unit 1 and smoke was also observed from Field Discharge Cubicle of Excitation System of Unit 1. LBB protection didn't operate in this case in CB failure condition. iii)During the same time, all other running units, i.e., 66MW Unit-2, 3 & 6 at Pong HEP tripped Generator Transformer Back up Earth Fault Protection operation due to system imbalance. iv)Further, at 22:58 hrs, 220kV Bus 1 & 2 at Pong(BB) were de-energized manually by opening all CBs of 220kV feeders to save Unit 1 as the B-ph limb of machine CB was stuck in closed position. v)As per PMU at Jalandhar(PG), no fault is observed in the system. However, fluctuation in voltage is observed. vi)As per SCADA, generation loss of approx. 238 MW at Pong HEP (BB) and no load loss is observed in HP control area.	1)66 MW Pong HPS - UNIT 2 2)66 MW Pong HPS - UNIT 3 3)66 MW Pong HPS - UNIT 6 4)220 KV Jessore(HP)-Pong(BB) (PG) Ckt 5)220 KV Pong(BB)-Dasuya(PS) (BBMB) Ckt-1 6)220 KV Pong(BB)-Dasuya(PS) (BBMB) Ckt-2 7)220 KV Jalandhar-Pong (BB) Ckt-1 8)220 KV Jalandhar-Pong (BB) Ckt-2 9)220 KV Bairasiul(NH)-Pong(BB) (PG) Ckt
8	GD-1	Jammu and Kashmir	04-06-2024 19:31	04-06-2024 20:26	00:55	61	0	0.111	0.000	54929	71784	i)Power flows from Alusteng(PG) to Drass(PG) to Kargil to Khalsti to Leh (radial connection). Generation of Chutak is connected to Kargil and generation of Nimoo bazgo is connected to Leh. ii)As reported, at 19:31 hrs, 220 KV Alusteng-Drass (PG) Ckt tripped on B-N phase to earth fault with fault distance of 115km from Alusteng(PG). iii)With the tripping of 220 KV Alusteng-Drass (PG) Ckt, complete blackout occurred at 220/66kV Drass(PG) and supply to Kargil, Khalsti and Leh also failed. iv)Generation of Chutak and Nimoo Bazgo tripped due to loss of evacuation path resulting in generation loss of approx. 37MW & 24MW at Chutak and Nimoo Bazgo respectively (as per SCADA). v)As per PMU at Amargarh(INDIGRID), B-N phase to earth fault is observed with fault clearing time of 80ms. vi)As per SCADA, no change in demand is observed in J&K control area.	1) 220 KV Alusteng-Drass (PG) Ckt
9	GI-2	Jammu and Kashmir	04-06-2024 13:05	04-06-2024 13:25	00:20	230	0	0.324	0.000	70947	82825	i)During antecedent condition, 130MW Unit-1, 2 & 3 at Dulhasti HEP were running and generating 127MW, 121MW and 126MW respectively (as per SCADA). Total generated power of 374MW was evacuating through 400 kV Dulhasti(NH)-Kishenpur(PG) (PG) Ckt-1 & 2. ii)As reported, at 13:05hrs, 400 kV Dulhasti(NH)-Kishenpur(PG) (PG) Ckt-1 tripped on R-Y phase to phase fault (exact reason and location of fault yet to be shared). As per DR at Dulhasti(NH), fault current was approx. Ir=~2.906kA and Iy=~2.539kA from Dulhasti(NH) and voltage dipped upto ~0.54 p.u. As per PMU at Kishepur(PG), fault current was approx. 5.5kA from Kishenpur(PG) and voltage dipped upto ~0.779 p.u. iii)At Dulhsti HEP(NHPC), there is a scheme which identifies a dead bus situation and issues trip command to running units connected to that bus to minimize the over-speed of unit due to a load throw off. iv)Due to this fault, undervoltage was sensed which was identified as dead bus condition by this scheme and it sent command to unit-controller to trip the connected units, i.e. unit-1 & 2. v)Due to this, 130MW Unit-1 and 2 at Dulhasti HEP tripped. vi)As per PMU at Kishenpur (PG), R-Y phase to phase fault is observed with fault clearing time of 80ms. vii)As per SCADA, generation loss of approx. 230MW is observed at Dulhasti HEP.	1)400 KV Dulhasti(NH)-Kishenpur(PG) (PG) Ckt-1 2)130MW Unit-1 at Dulhasti HEP 3)130MW Unit-2 at Dulhasti HEP
10	GI-1	Jammu and Kashmir	07-06-2024 16:29	07-06-2024 16:45	00:16	0	363	0.000	0.479	60503	75810	i)As reported, at 16:29hrs, 220/132kV 160MVA ICT 2 at Barn(J&K) tripped on over current earth fault protection operation (exact reason, location and type of fault yet to be shared). ii)Due to shifting of loading of 220/132kV 160MVA ICT 2 to 220/132kV 160MVA ICT 1 & 3, both ICTs also tripped on overloading. iii)As per PMU at Kishenpur(PG), R-B phase to phase fault with delayed fault clearing time of 2160 ms is observed. iv)As per SCADA, load loss of approx. 363MW occurred in J&K control area.	1)220/132kV 160MVA ICT 2 at Barn(J&K) 2)220/132kV 160MVA ICT 1 at Barn(J&K) 3)220/132kV 160MVA ICT 3 at Barn(J&K)

20.Details of Grid Events during the Month of June 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
11	GI-1	Punjab	07-06-2024 08:56	07-06-2024 09:12	00:16	0	168	0.000	0.248	58735	67824	i)During antecedent condition, 220 kV lines from Jalandhar(BB) to Alawalpur, Pong ckt-2, Jamalpur ckt-2, Jamsher ckt-1, 220/66kV 100MVA ICT-2, 220/132kV 100MVA ICT-4 & 220/132kV 90MVA ICT-2 were connected to 220kV Bus-2 and Butari, Pong ckt-1, Jamalpur ckt-1, Jamsher ckt-2, 220/66kV 100MVA ICT-1 and 220/132kV 100MVA ICT-1 & 3 were connected to 220kV Bus-1 at Jalandhar(BB) S/s. 220/132kV 90MVA ICT-2 was not in service. ii)As reported, at 08:56 hrs, R-N phase to earth fault occurred on 220 KV Jalandhar(BB)-Jamsher(PS) (BB) Ckt-1 with fault distance of 1.039km from Jalandhar(BB) end. As per DR, fault current was ~16.2kA from Jalandhar(BB) end and fault was sensed in zone-1 at Jalandhar(BB) end. iii)During fault clearance from Jalandhar(BB) end, CB did not open due to air leakage in its tripping block assembly. iv)Due to failure of CB opening of 220 KV Jalandhar(BB)-Jamsher(PS) (BB) Ckt-1 from Jalandhar(BB) end, LBB protection operated which led to tripping of all 220 kV lines (Alawalpur, Pong ckt-2, Jamalpur ckt-2 & Jamsher ckt-1), 220/66kV 100MVA ICT-2, 220/132kV 100MVA ICT-4 and Bus coupler between 220kV Bus-1 & Bus-2 connected to 220kV Bus-2 and 220kV Bus-2 became dead at Jalandhar(BB). v)As further reported, during the same time, 220/66kV 100MVA ICT-1 at Jalandhar(BB) (connected to Bus-1) also tripped on backup protection due overloading as the load of 220/66kV 100MVA ICT-2 shifted on ICT-1. vi)As per PMU at Jalandhar(PG), R-N phase to earth fault is observed with delayed fault clearing time of 240ms. vii)As per SCADA, change in demand of approx. 168 MW is observed in Punjab control area.	1)220 KV Jalandhar-Pong (BB) Ckt-2 2)220 KV Jalandhar-Jamalpur (BB) Ckt-2 3)220 KV Jalandhar(BB)-Jamsher(PS) (BB) Ckt-1 4)220KV Bus 2 at Jalandhar(BB) 5)220 KV Jalandhar(BB)-Alawalpur(PS) (PSTCL) Ckt 6)220/132kV 100MVA ICT-4 at Jalandhar(BB) 7)220/66kV 100MVA ICT-2 at Jalandhar(BB) 8)220/66kV 100MVA ICT-1 at Jalandhar(BB)
12	GD-1	Rajasthan	08-06-2024 21:03	08-06-2024 21:41	00:38	125	0	0.225	0.000	55520	75925	i)Generation of 220kV AHEJ4L PSS4(IP) station evacuates through 220 KV Adani RenewPark_SL_FGARH_FBTL (AREPRL)-AHEJ4L PSS 4 HB_FGRAH_FBTL (AHEJ4L) (AREPRL) Ckt. During antecedent condition, AHEJ4L PSS4(IP) station was generating approx. 125MW (as per PMU). ii)As reported, at 21:03hrs, 220 KV Adani RenewPark_SL_FGARH_FBTL (AREPRL)-AHEJ4L PSS 4 HB_FGRAH_FBTL (AHEJ4L) (AREPRL) Ckt tripped due to R-N phase to earth fault (exact reason, location and nature of protection operated yet to be shared). iii)Due to tripping of 220 KV Adani RenewPark_SL_FGARH_FBTL (AREPRL)-AHEJ4L PSS 4 HB_FGRAH_FBTL (AHEJ4L) (AREPRL) Ckt, AHEJ4L PSS4(IP) S/s lost its connectivity from grid and blackout occurred at 220kV AHEJ4L PSS4(IP) S/s. iv)As per PMU at Adani Fatehgarh Solar Park(IP), R-N phase to earth fault is observed with fault clearing time of 80ms. v)As per PMU, wind generation loss of approx. 125 MW is observed at AHEJ4L PSS4(IP).	1) 220 KV Adani RenewPark_SL_FGARH_FBTL (AREPRL)-AHEJ4L PSS 4 HB_FGRAH_FBTL (AHEJ4L) (AREPRL) Ckt
13	GI-2	Rajasthan	08-06-2024 19:53	08-06-2024 23:29	03:36	168	0	0.310	0.000	54212	72136	i)As reported, at 19:53hrs, due to heavy thunderstorm weather condition, B-ph jumper of dead end tower of 220 kV Akal-Lala (RS) line got earthed at distance of 46.77 meter from Akal(RS) S/s which caused B-N phase to earth fault on 220 kV Akal-Lala (RS) ckt with fault current of Ib=~25kA from Akal(RS) end. As per DR at Akal(RS) end of 220 kV Akal-Lala (RS) ckt, R-N phase to earth fault in zone-1 with fault current of Ir=~25.4kA is observed (phase sequence issue). ii)As reported, during the same time, due to very high fault current, 400/220 kV 315 MVA ICT-3 and 400/220 kV 500 MVA ICT-4 tripped instantaneously on High Set overcurrent protection operation at Akal (RS) S/s. 400 KV Akal-Jodhpur (RS) Ckt also tripped from Akal(RS) end on zone-5 (reverse) distance protection operation (not tripped from the remote end). Tie CB of 400/220kV 315MVA ICT-3 also opened along with tripping of 400/220kV 315MVA ICT-3 which separated 400kV Bus-1 & Bus-2. iii)As reported, at the same time, Bus coupler of 220kV Bus-1 & Bus-2 and all 220kV lines (Mulana, Rajgarh, Amarsagar, Giral, Bhensara(RS) ckt-1, Dangri ckt-1 & ckt-2) connected to 220kV Bus-2 at Akal(RS) S/s also tripped (exact reason of tripping yet to be shared). Due to tripping of Bus coupler, 220kV Bus-1 and Bus-2 separated at Akal(RS). iv)As reported, further at 19:55 hrs, due to bad weather conditions, Y-N phase to earth fault occurred on 400 KV Akal-Kankani (RS) ckt & 400 KV Akal-Jaisalmer2(Bhainsra) (RS) ckt (D/C lines on same tower) at a distance of approx. 3km from Akal(RS) S/s which led to tripping of both the mentioned lines (exact operation of protection yet to be shared). v)As reported, due to tripping of 400 KV Akal-Kankani (RS) ckt & 400 KV Akal- Jaisalmer2(Bhainsra) (RS) ckt, high voltage was observed on the 400 KV Akal-Barmer (RS) Ckt and line voltage reached up to 436kV on the same line and 400 KV Akal-Barmer (RS) Ckt tripped from Barmer(RS) end on over-voltage protection operation and also from Akal(RS) end on receiving DT from Barmer(RS) end. vi)Due to tripping of 400 KV Akal-Barmer (RS) Ckt, 400kV Bus-1 lost its connectivity from grid and 400kV Bus-1 and elements connected to 400kV Bus-1 (400/220 kV 500 MVA ICT-1 & 2, 220kV Bus-1 and 220kV lines connected to 220kV Bus-1) became dead at Akal(RS) S/s. vii)After all the above mentioned tripping events, only 400kV Bus-2 remain charged through 400kV Akal-Ramgarh (RS) D/C.	1)400 KV Akal-Jodhpur (RS) Ckt 2)400/220 kV 315 MVA ICT 3 at Akal(RS) 3)400/220 kV 500 MVA ICT 4 at Akal(RS) 4)220kV Akal-Lala (RS) ckt 5)220kV Akal-Mulana (RS) ckt 6)220kV Akal- Rajgarh (RS) ckt 7)220kV Akal- Amarsagar (RS) ckt 8)220kV Akal- Giral (RS) ckt 9)220kV Akal- Bhensara(RS) ckt-1 10)220kV Akal- Dangri (RS) ckt-1 11)220kV Akal- Dangri (RS) ckt-2 12)400 KV Akal-Kankani (RS) Ckt 13)400 KV Akal-Jaisalmer2(Bhainsra) (RS) Ckt 14)400 KV Akal-Barmer (RS) Ckt
14	GI-2	Rajasthan	09-06-2024 11:21	09-06-2024 13:05	01:44	2625	435	4.177	0.603	62838	72177	i)As reported, at 11:21 hrs, R-phase conductor of 400 KV Akal-Jaisalmer2(Bhainsra) (RS) Ckt broke at location no. 134 which caused R-B phase to phase fault on 400 KV Akal-Jaisalmer2(Bhainsra) (RS) Ckt. Line tripped from Akal(RS) end on zone-1 distance protection operation with fault current of Ir=~5.52kA & Ib=~5.57kA and fault distance of 44.86km from Akal(RS) end. Line tripped from Jaisalmer2(Bhainsra)(RS) end on zone-1 distance protection operation with fault distance of 9.6km from Jaisalmer2(Bhainsra)(RS) end. ii)Again, broken conductor of 400 KV Akal-Jaisalmer2(Bhainsra) (RS) Ckt fell on Suzlon 33kV feeder which is emanating from 220kV Bhensara (M/s Suzlon) S/s, which caused R-B phase to phase fault on the same feeder. On this fault, 220kV Akal-Bhensra(Suzlon) ckt tripped on zone-1 distance protection operation from Akal(RS) end. iii)During the same time, 400 KV Barmer(RS)-Rajwest(RW) (RS) Ckt tripped on R-B phase to phase fault (Ir=~2.0kA & Ib=~2.4kA) at a distance of 127.6 meters from Barmer(RS) end on zone-1 distance protection operation. Line didn't open from Rajwest(RS) end (Reason and location of fault yet to be shared). iv)During the same time, 400 KV Barmer(RS)-Jaisalmer2(Bhainsra) (RS) Ckt tripped on R-B phase to phase fault (Ir=~1.3kA & Ib=~1.1kA) from Barmer(RS) end on distance protection operation. As per DR, fault sensed in zone-3 from Barmer end. Line didn't open from Jaisalmer2(Bhainsra)(RS) end (Reason and location of fault yet to be shared). v)As per PMU at Jodhpur(RS), R-B phase to phase fault with fault clearing time of 80ms is observed. Voltage dipped upto 0.777 p.u. at Jodhpur(RS). vi)As per SCADA, change in total RE generation of approx. 2625 MW (ISTS Solar: ~1910 MW, Rajasthan Solar: ~715 MW) among which almost 97% generation revived within 3.5 minutes. vii)Due to a significant dip in RE generation, frequency dropped by 0.203 Hz (from 49.879 Hz to 49.676 Hz). viii)As per SCADA, change in Punjab demand of approx. 435 MW is observed (load loss details due to df/dt operation in Punjab yet to be shared by Punjab SLDC).	1) 400 KV Barmer(RS)-Jaisalmer2(Bhainsra) (RS) Ckt 2) 400 KV Barmer(RS)-Rajwest(RW) (RS) Ckt 3) 400 KV Akal-Jaisalmer2 (Bhainsra) (RS) Ckt 4) 220kV Akal-Bhensra(Suzlon) ckt
15	GD-1	Delhi	11-06-2024 14:10	11-06-2024 14:50	00:40	279	1601	0.400	1.931	69737	82920	i)During antecedent condition, 400/220kV 500MVA ICT-1, 2, 3 and 4 at Mandaula(PG) were carrying approx. 337MW, 338MW and 337MW respectively with total loading of 1347MW (as reported by CPCC1, Power Grid). 104 MW GTG-1 & 2 and 122 MW STG at Pragati were generating approx. 85MW, 85MW and 109MW respectively with total generation of 279MW (as reported by SLDC Delhi). 212 MAIN BAY - 400/220 kV 500 MVA ICT-3 at Mandaula(PG) was under planned shutdown and ICT-3 was charged from transfer bus through isolator. ii)As reported, at 14:10 hrs, fire was observed in isolator (transfer bus side) of 400/220 kV 500 MVA ICT-3 at Mandaula(PG). As per DR, Ir is reduced upto ~97.2A while Iy and Ib increased upto ~1.092kA and 1.107kA respectively and imbalance occurred in the system. iii)As reported by CPCC1, Power Grid, the sequence of the event is as follows: a.At 14:10:24hrs: 400/220 kV 500 MVA ICT 2 at Mandaula(PG) tripped on back-up earth fault protection operation due to system imbalance. As per DR, Ir=~615.8A, Iy=~488.8A and Ib=~498.3A before tripping of ICT-2. b.Due to tripping of ICT-2, loading of 400/220kV 500MVA ICT-1, 3 and 4 at Mandaula(PG) were increased to 457MW each. c.At 14:10:27hrs: 400/220 kV 500 MVA ICT 1 at Mandaula(PG) also tripped on back-up earth fault protection operation due to system imbalance. As per DR, Ir=~1.668kA, Iy=~1.214kA and Ib=~1.237kA before tripping of ICT-1. d.Due to tripping of both ICT-1 & 2, loading of 400/220kV 500MVA ICT-3 and 4 at Mandaula(PG) were increased to 696MW each. e.At 14:10:29hrs: 400/220 kV 500 MVA ICT 3 at Mandaula(PG) hand-tripped due to melting of isolator which led to heavy sparking. f.Due to unavailability of ICT-1, 2 & 3, loading of 400/220kV 500MVA ICT-4 at Mandaula(PG) was increased to 1454MW. g.At 14:10:30hrs: 400/220 kV 500 MVA ICT 4 at Mandaula(PG) tripped on back-up over-current protection operation due to excess over-loading. As per DR, maximum current recorded was: Ir=~2.09kA, Iy=~2.136kA and Ib=~2.124kA (max MW loading of approx. 1480MW as per DR). iv)As per PMU at Mandaula(PG), B-N phase to earth fault converted to Y-B-N double phase to earth fault with delayed fault clearing time of 2320ms is observed.	1)400/220 kV 500 MVA ICT 1 at Mandaula(PG) 2)400/220 kV 500 MVA ICT 2 at Mandaula(PG) 3)400/220 kV 500 MVA ICT 3 at Mandaula(PG) 4)400/220 kV 500 MVA ICT 4 at Mandaula(PG)

20.Details of Grid Events during the Month of June 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
16	GI-1	Jammu and Kashmir	13-06-2024 06:48	14-06-2024 08:59	26:11	0	100	0.000	0.136	55820	73692	i)220/132kV Hiranagar(J&K) has double main Bus arrangement at 220kV voltage side. ii)As reported, at 06:28hrs, 220 KV Samba(PG)-Hiranagar(PDD) (PG) Ckt-1 tripped on Y-N phase to earth fault. Fault sensed in zone-1, fault current Iy=~4.9kA and fault distance was 0.8km from Hiranagar end (Exact reason of fault yet to be recieved). iii)At the same time, 220 KV Samba(PG)-Hiranagar(PDD) (PDD JK) Ckt-2 also tripped from Hiranagar(J&K) end on overcurrent protection operation. iv)220 KV Samba(PG)-Hiranagar(PDD) (PG) Ckt-1 tripped from both ends(Samba and Hiranagar) and 220 KV Samba(PG)-Hiranagar(PDD) (PDD JK) Ckt-2 tripped from Hiranagar end only. v)As per PMU at Samba(PG), Y-N phase to earth followed by R-B phase to phase fault with fault clearing time of 80 msec is observed. vi)As per SCADA, load loss of approx. 360MW in J&K control area is observed. However, as per SLDC J&K, approx. 100MW load loss is observed in J&K control area.	1)220 KV Samba(PG)-Hiranagar(PDD) (PG) Ckt-1 2)220 KV Samba(PG)-Hiranagar(PDD) (PDD JK) Ckt-2
17	GI-1	Himachal Pradesh	14-06-2024 23:16	14-06-2024 23:32	00:16	0	296	0.000	0.370	58688	80098	i)220/132kV Kunihar(HP) S/s has double main Bus arrangement at 220kV side. ii)During antecedent condition, incoming power at Kunihar S/s was 365 MW through 220kV Kunihar-Jeori (HP) ckt (~192 MW) and 220kV Kunihar-Bhaba (HP) ckt (~173 MW). Outgoing power from Kunihar(HP) was 110MW through 220kV Kunihar-Baddi (HP) D/C and loading on 220/132kV 200MVA transformer bank-1 & 2 at Kunihar(HP) was 272MW (as reported). 220/132kV 100MVA transformer bank-3 was not in service. iii)As reported, at 23:16 hrs, 220/132kV 200MVA transformer bank-1 at Kunihar(HP) tripped on R-N phase to earth fault. Fault occurred due to burst of R-phase CT (Type of proection operated yet to be recieved). iv)As per PMU, R-N phase to earth fault is observed with fault clearing time of 80ms. v)At the same time, due to tripping of 220/132kV 200MVA transformer bank-1, the complete load shifted to 220/132kV 200MVA transformer bank-2 and the transformer-2 tripped due to overloading. vi)As reported, at the same time, 220kV Kunihar-Jeori (HP) ckt tripped on overvoltage protection operation (line voltage shoot upto 270kV). (HP has been communicated to share the voltage protection settings at Kunihar(HP) S/s) vii)During the same time, 220 kV Baddi-Upper Nangal(HP) Ckt also tripped (exact reason of tripping yet to be received). viii)As per SCADA, total change in demand of apprxx. 296MW in HP control area is observed.	1)220/132kV 200MVA ICT-1 at Kunihar(HP) 2)220/132kV 200MVA ICT-2 at Kunihar(HP) 3)220kV Kunihar-Jeori (HP) ckt 4)220 kV Baddi-Upper Nangal(HP) Ckt
18	GI-1	Himachal Pradesh	16-06-2024 15:56	16-06-2024 16:15	00:19	0	240	0.000	0.285	65695	84229	i)220/66kV Baddi(HP) has double main bus arrangement at 220kV side. ii)As reported, at 15:56 hrs, R-Y phase to phase fault occurred on 220kV Baddi(HP)-Pinjore(HV) (HPPTCL) Ckt-2 at a distance of 1.4km from Baddi(HP) end with fault current Ir=~13.5kA & Iy=~ 13.9kA (As per DR). 220kV Baddi(HP)-Pinjore(HV) (HPPTCL) Ckt-2 tripped on zone-1 distance protection operation from Baddi(HP) end (As per DR). (Relay flags not received from Pinjore end). iii)As per PMU at Panchkula(PG), R-Y phase to phase fault with fault clearance time of 120msec is observed. iv)At the same time, 220kV Baddi(HP)-Pinjore(HV) (HPPTCL) Ckt-1, 220 kV Baddi-Upperla Nangal(HP) Ckt & 220kV Baddi-Kunihar (HP) ckt also tripped. As per DRs from Baddi(HP) end, 220kV Baddi(HP)-Pinjore(HV) (HPPTCL) Ckt-1 & 220 kV Baddi-Upperla Nangal(HP) Ckt tripped on master trip relay operation (exact details of protection operated in these lines yet to be received). v)As per SCADA, change in demand of apprxx. 240MW in HP control area is observed.	1)220kV Baddi(HP)-Pinjore(HV) (HPPTCL) Ckt-1 2)220kV Baddi(HP)-Pinjore(HV) (HPPTCL) Ckt-2 3)220 kV Baddi-Upperla Nangal(HP) Ckt 4)220kV Baddi-Kunihar (HP) ckt-2
19	GI-2	Rajasthan	16-06-2024 18:20	16-06-2024 19:38	01:18	0	319	0.000	0.433	54344	73690	i)400/220kV Kankani(RS) has one and half breaker bus arrangement at 400kV side. ii)During antecedent condition, the loading of 400/220 kV 315 MVA ICT-1 and 400/220 kV 500 MVA ICT-2 at Kankani(RS) was approx. 223 MW and 341 MW respectively (As per SCADA). iii)As reported, at 18:20hrs, 400/220 kV 500 MVA ICT-2 at Kankani(RS) tripped on master trip relay maloperation due to issue in control cables (exact reason of tripping yet to be received). iv)Due to tripping of 400/220 kV 500 MVA ICT-2, complete load of ICT-2 shifted to 400/220 kV 315 MVA ICT-1 at Kankani(RS) which led to tripping of 400/220 kV 315 MVA ICT-1 on overloading. v)As per PMU at Jodhpur(RS), voltage fluctuation is observed in the system and no fault is observed. vi)As per SCADA, change in demand of approx. 319 MW in Rajasthan control area is observed.	1)400/220 kV 315 MVA ICT 1 at Kankani(RS) 2)400/220 kV 500 MVA ICT 2 at Kankani(RS)
20	GD-1	Himachal Pradesh	17-06-2024 18:25	17-06-2024 18:57	00:32	208	0	0.357	0.000	58300	79666	i)During antecedent condition, 96 MW Unit-1 & 2 at ADHPL(IP) were generating approx. 105 MW and 104 MW respectively as per SCADA. ii)As reported, at 18:25 hrs, due to inclement weather condition, 220KV Phozal(HP)-Nallagarh(PG) (ADHPL) Ckt tripped on Y-B-N double phase to earth fault with fault distance of 86.4km from Nallagarh(PG) end. As per DR, fault sensed in zone-1 with fault current of Iy=~2.2kA & Ib=~3.4kA at Nallagarh(PG) end. iii)As reported, at the same time, due to inclement weather condition, 220KV AD hydro(AD)-Nallagarh(PG) (ADHPL) Ckt tripped on Y-N phase to earth fault with fault distance of 176km and 75.5km from Nallagarh(PG) and AD hydro(IP) end respectively. As per DR, zone-1 distance protection operated from both ends with fault current of Iy=~1.7kA & Iy=~1.6kA at Nallagarh(PG) and AD hydro(IP) end. From Nallagarh(PG) end, line successfully reclosed which shows the transient nature of the fault. A/R didn't operate at AD hydro(IP) end (Proper functioning of A/R at AD hydro(IP) end need to ensured). iv)As per PMU at Nallagarh(PG), Y-B phase to phase fault with fault clearing time of 80ms is observed. v)Due to tripping of 220 KV Phozal(HP)-Nallagarh(PG) (ADHPL) Ckt and 220 KV AD hydro(AD)-Nallagarh(PG) (ADHPL) Ckt, island formed with 96MW Unit-1 & 2 at AD hydro(AD) and 220 KV AD hydro(AD)-Phozal(HP) (ADHPL) Ckt and generation-load imbalance occurred. After this, 96MW Unit-1 & 2 at AD hydro(AD) tripped due to over-speeding. vi)After tripping of 96MW Unit-1 & 2 at AD hydro(AD), 220 KV AD hydro(AD)-Phozal(HP) (ADHPL) Ckt got de-energized and complete blackout occurred at 220kV ADHPL(IP). vii)As per SCADA, no change in demand in Himachal Pradesh control area and generation loss of approx. 208 MW during the event at ADHPL(IP).	1)220 KV Phozal(HP)-Nallagarh(PG) (ADHPL) Ckt 2)220 KV AD hydro(AD)-Nallagarh(PG) (ADHPL) Ckt 3)96MW Unit-1 at AD hydro(AD) 4)96MW Unit-1 at AD hydro(AD)
21	GD-2	Haryana, Punjab, UP, Delhi, J&K, HP, Chandigarh, Uttarakhand & Rajasthan	17-06-2024 13:53	17-06-2024 14:32	00:39	5240	16500	7.089	18.454	73922	89410	i)At 13:53 Hrs Northern Region demand experienced a dip of around ~16,500 MW. The incident occurred immediately after tripping of 4 poles of HVDC Champa – Kurukshetra which was carrying ~4500MW from the Western Region to Northern Region. Further 765/400kV Aligarh (PGCIL) station faced major outages and 5 Nos. of 765 kV lines from Aligarh PG tripped. ii)The frequency post event rose to 50.68 Hz from 50.03 Hz. iii)Loads were restored progressively starting 14:30 hrs. Loads were gradually restored considering low voltages and high loading on 765 KV Agra-Gwalior lines. System load was restored to normal levels by 15.00 hrs. iv)In the Northern Region hydro generating units tripped at Bhakra, Karcham, Sainj, RSD (Total-1237 MW). Thermal generating units tripped at Lalitpur, Rajwest, Unchahar, Panipat (Total-1250 MW). v)Northern Region Renewable Generation (Solar) of 2800 MW approx. was also affected, however 1500 MW was generation was restored with 04 minutes. vi)Apart from above, there was no load loss reported from other regions of the country; however, tripping of 02 generation units in the Western Region (Mahan Energen – 2x600 MW) and 02 modules of OLTC Palatana (approx. 700 W) in the North Eastern Region were reported, all of which has since been revived.	1)765 KV Hapur(UP)-Rampur_PRSTL (UP) (GTL) Ckt-1 2)765 KV Aligarh(PG)-SIKAR_2 (PASTL) Ckt-1 & 2 3)800 KV HVDC Kurukshetra(PG) Pole-1, 2, 3 & 4 4)132 KV Mahendra Nagar(PG)-Tanakpur(NH) (PG) Ckt-1 5)250 MW Karcham Wangtoo HPS - UNIT 2 & 4 6)765 KV Jhatikara-Aligarh (PG) Ckt-1 7)765 KV Koteshwar-Meerut (PG) Ckt-2 8)765 KV Orai-Aligarh (PG) Ckt-1 & 2 9)400 KV Mahindergarh(APL)-Dhanoda(HV) (ATIL) Ckt-1 & 2 10)400 KV Mahindergarh(APL)-Bhiwani(PG) (PG) Ckt-3 11)765 KV Agra-Aligarh (PG) Ckt-1 12)765 KV Rampur_PRSTL -Ghatampur_TPS (UP) Ckt-1 13)250 MW Panipat TPS - UNIT 8 14)400 KV Koteswar(TH)-Koteshwar(PG) (PG) Ckt-1 15)765 KV Koteshwar-Meerut (PG) Ckt-1 16)400 KV Kishenpur-Moga (PG) Ckt-1 & 2 17)765 KV Kanpur_GIS-Aligarh (PG) Ckt-1 18)335 MW Rajwest (IPP) LTPS - UNIT 1 19)220 KV Baddi(HP)-Pinjore (HV) (HPPTCL) Ckt-2

20.Details of Grid Events during the Month of June 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
22	GI-1	Uttar Pradesh	18-06-2024 01:04	18-06-2024 01:40	00:36	0	146	0.000	0.178	59894	81806	i) 220kV Agra2(UP) (220kV Sikandra, Agra(UP)) has double main and transfer bus scheme at 220kV level and main and transfer bus scheme at 132kV level. ii) During antecedent condition, loading of 220/132kV 160MVA ICT-1 & 2 and 220/132kV 100MVA ICT-3 at Agra2(UP) were 83 MW, 83 MW and 47 MW respectively. iii) As reported, at 01:04 hrs, 132KV Agra2-railway TSS (patholi) Ckt 1 tripped on B-N phase to earth fault from Agra2(UP) end. Fault distance was 0.4km and zone-1 distance protection operated from Agra2(UP) end. iv) As per PMU at Agra(PG), Y-N phase to earth fault with delayed fault clearance time of 640ms is observed (phase sequence issue). v) As reported, at the same time, due to delayed fault clearance, 220/132kV 160MVA ICT-1 & 2 and 220/132kV 100MVA ICT-3 also tripped on overcurrent protection operation. vi) During the same time 220kV Agra2-Agra1 (UP) ckt also tripped from 400kV Agra1(UP) end (exact reason of tripping yet to be received). vii) Due to tripping of 220/132kV 160MVA ICT-1 & 2 and 220/132kV 100MVA ICT-3, other 132kV feeders and 132/33kV 63MVA ICT-4 & 5 became dead. viii) As per SCADA, change in demand of approx. 196 MW in UP control area. However, SLDC-UP reported 146MW load loss.	220kV Agra2-Agra1 (UP) Ckt 132KV Agra2-railway TSS (patholi) Ckt 1 220/132kV 160MVA ICT-1 at Agra2(UP) 220/132kV 160MVA ICT-2 at Agra2(UP) 220/132kV 100MVA ICT-3 at Agra2(UP)
23	GD-1	Rajasthan	18-06-2024 23:03	19-06-2024 10:09	11:06	0	180	0.000	0.219	61172	82109	i) 400/220kV Kalisindh(RS) has one and half breaker scheme at 400kV level and double main transfer bus scheme at 220kV level. ii) During antecedent condition, generation of 600 MW Kalisindh TPS – UNIT-1 & 2 were 551 MW & 568 MW respectively. This total power was evacuating through 400 KV Anta-Kalisindh (RS) Ckt-2 (~869MW) and 400/220kV 315 MVA ICT (~180MW). 220kV Jhalawar D/C was connected from 400/220kV 315 MVA ICT. 400 KV Anta-Kalisindh (RS) Ckt-1 was not in service (forced outage on R-N phase to earth fault from 18:16 hrs (18/06/2024)). iii) As reported, at 23:03hrs, due to heavy rain and stormy weather condition, 400 KV Anta-Kalisindh (RS) Ckt-2 tripped on R-Y phase to phase fault with fault distance of 19.98km and 52.87km from Anta and Kalisindh end respectively. From both ends zone-1 distance protection operated. Faults currents were Ir=~14kA & Iy=~15.3kA (As per DR) and Ir=~4.3kA & Iy=~2.9kA from Anta and Kalisindh end respectively. iv) As per PMU at Ajmer(PG), R-Y phase to phase fault with fault clearing time of 120msec is observed. v) Due to tripping of 400 KV Anta-Kalisindh (RS) Ckt-2, island formed with 600 MW Unit-1 & 2 at Kalisindh TPS(RS) and 220 KV Kalisindh-Jhalawar D/C and generation-load imbalance occurred. After this, 600 MW Unit-1 & 2 at Kalisindh TPS(RS) tripped due to over-speeding. vi) After tripping of 600 MW Unit-1 & 2 at Kalisindh TPS(RS), 400/220kV 315 MVA ICT and 220 KV Kalisindh-Jhalawar D/C got de-energized and complete blackout occurred at 400/220kV Kalisindh(RS). vii) As per SCADA, no change in demand is observed in Rajasthan control area. However, approx. 180 MW load loss is reported by SLDC- Rajasthan. viii) As per SCADA, loss in generation of approx. 1057 MW at Kalisindh TPS is observed. However, SLDC- Rajasthan reported approx. 1119 MW generation loss at Kalisindh TPS(RS).	1) 400 KV Anta-Kalisindh (RS) Ckt-2 2) 600 MW Kalisindh TPS - UNIT 1 3) 600 MW Kalisindh TPS - UNIT 2
24	GI-2	Rajasthan, Uttar Pradesh & Punjab	19-06-2024 12:42	19-06-2024 14:20	01:38	5530	1050	7.440	1.167	74323	89938	i) During antecedent condition, low voltage scenario was prevailing in mainly Rajasthan, Delhi and UP control area. As per SCADA, voltage at 400kV Bikaner(RS), Bhadla(RS), Bhinmal(RS) and Kankani(RS) were 377kV, 382kV, 379kV and 375kV respectively. i) As per PMU at Bhadla(PG), at 12:42:03:760 hrs, 3-phase to ground fault ia observed with fault clearing time of (exact location of the fault yet to be shared). Voltage dipped upto 0.835 p.u. at Bhadla(PG). ii) As per SCADA, total NR RE generation drop/loss was approx. 4930MW (ISTS Solar: ~3490 MW, Rajasthan Solar: ~843 MW, Rajasthan Wind: ~597 MW). iii) As per SCADA, total change in demand of approx. 1215 MW (Punjab: ~730 MW, UP: ~180 MW, Rajasthan: ~305 MW) is observed in NR control area. iv) As per PMU at Bassi(PG), frequency dropped by 0.409Hz (from 50.062 Hz to 49.653 Hz) due to significant dip in RE generation. Frequency recovered upto 49.865 Hz within 1 minute. v) As per details received from SLDCs, total load relief of approx. 1050 MW observed in NR region (Punjab: ~723 MW, UP: ~220MW, Rajasthan: ~107 MW) on df/dt operation. vi) Due to significant dip in RE generation (as RE generation failed to recover 90% of pre-fault active power within 1 sec and further inverters tripping on OV, LVRT/HVRT Non-compliant), over voltage (1.075pu at 400kV Bhadla(PG)) scenario occured immediately after the fault. vii) At the same time, 135 MW Rajwest (IPP) LTPS - UNIT 2, 4, 5, 6 and 8 also tripped due to "sudden change in speed protection" in turbine operated (protection logic: if 2 out of 3 sensors in turbine senses change in speed more than 20 rpm within 10ms then it sends tripping signal to turbine), as reported (further details yet to be received). viii) As per SCADA, generation loss of approx. 600 MW occurred at Rajwest(RS) LTPS.	1. 135 MW Rajwest (IPP) LTPS - UNIT 2 2. 135 MW Rajwest (IPP) LTPS - UNIT 4 3. 135 MW Rajwest (IPP) LTPS - UNIT 5 4. 135 MW Rajwest (IPP) LTPS - UNIT 6 5. 135 MW Rajwest (IPP) LTPS - UNIT 8
25	GI-1	Haryana	19-06-2024 22:05	19-06-2024 22:35	00:30	0	86	0.000	0.112	58638	76643	i) 220/132kV Safidon(HV) has double mail bus scheme at 220kV side. ii) During antecedent condition, 220kV lines from Safidon(HV) to Jind(HV) ckt-1, Mund(HV) ckt-1 & 2, PTPS(HV) ckt-1 & 220/132kV 100MVA ICT-2 connected to 220kV Bus-1 and 220kV lines from Safidon(HV) to Jind(HV) ckt-2, PTPS(HV) ckt-2 & 3 and 220kV 160MVA ICT-1 & 3 were connected to 220kV Bus-2 at Safidon(HV). iii) As reported, at 22:05 hrs, due to inclement weather, 220kV Safidon-Mund (HV) ckt-1 & 2 tripped on B-N phase to earth fault (Ib=~4.5kA) and R-Y-B three phase fault (Ir=~4.7kA, Iy=~2.3kA, Ib=~4.5kA) with fault distance of 24.5km and 36.3km respectively from Safidon(HV) end. Zone-2 distance protection operated for the lines from Safidon(HV) end. iv) As per PMU at Jind(PG), R-B phase to phase fault converted into R-Y-B three phase fault with fault clearing time of 680msec is observed. v) At the same time, due to high fault current, sparking occurred in bay (isolator) of 220kV Safidon-Mund (HV) ckt-1 & 2 at Safidon(HV) end due to which bus bar protection operated and all elements connected to 220kV Bus-1 (220kV lines to PTPS ckt-1, Jind ckt-1 and 220/132kV 100MVA ICT-2) tripped at Safidon(HV) S/s. vi) At the same time, due to inclement weather condition, 220 KV Jind(PG)-Mund (HV) (HVPNL) Ckt-1 & 2 also tripped. 220 KV Jind(PG)-Mund (HV) (HVPNL) Ckt-2 tripped on B-N phase earth fault with fault current of 1.9kA from Mund(HV) end. (type of protection operated and reason for tripping of 220 KV Jind(PG)-Mund (HV) (HVPNL) Ckt-1 yet to be received). vii) As per SCADA, 447 MW load loss is observed in Haryana control area. However, 86MW load loss is reported by SLDC-Haryana	1) 220 KV Safidon-Jind (HV) Ckt-1 2) 220 KV Safidon-Mund (HV) Ckt-1 3) 220 KV Safidon-Mund (HV) Ckt-2 4) 220 KV Safidon-PTPS (HV) Ckt-1 5) 220/132kV 100MVA ICT-2 at Safidon(HV) 6) 220 KV Jind(PG)-Mund (HV) (HVPNL) Ckt-1 7) 220 KV Jind(PG)-Mund (HV) (HVPNL) Ckt-2 8) 220 kV Mund-IOCL (HV) Ckt-1
26	GD-1	Rajasthan	20-06-2024 22:59	21-06-2024 05:54	06:55	0	0	0.000	0.000	57173	77776	i) 400/220kV Bikaner-2(PG) one and half breaker bus arrangement at 400kV side and double main & transfer bus arrangement at 220kV side. ii) During antecedent condition, no power generation at 220kV sub-stations connected from 400/220 kV 500 MVA ICT 1 & 2 at Bikaner-2(PG) (Grain Energy, Prerak Green & TP Saurya Banderwala). So loading of ICT-1 & 2 was approx. 0 MW at Bikaner-2(PG). 400/220kV 500 MVA ICT-3 and 220kV line to Serentica at Bikaner-2(PG) were not in service. iii) As reported, at 22:59hrs, due to inclement weather condition, 220 KV BTPSL_SL_BIK2_PG-Bikaner_2 (PBTSL) (BANDERWALA_TPSL) Ckt tripped on Y-N phase to earth fault. Fault distance was 5km and fault current was Iy=~16kA from Bikaner-2(PG) end. Zone-1 distance protection operated from Bikaner-2(PG) end. iv) As per PMU at Bikaner-2(PG), Y-N phase to earth fault is observed with fault clearing time of 80ms. v) As per SCADA SOE, during the dead time period of tripping of 220 KV BTPSL_SL_BIK2_PG-Bikaner_2 (PBTSL) (BANDERWALA_TPSL) Ckt, 400/220 kV 500 MVA ICT-1 & 2 at Bikaner-2(PG) also tripped due to Restricted earth fault protection operation. Through fault current of magnitude ~8kA fed by both the ICTs. Spill current of magnitude ~100mA was observed during the event while REF pickup current setting is 50mA for both the ICTs at Bikaner-2(PG) (As reported). vi) As 400/220 kV 500 MVA ICT-1 & 2 at Bikaner-2(PG) tripped, 220kV lines from Bikaner-2(PG) to Grain Energy, Prerak Green & TP Saurya Banderwala became dead and the respective sub-stations got blackout. vii) As per SCADA, no change in demand in Rajasthan control area and no generation loss is observed. viii) As per remedial action, stabilizing resistor value revised from 600 ohms to 915 ohms.	1) 220 KV BTPSL_SL_BIK2_PG-Bikaner_2 (PBTSL) (BANDERWALA_TPSL) Ckt 2) 400/220 kV 500 MVA ICT 1 at Bikaner_2 (PBTSL) 3) 400/220 kV 500 MVA ICT 2 at Bikaner_2 (PBTSL)

20.Details of Grid Events during the Month of June 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
27	GI-1	Delhi	20-06-2024 12:29	21-06-2024 14:06	25:37	0	219	0.000	0.265	71717	82570	i)220/66kV Park Street(DTL) has double main Bus arrangement at 220kV side. ii)As reported, at 12:29 hrs, 33kV Park Street-Baird ckt-2 tripped on fault on the same line (nature, reason and location of fault yet to be received). iii)As per PMU at Maharani Bagh(PG), R-Y-B three phase fault is observed with fault clearing time of 120msec respectively. iv)As reported, at the same time, 220/33kV 100MVA ICT-3 & 4 also tripped at Park Street(DTL). ICT-3 tripped on Buchholz relay operation and ICT-4 tripped on opening of incomer due to damage in B-phase pole of ICT-4 CB (exact reason of tripping of both ICTs yet to be received). v)As per SCADA, change in demand of approx. 219 MW is observed in Delhi control area.	1)220/33kV 100MVA ICT-3 at Park Street(DTL) 2)220/33kV 100MVA ICT-4 at Park Street(DTL)
28	GI-1	Rajasthan	21-06-2024 11:37	21-06-2024 13:29	01:52	0	744	0.000	0.912	66360	81554	i)220kV KTPS(RS) has double main bus arrangement at 220kV side. ii)During antecedent condition, power generation of 110 MW Unit-1 & 2, 210 MW Unit-3, 4 & 5 and 195 MW Unit-6 & 7 were 81MW, 95MW, 174MW, 150MW, 167MW, 171MW & 172MW respectively. 210 MW Unit-5, 220 KV KSTPS-Kota Sakatpura (RS) ckt-3 & station transformer (ST)-3 were connected to 220kV Bus-3 and 195 MW Unit-7 and 220 KV Kota(PG)-KTPS(RVUN) (RS) Ckt-1 were connected to 220kV Bus-5 at KTPS(RS). 220kV Bus-3 and Bus-5 were coupled through isolator only. iii)As reported, at 11:37hrs, due to inclement weather conditions, 220 KV KSTPS-Ranpur (RS) ckt tripped on R-Y phase to phase fault at a distance of 12.49km from KTPS(RS) end. Zone-1 distance protection operated from both ends. As per PMU, R-N followed by Y-N phase to earth fault is observed with fault clearing time of 120ms and 120ms. iv)As reported, at 11:39hrs, due to inclement weather conditions, 220 KV Kota(PG)-KTPS(RVUN) (RS) Ckt-1 tripped on B-N phase to earth fault (Ib~ 14.1kA & Ib~11.7kA from Kota(PG) and KTPS(RS) ends respectively) at a distance of 2.96km from Kota(PG) end. Zone-1 distance protection operated from Kota(PG) end. However, B-phase CB pole lagged in opening while clearing the fault from KTPS(RS) end which led to LBB protection operation at KTPS(RS). As per PMU, B-N phase to earth fault with delayed fault clearing time of 320msec is observed. v)Since 220kV bus-3 & bus-5 were coupled through isolator only, due to LBB operation all elements connected to 220kV bus-3 & bus-5 tripped (210 MW Unit-5, 220 KV KSTPS-Kota Sakatpura (RS) ckt-3, ST-3, 195 MW Unit-7 and 220 KV Kota(PG)-KTPS(RVUN) (RS) Ckt-1). vi)Due to tripping of ST-3, auxiliary supply of 110 MW Unit-1 and 210 MW Unit-3 & 4 disrupted which led to tripping of Unit-1, 3 & 4 at KTPS(RS) vii)At the same time, 220 KV Duni(RS)-Kota(PG) (RS) Ckt also tripped on R-N phase to earth fault (Ir~21kA & Ir~1.3kA from Kota(PG) and Duni(RS) end respectively) with fault distance of 75.2km from Kota(PG) end. Fault sensed in zone-1 from both ends. As per PMU, multiple R-N phase to earth fault with fault clearing time of 120ms, 120ms and 80ms. viii)As per SCADA, no change in demand in Rajasthan control area is observed.	1)220 KV KSTPS-Ranpur (RS) ckt 2)220 KV Kota(PG)-KTPS(RVUN) (RS) Ckt-1 3)220 KV KSTPS-Kota Sakatpura (RS) ckt-3 4)210 MW Unit-1 at KTPS(RS) 5)210 MW Unit-3 at KTPS(RS) 6)210 MW Unit-4 at KTPS(RS) 7)210 MW Unit-5 at KTPS(RS) 8)195 MW Unit-7 at KTPS(RS) 9)220 KV Duni(RS)-Kota(PG) (RS) Ckt
29	GD-1	Punjab	21-06-2024 10:38	21-06-2024 11:24	00:46	0	320	0.000	0.400	63575	80089	i)220/66kV Mohali(PS) and 220/66kV Majra(PS) S/s have double main bus arrangement at 220kV side. ii)During antecedent condition, 220kV lines from Mohali(PS) to Nallagarh(PG) D/C, Majra(PS), Mohali2(PS), 220/66kV 160 MVA ICT-1 & 220/66kV 100 MVA ICT-2 were connected to 220kV Bus-1 and 220kV lines from Mohali(PS) to Rajpura D/C, GGSTP, Khrah, Derab, Ganguwal & 220/66kV 160MVA ICT-3 were connected to 220kV Bus-2 at 220kV Mohali(PS) S/s. 220kV bus coupler was in open condition at Mohali(PS) S/s. Incoming power at 220kV Bus-1 Mohali(PS) was only through 220 KV Nallagarh(PG)-Mohali(PS) (PS) D/C and 220kV Mohali-Majra (PS) ckt and incoming power at Majra(PS) was only through 220 KV Ganguwal(BB)-Majra(PS) (PS) Ckt. iii)As reported, at 10:38 hrs, 220 KV Nallagarh(PG)-Mohali(PS) (PS) D/C tripped only from Nallagarh(PG) end on receiving DT from Mohali(PS) end (exact reason for DT received at Nallagarh(PG) end yet to be received). iv)As per PMU at Nallagarh(PG), fluctuation in voltage, no fault is observed in the system. v)As 220 KV Nallagarh(PG)-Mohali(PS) (PS) D/C tripped, the complete load which connected to 220kV Bus-1 Mohali(PS) shifted to 220kV Mohali-Majra (PS) ckt due to which 220 KV Ganguwal(BB)-Majra(PS) (PS) Ckt tripped on overcurrent protection operation due to overloading. vi)Due to tripping of 220 KV Nallagarh(PG)-Mohali(PS) (PS) D/C and 220 KV Ganguwal(BB)-Majra(PS) (PS) Ckt, all elements connected to 220kV Bus-1 at Mohali(PS) S/s (220kV lines from Mohali(PS) to Majra(PS), Mohali2(PS), 220/66kV 160 MVA ICT-1 & 220/66kV 100 MVA ICT-2) and complete 220kV Majra(PS) S/s became dead. This resulted into blackout of 220/66kV Majra(PS) S/s. vii)As per SCADA, load loss of approx. 320 MW in Punjab Control area.	1)220 KV Nallagarh(PG)-Mohali(PS) (PS) Ckt-1 2)220 KV Nallagarh(PG)-Mohali(PS) (PS) Ckt-2 3)220 KV Ganguwal(BB)-Majra(PS) (PS) Ckt
30	GD-1	Delhi	21-06-2024 00:54	21-06-2024 01:34	00:40	0	173	0.000	0.227	56253	76070	i)220/66kV Naraina(DTL) has double main bus arrangement at 220kV side. ii)As reported, at 00:54 hrs, 220/33kV 100MVA ICT-1, 2 and 3 at Naraina(DTL) tripped on O/C protection operation due to B-Ph cable end terminal of 33kV Bus-2 damaged at Naraina(DTL). iii)Due to tripping of all the three ICTs at Naraina(DTL), complete blackout occurred at 220/33kV Naraina(DTL) S/s. iv)As per PMU at Dwarka(PG), B-N phase to earth fault converted to 3-phase to earth fault is observed with delayed fault clearing time of 600 ms. v)As per SCADA, change in demand of approx. 360 MW is observed in Delhi control area. As reported by SLDC-Delhi, load loss of approx. 173 MW occurred.	1)220/33kV 100MVA ICT-1 at Naraina(DTL) 2)220/33kV 100MVA ICT-2 at Naraina(DTL) 3)220/33kV 100MVA ICT-3 at Naraina(DTL)
31	GI-2	Haryana and Punjab	23-06-2024 09:11	23-06-2024 10:24	01:13	0	880	0.000	1.208	58647	72828	i)During antecedent condition, 800kV HVDC Champa-Kurukshetra was carrying total 5687 MW (approx. 1415 MW, 1425 MW, 1426 MW and 1421 MW by Pole 1, 2, 3 and 4 respectively). ii)As reported at 09:11 hrs, 800 kV HVDC Kurukshetra (PG) Pole-01 blocked on T-zone protection operation at Kurukshetra end. Noise in Ihl current of Pole-1 Lane-1 led to latching of T-zone protection. iii)During the same time, 800 kV HVDC Kurukshetra (PG) Pole-03 also blocked on CAT-B sequence initiated by parallel Pole-01 due to latching of T-zone protection. iv)As further reported, sequence of event is as follows: a.09:11:46:724 hrs –Pole-1 Lane 1 Main 1 & 2 T-zone Latched. b.09:11:46:727 hrs–CAT-B latched in Pole-1 due to T-zone Protection. c.09:11:46:728 hrs– CAT-B latched in Pole-3 due to T-zone Protection in Pole-1. v)Due to tripping of two poles (Pole-01 and Pole-03), power order reduced from 5687 MW to 2683 MW. vi)As per PMU, no fault is observed in the system. However, fluctuation in voltage was observed. vii)At the same time, df/dt protection operated in Punjab which led to load throw-off of approx. 880 MW (details of df/dt operation at Punjab is attached in Annexure). viii)As per SCADA, change in demand of approx. 610MW is observed in Punjab control area.	1)800 kV HVDC Kurukshetra(PG) Pole-01 2)800 kV HVDC Kurukshetra(PG) Pole-03
32	GI-1	Delhi	23-06-2024 15:35	23-06-2024 16:15	00:40	0	312	0.000	0.395	63584	79056	i)400/220/66kV Tughlakabad(DTL) has one and half bus arrangement at 400kV side and double main bus arrangement at 220kV side. 220KV Tughlakabad-BTPS (DTL) D/C and 220KV Tughlakabad-Okhla (DTL) D/C have common towers up to some distance. ii)As reported, at 15:35 hrs, 220KV Tughlakabad- BTPS (DTL) D/C and 220KV Tughlakabad-Okhla (DTL) D/C tripped on R-Y-B three phase fault. Optical ground wire (OPGW) broke down and fell on the mentioned lines which caused multiple R-Y-B three phase faults (type of protection operated and location of fault yet to be received). iii)As per PMU at Maharani Bagh(PG), multiple R-Y-B three phase faults are observed with delayed fault clearance time of 400msec. iv)At the same time 220KV Tughlakabad-Masjid Moth (DTL) Ckt-1 also tripped from Masjid Moth(DTL) end (exact reason of tripping and type of protection operated yet to be received). v)As per SCADA, change in demand of approx. 312 MW is observed in Delhi control area.	1)220KV Tughlakabad- BTPS (DTL) Ckt-1 2)220KV Tughlakabad- BTPS (DTL) Ckt-2 3)220KV Tughlakabad-Okhla (DTL) Ckt-1 4)220KV Tughlakabad-Okhla (DTL) Ckt-2 5)220KV Tughlakabad-Masjid Moth (DTL) Ckt-1

20.Details of Grid Events during the Month of June 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
33	GD-1	Delhi	27-06-2024 09:48	27-06-2024 10:39	00:51	0	214	0.000	0.333	57905	64327	i)220kV Gopalpur(DTL) has double main bus arrangement at 220kV side. ii)During antecedent condition, 220kV lines from Gopalpur(DTL) to Mandaula(PG) ckt-1, South Wazirabad ckt-2, Subzimandi ckt-2 and 220/66kV 100MVA ICT-2 & 220/66kV 160MVA ICT-4 were connected to 220kV Bus-1 and 220kV lines from Gopalpur(DTL) to Mandaula(PG) ckt-2, South Wazirabad ckt-1, Subzimandi ckt-1 and 220/33kV 100MVA ICT-1 & 3 were connected to 220kV Bus-2 at Gopalpur(DTL) S/s. 220kV lines from Gopalpur(DTL) to South Wazirabad D/C and Subzimandi ckt-2 were not in service. 220kV Gopalpur-Mandola Ckt-1 & 2 were feeding the load of 220kV Gopalpur(DTL) S/s and further connected to 220kV Subzimandi(DTL) S/s through 220kV Gopalpur-Subzimandi ckt-1 & 2. iii)As reported, at 09:48 hrs, B-N phase to earth fault occurred on 220kV Bus-2 with fault current of approx. 7kA which caused bus bar protection operation at 220 kV Bus-1 & 2 at Gopalpur(DTL) (exact reason of fault and reason for bus bar protection on 220kV bus-1 yet to be shared). iv)Due to this, all the elements connected to 220kV Bus-1 and Bus-2 tripped along with the 220kV bus coupler at Gopalpur(DTL). But 220 KV Mandola(PG)-Gopalpur(DTL) (DTL) Ckt-2 didn't trip from Gopalpur(DTL) end, it tripped from Mandaula(PG) end only. v)As reported, Mandola(PG)-Gopalpur(DTL) (DTL) Ckt-2 didn't trip from Gopalpur end due to non-availability of bus isolator status in bus bar protection relay. vi)As per DR at Mandaula(PG) end, 220 KV Mandola(PG)-Gopalpur(DTL) (DTL) Ckt-1 tripped from Gopalpur(DTL) end only with fault current of ~5.9kA from Mandaula(PG) end; fault sensed in zone-3 from Mandaula(PG). vii)As per DR at Mandaula(PG) end, 220 KV Mandola(PG)-Gopalpur(DTL) (DTL) Ckt-2 tripped from Mandaula(PG) end only with fault current of ~7.4kA and fault distance of 27.5 km from Mandaula(PG) end (as reported); fault sensed in zone-2 from Mandaula(PG). viii)As 220KV Mandola(PG)-Gopalpur(DTL) (DTL) D/C tripped, Gopalpur(DTL) and Subzimandi(DTL) lost their connectivity from grid which led to blackout of 220kV Gopalpur(DTL) S/s and 220kV Subzimandi(DTL) S/s. ix)As per PMU at Mandaula(PG), B-N phase to earth fault is observed with delayed fault clearing time of 280msec respectively. x)As per SCADA, change in demand of approx. 123 MW is observed in Delhi control area. However, 214 MW load loss is reported by SLDC-Delhi.	1.220 KV Mandola(PG)-Gopalpur(DTL) (DTL) Ckt-1 2.220 KV Mandola(PG)-Gopalpur(DTL) (DTL) Ckt-2 3.220 KV Subzimandi-Gopalpur(DTL) Ckt-1 4.220 KV Subzimandi-Gopalpur(DTL) Ckt-2 5.220/33kV 100MVA ICT-1 at Gopalpur(DTL) 6.220/66kV 100MVA ICT-2 at Gopalpur(DTL) 7.220/33kV 100MVA ICT-3 at Gopalpur(DTL) 8.220/66kV 160MVA ICT-4 at Gopalpur(DTL)
34	GI-2	Haryana	27-06-2024 21:17	27-06-2024 21:54	00:37	0	625	0.000	0.800	54726	78141	i)During antecedent condition, 800 kV HVDC Kurukshetra(PG) Pole-1, 2, 3 & 4 were carrying 1125 MW each and hence total 4500 MW power was flowing from Champa to Kurukshetra. ii)As reported at 21:17 hrs, 800 kV HVDC Kurukshetra (PG) Pole-2 & 4 blocked on CLD HV Cable Protection and Pole-1 & 3 blocked on latching of SYS Fail in Pole-3. iii)Pole-4 Lane-2 (M1 & M2) went into fault due to latching of major fault without reporting any further alarms. DC Line Fault restart sequence operated once in Pole-4 along with activation of CAT-D Alarm. Simultaneously, CLD Minor Fail alarm latched in Pole-4 Lane-1. As Pole-4 Lane-2 was faulty, the Lane changeover was not successful & controls generated for blocking of Pole-4 along with generation of CAT-B alarm which subsequently blocked its parallel Pole i.e. Pole-2. iv)During the same time, SYS Fail latched onto Pole-3 and as both the Lanes became unavailable, Pole-3 got blocked and generated CAT-B Alarm which subsequently blocked its parallel Pole i.e. Pole-1. v)Due to tripping of all four poles, power order reduced from 4500 MW to 0 MW. vi)As per PMU, no fault is observed in the system. However, fluctuation in voltage was observed. vii)As per SCADA, change in demand of approx. 400 MW and 225 MW in Rajasthan and UP control area are observed. However, Rajasthan and UP reported no df/dt operation during the event.	1) 800 kV HVDC Kurukshetra(PG) Pole-01 2) 800 kV HVDC Kurukshetra(PG) Pole-02 3) 800 kV HVDC Kurukshetra(PG) Pole-03 4) 800 kV HVDC Kurukshetra(PG) Pole-04
35	GI-2	Rajasthan	30-06-2024 11:14	30-06-2024 14:10	02:56	328	177	0.528	0.260	62115	67952	i)400/220kV Bhadla(RS) has double main and transfer bus arrangement at 220kV side. ii)During antecedent condition, loading of 400/220 kV 500 MVA ICT-1, 2 & 3 were approx. 380 MW, 373 MW & 379 MW respectively. 220 kV Bhadla- Adani REPRL D/C & 220 kV Bhadla-Saurya Urja-2 Ckt were carrying approx. 215 MW & 280 MW respectively (reported data). iii)As reported, at 11:14hrs, R-phase jumper of LV bushing of 400/220 kV 500 MVA ICT-1 at Bhadla(RS) snapped which led to tripping of ICT-1 on directional overcurrent earth fault protection operation. iv)As per "SPS for Transformers at 400KV Bhadla (RVPN) substation", overloading of remaining ICTs after tripping of any of the three 400/220kV 500MVA ICTs on fault/protection operation will lead to operation of SPS as per logic described in schematic diagram for SPS at Bhadla(RS) (attached in Annexure). v)As per logic, tripping commands for 220kV lines and/or 220kV Bus-sectionalizer-II are to be taken from overload relay/ over-current back up relay of 400kV and/or 220kV side of ICTs considering 100% loading of the ICTs (500MVA) with appropriate time delay (ICT-1: 4.5 sec, ICT-2: 3.75 sec and ICT-3: 3 sec). vi)As ICT-1 tripped, the load of ICT-1 shifted to ICT-2 & 3 which led to overloading (more than 100%) of ICT-2 & 3 at Bhadla(RS). Due to less time grading of ICT-3 (3 sec), as per SPS logic, group-3 & 4, i.e., 220 kV Bhadla- Adani REPRL Ckt-1 & 2 and 220 kV Bus sectionalizer-II to evacuate power of Saurya Urja-II (300 MW) tripped on 86 master trip relay operation (correct operation of SPS). vii)As per PMU at Bhadla(PG), no fault in system, however fluctuation in voltage is observed. viii)As per SCADA, change in demand of approx. 177MW in Rajasthan control area and loss in NR solar generation of approx. 328MW is observed.	1)400/220 kV 500 MVA ICT-1 at Bhadla(RS) 2)220 kV Bhadla- Adani REPRL Ckt-1 3)220 kV Bhadla- Adani REPRL Ckt-2 4)220 kV Bus sectionalizer-II to evacuate power of Saurya Urja-II (300 MW)
36	GD-1	Delhi	30-06-2024 10:12	30-06-2024 10:22	00:10	0	177	0.000	0.271	59280	65322	i)220kV Gopalpur(DTL) has double main Bus arrangement at 220kV side. ii)During antecedent condition, 220kV lines from Gopalpur(DTL) to Mandaula(PG) ckt-1, South Wazirabad ckt-2, Subzimandi ckt-2 and 220/66kV 100MVA ICT-2 & 220/66kV 160MVA ICT-4 were connected to 220kV Bus-1 and 220kV lines from Gopalpur(DTL) to Mandaula(PG) ckt-2, South Wazirabad ckt-1, Subzimandi ckt-1 and 220/33kV 100MVA ICT-1 & 3 were connected to 220kV Bus-2 at Gopalpur(DTL) S/s. 220kV lines from Gopalpur(DTL) to South Wazirabad D/C and Subzimandi ckt-2 were not in service. 220kV Gopalpur-Mandola Ckt-1 & 2 were feeding the load of 220kV Gopalpur(DTL) S/s and further connected to 220kV Subzimandi(DTL) S/s through 220kV Gopalpur-Subzimandi ckt-1 & 2. iii)As reported, at 10:12 hrs, R-N phase to earth fault occurred at Gopalpur(DTL) (exact location and reason of fault yet to be received). As per PMU at Mandaula(PG), R-N phase to earth fault is observed with delayed fault clearing time of 800msec respectively. iv)As reported, bus bar protection kept off with time delay setting of 160msec for zone-4 distance protection at Gopalpur(DTL) to avoid any maloperation of bus bar protection relay. On R-N phase to earth fault, 220 KV Mandola(PG)-Gopalpur(DTL) (DTL) D/C didn't trip from Gopalpur end (Reason for not tripping of 220 KV Mandola(PG)-Gopalpur(DTL) (DTL) Ckt D/C yet to be received). v)As per DR, 220 KV Mandola(PG)-Gopalpur(DTL) (DTL) D/C tripped from Mandaula(PG) end only. Ckt-1 tripped on zone-3 distance protection operation and ckt-2 tripped on overcurrent earth fault protection. vi)As further reported, 220/33kV 100MVA ICT-1 & 3, 220/66kV 100MVA ICT-2 & 220/66kV 160MVA ICT-4 at Gopalpur(DTL) tripped on overcurrent protection operation and 220/33kV 100MVA ICT-1 & 2 tripped at Subzimandi(DTL) on restricted earth fault protection operation. vii)As 220KV Mandola(PG)-Gopalpur(DTL) (DTL) D/C tripped, Gopalpur(DTL) and Subzimandi(DTL) lost their connectivity from grid which led to blackout of 220kV Gopalpur(DTL) S/s and 220kV Subzimandi(DTL) S/s. viii)As per SCADA, change in demand of approx. 171 MW in Delhi control area (as per SCADA). However, 177 MW load loss is reported by SLDC-Delhi.	1)220 KV Mandola(PG)-Gopalpur(DTL) (DTL) Ckt-1 2)220 KV Mandola(PG)-Gopalpur(DTL) (DTL) Ckt-2 3)220/33kV 100MVA ICT-1 at Gopalpur(DTL) 4)220/66kV 100MVA ICT-2 at Gopalpur(DTL) 5)220/33kV 100MVA ICT-3 at Gopalpur(DTL) 6)220/66kV 160MVA ICT-4 at Gopalpur(DTL) 7)220/33kV 100MVA ICT-1 at Subzimandi(DTL) 8)220/33kV 100MVA ICT-2 at Subzimandi(DTL)

20.Details of Grid Events during the Month of June 2024 in Western Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	WR	01-06-2024 00:26	01-06-2024 01:07	00:41	341	-	0.37%	-	91677	69201	At 00:26 Hrs / 01-06-2024, 220 kV Bhuj-Baranda-1 tripped on 3 phase to ground fault due to falling of spare conductor of 220 kV Bhuj-Kotda Madh on 220 kV Bhuj-Baranda-1 near Bhuj gantry . Simaltaneously 220 kV Bhuj-Gadhsisa-1 tripped due to maloperation of relay. Generation loss of 196 MW and 145 MW occurred at Gadhsisa (Renew Power) and Baranda (Avikiran) respectively due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV Bhuj-Baranda 2. 220 kV Bhuj-Gadhsisa
2	GD-1	WR	05-06-2024 01:28	05-06-2024 03:07	01:39	-	-	-	-	85001	64030	At 01:28 Hrs / 05-06-2024, 220 kV Raipur-Serisha tripped from Serisha end only due to relay maloperation at Serisha end. Maloperation of relay is under investigation by Serisha. No generation loss occurred at Serisha Solar Power plant (due to night hours) during above tripping.	Tripping of following Elements: 1. 220 kV Raipur-Serisha 2. 220/33 kV Serisha-ICT-1 3. 220 kV Serisha-Main Bus
3	GD-1	WR	05-06-2024 23:07	06-06-2024 19:51	20:44	149.25	-	0.18%	-	84323	67866	At 23:07 Hrs / 05-06-2024, 220 kV Bhuj-Baranda-1 tripped on 3 phase to ground fault due to falling of spare conductor of 220 kV Bhuj-Kotda Madh on 220 kV Bhuj-Baranda-1 near Bhuj gantry. B phase wavetrap of 220 kV Bhuj-Baranda blasted at Bhuj end. Simaltaneously, 220 kV Bhuj-Gadhsisa-1 tripped due to relay maloperation. Fault was in Z2 but Z2 didn't pickup and line tripped. Generation loss of 99 MW and 50 MW occurred at Gadhsisa (Renew Power) and Baranda (Avikiran) respectively due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV Bhuj-Baranda 2. 220kV Bhuj-Gadhsisa
4	GD-1	WR	09-06-2024 06:03	09-06-2024 06:38	00:35	-	75	-	0.13%	73050	59391	At 06:03 hrs / 09-06-2024, 220 kV Vasai(TP)-Nalasopara (220 kV Padghe-Nalasopara TAP at Vasai) tripped on B-E fault. Prior to the tripping, 220 kV Boisar(PG)-Panchali was under planned shutdown. 220 kV Nalasopara is being fed through two links, Padghe-Vasai(TP)-Nalasopara and Boisar(PG)-Panchali-Palghar-Nalasopara. With the tripping of 220 kV Vasai(TP)-Nalasopara, Nalasopara substation became dark. Heavy rains and thunderstorms were reported at the time of tripping Load loss of 75 MW occurred at Nalasopara during the event.	Tripping of following Elements: 1. 220 kV Vasai(TP)-Nalasopara
5	GD-1	WR	12-06-2024 13:10	12-06-2024 15:31	02:21	143	-	0.18%	-	77613	62443	At 13:10 hrs / 12-06-2024, 220 kV Nakhatrana-Dedhiya tripped on earthfault (as infromed in real time by Nakhatrana (Adani). However as seen from relay and PMU no fault indication is present the tripping was due to maloperation. Generation loss of 143 MW occurred at Dedhiya (Adani) wind power plant due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV Nakhatrana-Dedhiya
6	GD-1	WR	13-06-2024 18:16	13-06-2024 18:27	00:11	1095	350	1.37%	0.57%	79926	61493	At 18:16 hrs / 13-06-2024, 400 kV JPL Stage-1-Tamnar tripped on Z-1 protection operation due to flashover occurred in tie breaker of 400 kV JPL Stage-1-Tamnar and 400 kV JPL Stage-1-JPL Stage-2 due to lightening strike. At the same time 400 kV JPL Stage-1-Raipur-1&2 tripped, no fault signature was present as seen from disturbance recorders. All four units at JPL stage-1 (250MW) tripped on over frequency protection operation. Generation loss of 1095 MW occurred at JPL (Jindal) Stage-1 and JSDCPP. Load loss of 350 MW occurred in Gerwani of Chhattisgarh control area.	Tripping of following Elements: 1. 400 kV JPL Stage-1-Raipur-1&2 2. 400 kV JPL Stage-1-Tamnar 3. 400 kV JPL Stage-1-JPL Stage-2 4. 220 kV JPL Stage-1-Gerwani 5. 220 kV JPL Stage-1-DCPP 6. JPL Stage-1 Unit-1,2,3&4 (250 MW) 7. JSDCPP Unit 3&4 (135 MW)
7	GD-1	WR	17-06-2024 11:32	17-06-2024 14:46	03:14	1076	250	1.37%	0.38%	78604	65399	At 11:32 hrs / 17-06-2024, 400/33kV Korba NTPC Tie Transfomrer-1 tripped on directional Earth fault protection operation. This led to the tripping of 33 kV Bus-A at Korba(NTPC), which was feeding the auxilaries of Korba(NTPC)-Unit-1,2&3 (210 MW). Prior to the tripping 220 Korba(W)-Bus-1 was under planned shutdown and 220 Korba(W)-Bus-2 tripped while shifting of feeders on 220 Korba(W)-Bus-2. As per PMU plot fault persisted for 440 msec due to which 400/33kV Korba NTPC Tie Transfomrer-1 tripped on directional Earth fault. As informed by NTPC Korba no indication in relays of 400 kV Korba(NTPC)-Korba(W) was present. 220 Korba(W) substation became dark. Simaltaneously, 220 kV Amarkantak-Annupur-2 tripped on overcurrent protection operation. Prior to the tripping, 220 kV Amarkantak-Annupur-1 was under planned shutdown. Due to tripping of units at Korba(NTPC), Korba(E), DSPM and 220 kV Korba-Churri-2, the load of Kotmikala was fed through 220 kV Annupur, resulting in overloading of 220 kV Amarkantak-Annupur-2. With these Annupur and Kotmikala became dark (Elements handtripped due to no load). 220 kV Kotmikala-Churi-1&2 were out to control loading on 220 kV Annupur-Amarkantak-2. Generation loss of 500 MW at Korba(NTPC), 136 MW at Korba(W) and 440 MW at DSPM(Korba East Extension) occurred due to the event. Load loss of 250 MW occurred at Kotmikala, Churri.	Tripping of following Elements: 1. 220 kV Annupur-Amarkanak-2 2. 400/33 kV Korba(NTPC) Tie Transformer-1 3. Korba(NTPC)-Unit-1,2&3 (210 MW) 4. 220 kV Korba(W)-Bus-2 5. 400/220 kV Korba(W)-ICT-1 6. 220 kV Churri-Korba(W)-1&2 7. 220 kV Korba(W)-Mopka-1 8. 220 kV Korba(E)-Korba(W)-1&2 9. 220 kV DSPM-Korba(W) 10. 220 kV Churri-Kotmikala-1&2
8	GD-1	WR	17-06-2024 13:54	17-06-2024 14:45	00:51	1117	-	1.41%	-	79451	65856	At 13:54 hrs / 17-06-2024, Mahan Energen-Unit-1&2 (600 MW) tripped on overfrequency operation. At the time of tripping grid frequency was around 50.63 Hz. At 13:53 Hrs / 17-06-2024, net load loss of 9725 MW occurred in the Grid. The incident occurred immediately after tripping of both bipoles of HVDC Champa-Kurukshetra which was carrying 4,500 MW from the WR to NR. After tripping of the HVDC link, low voltages were observed across the Northern region and multiple lines and generation tripping occurred. Generation loss of 1117 MW occurred at Mahan Energen (Adani) Thermal Plant during the event. No load loss occurred in Western Region.	Tripping of following Elements: 1. Mahan Energen-Unit-1&2 (600 MW) 2. HVDC Champa-Kurukshetra Pole-1,2,3&4
9	GI-2	WR	18-06-2024 03:15	18-06-2024 19:01	15:46	200	-	0.25%	-	79655	58460	At 03:15 hrs / 18-06-2024, Y phase CT failure of Main bay of 400 kV RGPPL Dabhol-Nagothane-2 at RGPPL Dabhol resulted in Bus bar protection operation of 400 kV RGPPL Dabhol-Bus-1. 400 kV RGPPL Dabhol-Nagothane-2 and Main bays connected to 400 kV RGPPL Dabhol-Bus-1 tripped. RGPPL-Dabhol-GT-3X tripped as both Main and Tie bay of RGPPL Dabhol-Unit-3B which was in Dia of RGPPL Dabhol-GT-3X were open due to unit outage. As the Module-3 was running in half module mode RGPPL-Dabhol-Unit-3A also tripped. Generation loss of 200 MW occurred at RGPPL Dabhol Gas Power Plant during the event.	Tripping of following Elements: 1. 400 kV RGPPL Dabhol-Nagothane-2 2. 400 kV RGPPL Dabhol-Bus-1 3. RGPPL Dabhol-GT-3X 4. RGPPL Dabhol-Unit-3A
10	GI-2	WR	22-06-2024 08:11	22-06-2024 10:05	01:54	-	-	-	-	72416	57541	At 08:11 Hrs / 22-06-2024, 765 kV Padghe-Bus-1 tripped due to DC earth fault in the system. 765/400 kV Padghe-ICT-1 which was in dia with 765 kV Padghe-Bus Reactor tripped. 765/400 kV Padghe-ICT-2 tripped due to CSD relay malfunctioning due to DC earth fault. No load or generation loss occurred during the event.	Tripping of following Elements: 1. 765 kV Padghe-Bus-1 2. 765/400 kV Padghe-ICT-1&2 3. 765 kV Padghe-Bus Reactor
11	GD-1	WR	22-06-2024 22:52	23-06-2024 04:38	05:46	59	-	0.07%	-	80553	61757	At 22:52 Hrs / 22-06-2024, 220 kV Bhuj-Gadhsisa tripped on R-E fault due to falling of spare conductor of 220 kV Bhuj-Kotda Madh on 220 kV Bhuj-Gadhsisa between location 4&5. Generation loss of 59 MW occurred at Gadhsisa (Renew Power) Wind Power plant due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV Bhuj-Gadhsisa
12	GD-1	WR	23-06-2024 21:01	24-06-2024 00:44	03:43	91	-	0.12%	-	76659	57086	At 21:01 Hrs / 23-06-2024, 220 kV Jamkhambaliya-Khakharda tripped due to Gas density cable got damaged at Jamkhambaliya end. Generation loss of 91 MW occurred at Khakharda (Apraava) Wind Power Plant due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV Jamkhambaliya-Khakharda
13	GD-1	WR	24-06-2024 16:36	24-06-2024 18:30	01:54	57	-	0.07%	-	76659	57086	At 16:36 Hrs / 24-06-2024, 765 kV Khavda-PS-Bhuj-1 tripped on persisting Y-E fault (Auto recloser not attempted at Khavda end). Prior to the event 765 kV Khavda-PS-Bhuj-2 tripped on persisting Y-E fault. With these trippings no evacuation path was present for generation at Khavada PSS-1 and Khavda-PSS-2. Generation loss of 27 MW occurred at Khavda-PS connected plants due to loss of evacuation path.	Tripping of following Elements: 1. 765 kV Khavda-PS-Bhuj-1&2
14	GD-1	WR	28-06-2024 15:09	28-06-2024 23:12	08:03	66	-	0.09%	-	72391	58806	At 15:09 Hrs / 28-06-2024, 220 kV Bhuj-Gadhsisa tripped on B-E fault due to falling of CT connector at Bhuj End. Generation loss of 66 MW occurred at Gadhsisa (Renew Power) Wind Power plant due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV Bhuj-Gadhsisa
15	GD-1	WR	28-06-2024 17:56	28-06-2024 19:20	01:24	113	-	0.15%	-	73850	57233	At 17:56 Hrs / 28-06-2024, 220 kV Bhuj-Kotda Madh tripped on B-E fault. As informed by Kotda Madh, the tripping was done by some miscreants. Generation loss of 113 MW occurred at Kotda Madh (Alfanar) Wind Power plant due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV Bhuj-Kotda Madh
16	GD-1	WR	30-06-2024 23:37	01-07-2024 02:16	02:39	28.5	-	0.04%	-	72774	57776	At 15:09 Hrs / 28-06-2024, 220 kV Bhuj-Gadhsisa on B-E fault. During patrolling no abnormalities were found. Generation loss of 28.5 MW occurred at Gadhsisa (Renew Power) Wind Power plant due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV Bhuj-Gadhsisa

20.Details of Grid Events during the Month of June 2024 in Southern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped	Energy Unserved due to Generation loss (MU)	Energy Unserved due to Load loss (MU)
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)				
1	GD-1	Andhra Pradesh	05-06-2024 13:02	05-06-2024 13:09	00:07	0	292	0.00%	0.57%	48215	50834	Complete Outage of 220kV/132kV Kakinada SS, 220kV/132kV Paravada SS, 220kV Anrak SWS of APTRANSCO: Due to the tower collapse of 220kV VSS-Parawada, 220kV Parawada and 220kV Anrak SwS are being radially connected from 220kV/132 Kakinada SS through 220kV Kakinada-Bommuru and 220kV Kakinada Samalkota. As per the reports submitted, the triggering incident was a high resistance B-N fault in the 220kV Kakinada-Bommuru line. The fault was sensed in zone-1 at Bommuru end the line tripped. The fault was not cleared from Kakinada end and instead 220kV Kakinda Samalkot line tripped at Samalkot end on zone-3 protection. Subsequently, with the tripping of 132kV source lines from Kakinada on OC protection, the fault was cleared. This led to complete outage of 220kV/132kV Kakinada SS which further led to complete outage of 220kV/132kV Paravada SS and 220kV Anrak SWS.	i. 220KV Kakinada-Bommuru ii. 220KV Kakinada-Samalkot	0	Not furnished
2	GD-1	Andhra Pradesh	07-06-2024 23:31	09-06-2024 08:45	33:14	0	0	0.00%	0.00%	42325	41305	Complete Outage of 400kV RYTPP of APGENCO: 400kV RYTPP is connected radially through 400KV Kalliki-RYTPP-1&2. As per the reports submitted, the triggering incident was tripping of Rayalaseema U6 on generator protection operation at 22:00Hrs. With the tripping of the unit, high voltages were observed and 400KV Kalliki-RYTPP-2 tripped on over voltage at RYTPP end and DT was sent to remote end and subsequently at 23:31 400KV Kalliki-RYTPP-1 tripped on over voltage at RYTPP end and DT was sent to remote end. Tripping of both the lines led to the complete outage of 400kV RYTPP.	i. 400kV RYTPP Kalliki Line-1&2	0	0
3	GD-1	Andhra Pradesh	10-06-2024 10:33	10-06-2024 10:48	00:15	0	100	0.00%	0.23%	48175	43730	Complete Outage of 220kV/132kV Parawada SS of APTRANSCO: During antecedent conditions, 220kV VSS Parawada Line-2 was under outage. 220kV Parawada Kakinda and 220kV Parawada Anrak lines were under LC. 220kV/132kV Parawada SS was being radially through 220kV Parawada VSS Line-1 and 132kV Parawada Brandix line. As per the reports submitted, the triggering incident was RB-N fault in 220kV VSS Parawada Line-1 and the line tripped. Subsequently, 132kV Parawada Brandix line tripped on over current. This led to complete outage of 220kV/132kV Parawada SS.	i. 220kV VSS Parawada Line-1 ii. 132kV Parawada Brandix	0	Not furnished
4	GD-1	Tamil Nadu	11-06-2024 11:40	11-06-2024 12:26	00:46	570	250	1.13%	0.54%	50233	45893	Complete Outage of 230kV/110kV Karaikudi SS, 230kV/110kV Kavanur SS, 230kV/110kV Valathur SS, 230kV/110kV Tiruchuli SS and Tripping of 230kV Bus-2 of 400kV/230kV Bus-2 of Kamudhi SS of TANTRANSCO: During antecedent conditions, 400kV/230kV Kamudi SS was operating with bus split conditions at 230kV level with 230kV/110kV Tiruchuli SS and 230kV Karaikudi lines on 230kV Bus-2 where 230kV/110kV Tiruchuli SS is being radially connected. 230kV Kavanur Ottapidram and 230kV Valathur Alagar Kovil lines were open. 230kV/110kV Karaikudi SS is operating with single bus configuration at 230kV level. As per the reports submitted, the triggering incident was Y-Phase bus jumper failure in 230 Bus at Karaikudi SS. Immediately, BBP operated and all elements connected to the bus tripped. Subsequently, 110kV feeder connected to 230kV/110kV Karaikudi SS tripped on over durrent protection. This resulted in complete outage of 230kV/110kV Karaikudi SS which further led to complete outage of 230kV/110kV Kavanur SS, 230kV/110kV Valathur SS, 230kV/110kV Tiruchuli SS and	i. 230kV Karaikudi Kavanur Line-1,2,3&4 ii. 230kV Karaikudi Valuthur iii. 230kV Karaikudi N T Kudi Line-1&2 iv. 230kV Karaikudi karaikudi_PG-1&2 v. 230kV/110kV Karaikudi Transformer-1&2	Not furnished	Not furnished
5	GD-1	Andhra Pradesh	25-06-2024 12:24	25-06-2024 12:41	00:17	288	51	0.53%	0.10%	54592	51114	Complete Outage of 220kV Dhone SWS, 220kV/11kV Kambalapadu SS, 220kV/11kV Lakkasagaram SS, 220kV/33kV/11kV Krishnagiri SS, 220kV/11kV Settypalli SS, 220kV/33kV/11kV Nansuralla SS, 220kV/33kV Aspiri SS, 220kV/33kV Molagaveli SS, 220kV/11kV Pendekolu SS of APTRANSCO : 220kV/11kV Kambalapadu SS, 220kV/11kV Lakkasagaram SS, 220kV/33kV/11kV Krishnagiri SS, 220kV/11kV Settypalli SS, 220kV/33kV/11kV Nansuralla SS, 220kV/33kV Aspiri SS, 220kV/33kV Molagaveli SS, 220kV/11kV Pendekolu SS are being radially connected to 220kV Dhone SWS. During antecedent conditions, 220kV Dhone Bus-1 was under LC and all elements were connected to 220kV Bus-2. As per the reports submitted, the triggering incident was R-Y fault in 220kV Dhone Bus-2. Immediately, Bus-2 BBP operated and all elements connected to 220kV Bus-2 tripped. This led to complete outage of 220kV Dhone SWS, which inturn led to complete outage of 220kV/11kV Kambalapadu SS, 220kV/11kV Lakkasagaram SS, 220kV/33kV/11kV Krishnagiri SS, 220kV/11kV Settypalli SS, 220kV/33kV/11kV Nansuralla SS, 220kV/33kV Aspiri SS, 220kV/33kV Molagaveli SS, 220kV/11kV Pendekolu SS.	i. 220kV Dhone Sattenapally Line-1&2 ii. 220kV Dhone Gooty Line-1&2	Not furnished	Not furnished
6	GD-1	Karnataka	26-06-2024 09:34	26-06-2024 09:54	00:20	0	209	0.00%	0.43%	49698	49101	Tripping of 220kV Bus-1 of 220kV/110kV Kemar SS and Complete Outage of 220kV/110kV Kavoor SS and 220kV/110kV MSEZ SS: During antecedent conditions, 220kV/110kV Kemar is operating with Bus split condition at 220kV level. 220kV/110kV Kavoor SS and 220kV/110kV MSEZ SS are being radially connected to 220kV Bus-1 of Kemar SS. As per the reports submitted, the triggering incident was RYB_N fault in 220kV UPCL Kemar Line-1&2 and the lines tripped. Tripping of both lines led to loss of power supply to 220kV Kemar Bus-1 which inturn led to complete outage of 220kV/110kV Kavoor SS and 220kV/110kV MSEZ SS.	i. 220kV UPCL Kemar Line-1&2	0	Not furnished
7	GI-1	Andhra Pradesh	01-06-2024 01:06	01-06-2024 03:13	02:07	0	0	0.00%	0.00%	47147	49762	Tripping of 220kV Bus-2 of Vizag Switching Station of APTRANSCO: As per the reports submitted, the triggering incident was R-N fault in 220kV VSS Parawada line. At VSS end, the fault was sensed in zone-1 and the line tripped. However, b-pole of the breaker failed to open causing LBB to operate and all lines connected to 220kV Bus-2 tripped.	i. 220kV VSS Parawada Line ii. 400kV/220kV Gazuwaka ICT-1&2 iii. 220kV VSS Kalapaka-II iv. 220kV VSS MRS-II v. 220kV Gangavaram	0	0
8	GI-1	Andhra Pradesh	03-06-2024 07:54	03-06-2024 09:27	01:33	0	0	0.00%	0.00%	37934	41246	Tripping of 220kV Bus-1 of 220kV Srisailam RB Generating station of APGENCO: As per the reports submitted, the triggering incident was L-G fault in 220kV Srisailam RB Nagarjun Sagar line. At Srisailam end, the fault was sensed in zone-1 but breaker failed to open. LBB operated and all elements connected to 220kV Bus-1 tripped.	i. 220kV Srisailam RB Nagarjun Sagar line ii. 220kV Srisailam RB Bilakala Guduru line iii. 220kV Srisailam RB Markapur line iv. 220kV Srisailam RB Tallapalli line-1 v. 220kV Srisailam Domalapenta line	0	0
9	GI-1	Andhra Pradesh	06-06-2024 00:59	06-06-2024 03:53	02:54	0	0	0.00%	0.00%	36752	41253	Tripping of 220kV Bus-2 of 220kV Tallapalli SWS of APTRANSCO: As per the reports submitted, the triggering incident was maloperation of LBB in LV side of 400kV/220kV Nagarjun Sagar ICT-1& 2. This resulted in the tripping of 220kV Bus-2 of Tallapalli SWS.	1. 220kV Tallapalli NagarjunaSagar-2 2. 220kV Tallapalli NagarjunaSagar-3 3. 220kV Tallapalli Srisailam-2 4. 220kV Tallapalli Inupurajupalli -2 5. 400kV/220kV Tallapally ICT-1&2	0	0
10	GI-1	Tamil Nadu	06-06-2024 08:11	06-06-2024 08:36	00:25	0	0	0.00%	0.00%	36296	43109	Tripping of 230kV Bus of 230kV/110kV Karaikudi SS of TANTRANSCO: As per the reports submitted, the triggering incident was Y-N fault in 230kV Karaikudi NT Kudy Line-2. At Karaikudi end, the breaker failed to open causing LBB to operate. Immediately all elements connected to 230kV Bus tripped. 110kV was intact during the event.	i. 230kV Karaikudi Kavanur Line-1,2&3 ii. 230kV Karaikudi NT Kudi Line-1&2 iii. 230kV Karaikudi Valuthur iv. 230kV karaikudi Karaikudi_PG Line-1&2 v. 230kV/110kV Karaikudi Auto Transformer-1&2	0	0
11	GI-1	Tamil Nadu	13-06-2024 15:57	13-06-2024 16:12	00:15	0	0	0.00%	0.00%	47546	46711	Tripping of 110kV Bus-1&2 of 400kV/230kV/110kV Alamathy SS of TANTRANSCO: As per the reports submitted, the triggering incident as B-N fault in 110kV Bus-1 of 400kV/230kV/110kV Alamathy SS. Immediately, 110kV Bus-1 BBP operated and all elements connected to the bus tripped which include 400kV/110kV Aamathy ICT-2. Subsequently, 400kV/110kV Aamathy ICT-1 tripped on over loading leading to loss of power supply to 110kV Bus-2 of 400kV/230kV/110kV Alamathy SS.	i. 400kV/110kV Alamathy ICT-1&2	0	0
12	GI-1	Tamil Nadu	16-06-2024 23:31	17-06-2024 00:46	01:15	0	0	0.00%	0.00%	44285	43775	Tripping of 110kV Bus-1&2 of 400kV/110kV Rasipalyam SS of TANTRANSCO: As per the reports submitted, the triggering incident as Y-N fault in 110kV Bus-1&2 of 400kV/110kV Rasipalyam SS. Immediately, 110kV Bus-1&2 BBP operated and all elements connected to the buses tripped including 400kV/110kV Rasipalyam ICT-1,2&3.	i. 400kV/110kV Rasipalyam ICT-1,2&3	0	0
13	GI-2	Telangana	17-06-2024 12:36	17-06-2024 13:43	01:07	0	0	0.00%	0.00%	53264	53132	Tripping of 400kV Bus-1 & Bus-2 of 400kV/220kV Asupaka SS of TSTRANSCO: As per the reports submitted, the triggering incident was B-N fault in 400kV Khammam Asupaka Line-1. immediately, 400kV/220kV Asupaka ICT-1 tripped on overflux protection. This led to loss of power supply to 400kV Bus-1 & Bus-2 of 400kV/220kV Asupaka SS.	i. 400kV Khammam Asupaka Line-1 ii. 400kV/220kV Asupaka ICT-1	0	0
14	GI-1	Karntaka	18-06-2024 13:14	18-06-2024 15:58	02:44	93	0	0.18%	0.00%	50704	50483	Tripping of 220kV Bus-2 of 220kV Sharavathy PH of KPCL: As per the reports submitted, the triggering incident was B-N fault in 220kV Bus-2 and BBP operated tripping all elements connected to 220kV Bus-2 of 220kV Sharavathy PH.	i. Sharavathy Uni-6,7,8 & 10 ii. 220kV Sharavathy Sirsi Line-1&2	Not furnished	0

Wind loss-29

20.Details of Grid Events during the Month of June 2024 in Eastern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Budhipadar, IB-TPS	05.06.2024 04:11	05.06.2024 04:54	00:43	350	0	1.17%	0.00%	30020	25838	At 04:11 Hrs on 05.06.2024 , R-ph CT of 220kV IBTPS-Budhipadar #3 burst at Budhipadar end and all emanating lines from Budhipadar tripped. Both units of IB TPS tripped due to loss of evacuation path, leading to generation loss of 350MW. Power was restored via charging of 220kV Budhipadar-Lapanga ckt#2 at 04:54 Hrs.	220kV Budhipadar IB 1,2,3,4 220kV Budhipadar-Korba 1,2 220KV-Budipada-Raigarh- 1 220kV Budhipadar-Tarkera 1,2 (Bamra,Kaurmunda) 220kV Budhipadar-Aditya Alumina 1,2 220kV Budhipadar Lapanga 1,2 220kV Budhipadar-Vedanta 1,2 220kV Budhipadar-BPSL 1,2 220kV Budhipadar-CONCAST 1,2 210 MW IBTPS 1,2
2	GD-1	Tenughat, Govindpur	13.06.2024 16:55	13.06.2024 17:08	00:13	00:00	140	0.50%	0.49%	28515	28582	At 16:55 hrs on 13.06.2024, 220KV Dumka-Govindpur D/C tripped on R_N Fault . 220 kV Tenughat-Biharsharif was already under breakdown. Consequently, Tenughat unit -2 tripped (as unit -1 was under forced shutdown) due to loss of evacuation path and around 142 MW generation loss occurred. Around 140 MW load loss in Govindpur, Chandankyari, Jainamod area reported.	220 Kv Tenughat -Govindpur-1 220 KV Tenught -Govindpur-2 220kV Bus #1,2 at Tenughat 220kV Bus #1,2 at Govindpur 210 MW U#2 at Tenughat
3	GD-1	Budhipadar , IB-TPS	13.06.2024 19:11	13.06.2024 20:46	01:35	320	0	1.07%	0.00%	29775	29345	At 19:11 hrs on 13.06.2024, 220 kV Bus-2 at Budhipadar tripped due to operation of bus bar protection after failure of R_ph CB of 220 kV Budhipadar-Raigarh-1 at Budhipadar. All associated feeders connected to Bus-2 tripped. At the same time, 320 MW generation loss occurred at IB TPS due to tripping of both running units leading to generation loss of 320 MW.	220kV Budhipadar IB 1,2,3,4 220kV Budhipadar-Korba 2 220KV-Budipada-Raigarh- 1 220KV-Budipada -Kaurmunda-1 220kV Budhipadar Lapanga 2 220kV Budhipadar-Vedanta 1,2 220kV Budhipadar-BPSL 1,2 210 MW IBTPS 1,2 220kV Bus-2 at Budhipadar
4	GD-1	Baruipur	14.06.2024 14:35	14.06.2024 14:41	00:06	0	138	0.00%	0.46%	29864	29839	At 14:35 Hrs on 14.06.2024, 220 kV Newtown-Subhasgram (PG)- Baruipur tripped due to fault in R-ph , 220 kV Baruipur-Subhasgram (WB) was already in opened condition to control loading of ICTs at Subhasgram (PG) which led to a total load loss of 138 MW at Baruipur S/s (Renia, Behala, Sirakol & Baruipur Area). Power to the downstream network was made available through 132 kV Lakhikantapur- Sirakol at 14:41 Hrs. 220 kV Bus was charged at 14:48 Hrs through 220 kV Baruipur- Subhasgram (WB)	220 kV Baruipur-Subhasgram (PG)- Newtown
5	GD-1	New Melli , Jorethang & Tashiding	19.06.2024 06:38	19.06.2024 09:56	03:18	200	0	0.65%	0.00%	30782	30582	At 06:38 hrs.on 19.06.2024, 220 kV Rangpo- New Melli-1 tripped due to phase to Phase fault (Y-B Ph) , 220 kV Rangpo- New Melli-2 was under planned shutdown since 21/05/2024 .Consequently due to loss of evacuation path, Tashiding unit #1 & unit #2 and Jorethang unit #1 & unit #2 tripped causing Generation Loss around 200 MW (Tashiding-105 MW, Jorethang- 95 MW).	220 kV Rangpo- New Melli-1
6	GD-1	GMR (STU), Meramundali-B	20.06.2024 19:18	20.06.2024 21:15	01:57	252	0	0.81%	0.00%	30964	26743	At 19:18 Hrs on 20.06.2024, 400kV-GMR-MERAMUNDALI-B tripped due to R_N fault which led to tripping of GMR Unit #3 (350 MW) due to loss of evacuation path (GMR Unit #3 is connected to Meramundali-B bus only through single line, Bus Split at 400kV-GMR S/S). Total generation loss of around 252 MW occurred.	400 kV GMR-Meramundali B-1 350 MW U#3 at GMR

20.Details of Grid Events during the Month of June 2024 in North-East Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD I	Rengpang area of Manipur Power System	03-06-2024 13:09	03-06-2024 16:24	03:15	0	1	0.00%	0.05%	2947	2099	Rengpang area of Manipur power system is connected to the rest of the grid via 132 kV Loktak-Rengpang line and 132 kV Jiribam-Rengpang line. 132 kV Jiribam-Rengpang line is under outage since 18:18 Hrs of 17.11.2023. At 13:09 Hrs of 03-06-2024, 132 kV Loktak-Rengpang line tripped. Due to tripping of this element, Rengpang area of Manipur power system got separated from rest of the grid due to no source available in this area. Power was extended to Rengpang area of Manipur power system by charging 132 kV Loktak-Rengpang Line at 16:24 Hrs of 03-06-2024.	132 kV Loktak-Rengpang line
2	GD I	Churachandpur and Thanlon area of Manipur Power system	03-06-2024 13:59	03-06-2024 14:50	00:51	0	40	0.00%	1.94%	3271	2063	Churachandpur and Thanlon areas of Manipur power system is connected to the rest of the grid via 132 kV Ningthoukhong-Churachandpur I & II lines. Prior to the event, 132 kV Ningthoukhong-Churachandpur I line is under outage since 18:18 Hrs of 17.11.2023. At 13:59 Hrs of 03-06-2024, 132 kV Loktak-Ningthoukhong & 132 kV Ningthoukhong-Churachandpur II lines tripped. Due to tripping of these elements, Churachandpur and Thanlon areas of Manipur power system got separated from rest of the grid due to no source available in these areas. Power supply was extended to Churachandpur and Thanlon area of Manipur Power System by first charging 132kV-Loktak-Ningthoukhong line at 14:23 Hrs of 03.06.2024 and then charging 132kV-Ningthoukhong-Churachandpur II line at 14:50 Hrs of 03.06.2024.	132 kV Loktak-Ningthoukhong & 132 kV Ningthoukhong-Churachandpur II lines
3	GD I	Part load of Bornagar area of Assam Power system	05-06-2024 01:07	05-06-2024 01:15	00:08	0	27	0.00%	1.16%	2908	2326	Bornagar area of Lower Assam Power System was connected with rest of NER Grid through 132 kV Dhaligaon- Bornagar line and 132 KV Bornagar – Nathkuchi line. Due to bus split at Bornagar, half load of Bornagar is feeding through 132 kV Dhaligaon- Bornagar line and half load is feeding through 132 kV Bornagar – Nathkuchi line. At 01:07 Hrs of 05-06-2024, 132 kV Bornagar- Nathkuchi line tripped. Due to tripping of thiselement, partial load loss occurred at Bornagar area of Assam Power System. Power supply was extended to Bornagar area of Assam Power System by charging 132 kV Bornagar-Nathkuchi line at 01:15 Hrs of 05.06.2024.	132 kV Bornagar- Nathkuchi line
4	GD I	Ningthoukhong area of Manipur Power System	06-06-2024 16:13	06-06-2024 16:49	00:36	0	36	0.00%	1.28%	2661	2808	Ningthoukhong area of Manipur Power System was connected with rest of NER Grid via 132kV Loktak-Ningthoukhong line, 132kV-Ningthoukhong-Churachandpur D/C and 132kV-Ningthoukhong-Imphal lines. 132kV-Ningthoukhong-Churachandpur-1 was under outage since 21:14 Hrs of 29-04-2024. PSD was taken for 132kV-Ningthoukhong Churachandpur-2 for Replacement of disc insulators for 132 kV Ningthoukhong-Churachandpur-1 line. At 16:13 Hrs of 06-06-2024, while returning from shutdown after rectification work, while charging the line 132kV-Loktak-Ningthoukhong and 132kV Imphal-Ningthoukhong line tripped resulting in blackout of Ningthoukhong Area of Manipur Power System. Power supply was extended to Ningthoukhong area of Manipur Power System by charging 132 kV Loktak-Ningthoukhong line at 16:49 Hrs of 06.06.2024.	132 kV Loktak-Ningthoukhong & 132 kV Imphal-Ningthoukhong lines
5	GD I	Blackout in Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh, Thoubal old, Thoubal New and Kongba area of Manipur Power System and Tamu load of Myanmar power system	08-06-2024 18:40	08-06-2024 19:18	00:38	0	120	0.00%	3.74%	3231	3205	Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh, Thoubal old, Thoubal New and Kongba area of Manipur Power System and Tamu load of Myanmar power system were connected with rest of NER Grid through 132 kV Loktak Ningthoukhong line, 132 kV Ningthoukhong –Churachandpur D/C lines, 132 kV Yiangangpokpi -Kongba D/C lines. 400kV-Imphal(PG)-Thoubal-1 line was under outage since 13:32 Hrs of 18.10.2021, 400kV-Imphal(PG)-Thoubal-2 line was under outage since 15:05 Hrs of 24.04.2024. At 18:32 Hrs of 08-06-2024, 132 kV Loktak-Ningthoukhong line, 132 kV Ningthoukhong – Churachandpur D/C lines, 132 kV Yiangangpokpi -Kongba D/C lines tripped. Due to tripping of these elements, Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh, Thoubal old, Thoubal New and Kongba area of Manipur Power System and Tamu load of Myanmar power system were isolated from NER Grid and collapsed due to no source available in these areas Power was extended to Manipur power system at 19:18 Hrs of 08.06.2024.	132 kV Loktak-Ningthoukhong line, 132 kV Ningthoukhong – Churachandpur D/C lines, 132 kV Yiangangpokpi -Kongba D/C lines
6	GD I	Blackout in Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh, Thoubal old, Thoubal New, Yiangangpokpi, Hundung, Yurembam, Rengpang and Kongba areas of Manipur Power System, Loktak Generating S/S and Tamu load of Myanmar power	09-06-2024 13:53	09-06-2024 14:39	00:46	70	60	2.75%	2.71%	2545	2213	Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh, Thoubal old, Thoubal New, Yiangangpokpi, Hundung, Yurembam, Rengpang and Kongba areas of Manipur Power System, Loktak Generating S/S and Tamu load of Myanmar power systemwere connected with rest of NER Grid through 132 kV Loktak-Jiribam & 132 kV Loktak Ningthoukhong line. 132 kV Imphal-Ningthoukhong went under planned shutdown at 10:17 Hrs & 132 kV Imphal-Loktak Line went under planned shutdown at 10:31 Hrs, 132 kV Jiribam(MA)-Rengpang line and 132 kV Imphal(Yurembam) - Karong line were under outage. At 13:53 Hrs of 09-06-2024, 132 kV Loktak-Jiribam & 132 kV Loktak-Ningthoukhong line tripped. Due to tripping of these elements, Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh, Thoubal old, Thoubal New, Yiangangpokpi, Hundung, Yurembam, Rengpang and Kongba areas of Manipur Power System, Loktak Generating S/S and Tamu load of Myanmar power system isolated from NER Grid and collapsed due to load generation mismatch in these areas. Power was extended to Manipur power system at 14:39 Hrs of 09.06.2024.	132 kV Loktak-Jiribam & 132 kV Loktak-Ningthoukhong line

20.Details of Grid Events during the Month of June 2024 in North-East Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
7	GD I	Zhadima, Chiephbozou & Wokha areas of Nagaland Power System	10-06-2024 22:31	11-06-2024 10:52	12:21	0	1	0.00%	0.03%	2915	3245	Zhadima, Chiephbozou & Wokha areas of Nagaland Power System is connected with rest of NER Grid through 132 kV Sanis-Wokha and 132 kV Kohima-Zhadima lines. Prior to the event, 132 kV Sanis-Wokha line was under planned shutdown since 08:03 Hrs of 10.06.2024. At 22:31 Hrs of 10-06-2024, 132 kV Kohima-Zadima line tripped. Due to tripping of this element, Zadima, Chiephebozou and Wokha areas of Nagaland power system got separated from rest of the grid due to no source available in these areas. Power was extended to Zadima, Chiephebozou and Wokha areas of Nagaland power system by charging 132 kV Kohima-Zadima line at 10:52 Hrs of 11-06-2024.	132 kV Kohima-Zadima line
8	GD I	Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh, Thoubal old, Thoubal New and Kongba area of Manipur Power System and Tamu load of Myanmar power system	12-06-2024 10:33	12-06-2024 10:57	00:24	0	45	0.00%	1.69%	2518	2657	Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh, Thoubal old, Thoubal New and Kongba area of Manipur Power System and Tamu load of Myanmar power system are connected with rest of NER Grid through 132 kV Ningthoukhong – Churachandpur D/C lines & 132 kV Yiangangpokpi - Kongba D/C lines. 400kV-Imphal(PG)-Thoubal-1 line was under outage since 13:32 Hrs of 18.10.2021, 400kV-Imphal(PG)-Thoubal-2 line was under outage since 15:05 Hrs of 24.04.2024. 132 kV Ningthoukhong –Churachandpur 1 line was under tripped condition since 08.06.2024. At 10:33 Hrs of 12-06-2024, 132 kV Loktak-Ningthoukhong line, 132 kV Ningthoukhong – Churachandpur 2 line, 132 kV Yiangangpokpi -Kongba D/C lines tripped. Due to tripping of these elements, Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh, Thoubal old, Thoubal New and Kongba area of Manipur Power System and Tamu load of Myanmar power system were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was restored to Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh, Thoubal old, Thoubal New and Kongba area of Manipur Power System and Tamu load of Myanmar power system by charging 132 kV Yiangangpokpi -Kongba D/C lines at 10:57 Hrs of 12-06-2024.	132 kV Loktak-Ningthoukhong line, 132 kV Ningthoukhong – Churachandpur 2 line, 132 kV Yiangangpokpi -Kongba D/C lines
9	GD I	Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh areas of Manipur Power System and Tamu load of Myanmar power system	13-06-2024 08:08	13-06-2024 08:54	00:46	0	40	0.00%	2.18%	2273	1838	Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh areas of Manipur Power System and Tamu load of Myanmar power system are connected with rest of NER Grid through 132 kV Ningthoukhong – Churachandpur D/C lines. 400kV-Imphal(PG)-Thoubal-1 line was under outage since 13:32 Hrs of 18.10.2021, 400kV-Imphal(PG)-Thoubal-2 line was under outage since 15:05 Hrs of 24.04.2024. 132 kV Ningthoukhong – Churachandpur 1 line was under tripped condition since 08.06.2024. 132 kV New Thoubal – Kakching & 132 kV Old Thoubal – Kakching were under outage condition prior to event. At 08:08 Hrs of 13-06-2024, 132 kV Ningthoukhong – Churachandpur 2 line tripped. Due to tripping of these elements, Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh areas of Manipur Power System and Tamu load of Myanmar power system were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was restored to Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Moreh area of Manipur Power System and Tamu load of Myanmar power system by charging 132 kV Ningthoukhong – Churachandpur 2 line at 08:54 Hrs of 13-06-2024.	132 kV Ningthoukhong – Churachandpur 2 line
10	GD I	Leshka generating station of Meghalaya Power System	13-06-2024 01:10	13-06-2024 01:19	00:09	0	36	0.00%	1.30%	2392	2771	Leshka HEP of Meghalaya Power System was connected with rest of NER Grid via 132 kV Leshka - Khleihriat D/C lines. At 01:10 Hrs of 13-06-2024, 132 kV Leshka - Khleihriat D/C lines tripped. Due to tripping of these lines, Leshka HEP of Meghalaya Power System was isolated from NER Grid and collapsed due to loss of evacuation path. Power supply was extended to Leshka HEP of Meghalaya Power System by charging 132 kV Myntdu Leshka - Khleihriat 1 line at 01:19 Hrs.Subsequently, 132 kV Myntdu Leshka - Khleihriat 2 line was charged at 01:21 Hrs.	132 kV Leshka - Khleihriat D/C lines
11	GD I	Bilasipara and Gauripur areas of Assam Power System	13-06-2024 05:02	13-06-2024 05:18	00:16	0	120	0.00%	7.01%	2248	1711	Bilasipara and Gauripur areas of Assam Power System are connected to NER Power system via 132 kV Bilasipara – Kokrajhar D/C lines. 132 kV Gauripur – Gosaigaon line is kept opened for load segregation purpose. At 05:02 Hrs of 13-06-2024, 132 kV Bilasipara – Kokrajhar D/C lines tripped. Due to tripping of these lines, Bilasipara and Gauripur areas of Assam Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power was extended to Bilasipara and Gauripur areas of Assam Power System by charging 132 kV Bilasipara – Kokrajhar D/C lines at 05:18 Hrs of 13-06-2024.	132 kV Bilasipara – Kokrajhar D/C lines

20.Details of Grid Events during the Month of June 2024 in North-East Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
12	GD II	BTPS, Kokrajhar, Bilasipara, Gauripur, Gossaingaon, Dhaligaon, APM, Barpeta, Nalbari, Barnagar, Nathkuchi, Kamalpur, Sipajhar, BGR, Railway TSS areas of Assam Power System	14-06-2024 22:32	14-06-2024 22:36	00:04	0.79	410	0.02%	13.82%	3344	2966	BTPS, Kokrajhar, Bilasipara, Gauripur, Gossaingaon, Dhaligaon, APM, Barpeta, Nalbari, Barnagar, Nathkuchi, Kamalpur, Sipajhar, BGR, Railway TSS areas of Assam Power System was connected to NER Power system via multiple elements connected to 220 kV BTPS GSS. At 22:32 Hrs of 14-06-2024, all feeders connected to Bus 1 and Bus 2 of 220 kV BTPS GSS tripped due to operation of Bus bar protection. Due to tripping of these elements, BTPS, Kokrajhar, Bilasipara, Gauripur,Gossaigaon, Dhaligaon, APM, Barpeta, Nalbari, Barnagar, Nathkuchi, Kamalpur, Sipajhar, BGR, Railway TSS areas of Assam Power System was isolated from NER Grid and collapsed due to no source available in these areas. BTPS Bus-2 and its associated elements were charged first via 220 KV BTPS-Salakati 2 at 22:36 Hrs of 14.06.2024. Subsequently, BTPS Bus 1 was charged at 00:31 Hrs of 15-06-2024 by charging 220 kV BTPS-Salakati 1 line and Power supply was restored at BTPS, Kokrajhar, Bilasipara, Gauripur, Gossaingaon, Dhaligaon, APM, Barpeta, Nalbari, Barnagar, Nathkuchi, Kamalpur, Sipajhar, BGR, Railway TSS areas of Assam Power System.	BTPS Bus-1 & 2
13	GD I	Churachandpur Thanlon, Elangkangpokpi, Kakching, Chandel, Thoubal Old, New Thoubal and Kongba area of Manipur Power System	15-06-2024 06:11	15-06-2024 07:03	00:52	0	45	0.00%	2.19%	2432	2059	Churachandpur Thanlon, Elangkangpokpi, Kakching, Chandel, Thoubal Old, New Thoubal and Kongba area of Manipur Power System were connected with rest of NER Grid via 132kV Ningthoukhong-Churachandpur-1 & 2 and 132kV Kongba – Yanganpokpi 1 & 2 lines. At 06:11 Hrs of 15-06-2024, 132 kV Loktak-Ningthoukhong line, 132 kV Ningthoukhong-Churachandpur-1 & 2 and 132 kV Kongba – Yanganpokpi 1 & 2 lines tripped. Due to tripping of these elements, Churachandpur, Thanlon, Elangkangpokpi, Kakching, Chandel, Thoubal Old, New Thoubal and Kongba areas of Manipur Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Churachandpur and Thanlon area of Manipur Power System by first charging 132kV-Loktak-Ningthoukhong line at 06:43 Hrs of 15.06.2024 and then charging 132 kV Ningthoukhong-Churachandpur-2 line at 07:03Hrs of 15.06.2024.	132 kV Loktak-Ningthoukhong line, 132 kV Ningthoukhong-Churachandpur-1 & 2 and 132 kV Kongba – Yanganpokpi 1 & 2 lines
14	GD I	Nangalbibra, Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power System	15-06-2024 04:52	15-06-2024 05:14	00:22	22	30	0.86%	1.47%	2564	2040	Nangalbibra, Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power System are connected to NER Power system via 132 kV Agia-Nangalbibra, 132 kV Mendipathar-Nangalbibra & 132 kV Nangalbibra-Nongstoin lines. Prior to the event, 132 kV Agia Nangalbibra was under tripped condition since 03:32 Hrs of 15-06-2024 and 132 kV Mendipathar – Nangalbibra line was under tripped condition since 03:48 Hrs of 15-06-2024. At 04:52 Hrs of 15-06-2024, 132 kV Nangalbibra-Nongstoin line tripped. Due to tripping of this element, Nangalbibra, Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power System were isolated from NER Grid and collapsed due to load generation mismatch in these areas. Power was extended to Nangalbibra, Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power System by charging 132 kV Nangalbibra-Mendipathar line at 05:14 Hrs of 15.06.2024.	132 kV Nangalbibra-Nongstoin line
15	GD I	Serchhip area of Mizoram Power System	15-06-2024 13:46	15-06-2024 14:13	00:27	0	17	0.00%	0.73%	2602	2331	Serchhip area of Mizoram Power System was connected to NER Power system via 132 kV Zuangtui – Serchhip line. 132 kV Serchhip – Lunglei was kept open on system requirement. At 13:46 Hrs of 15-06-2024, 132 kV Zuangtui – Serchhip line tripped. Due to tripping of the element, Serchhip area of Mizoram Power System was isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Serchhip area of Mizoram power system by charging 132 kV Zuangtui - Serchhip line at 14:13 Hrs of 15-06-2024.	132 kV Zuangtui – Serchhip line
16	GD I	Dhaligaon, Barnagar, Barpeta, APM, Gossaigaon areas of Assam Power System	16-06-2024 22:40	16-06-2024 23:06	00:26	0	110	0.00%	4.31%	3457	2553	Dhaligaon, Part load of Barnagar, Barpeta, APM, Gossaigaon areas of Assam Power System were connected to NER Power system via 132 kV Dhaligaon-BTPS D/C lines. At 22:40 Hrs of 16-06-2024, 132 kV Dhaligaon-BTPS D/C lines tripped. Due to tripping of these lines, Dhaligaon, Part load of Barnagar, Barpeta, APM, Gossaigaon areas of Assam Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Dhaligaon, Part load of Barnagar, Barpeta, APM, Gossaigaon areas of Assam Power System by charging 132kV BTPS-Dhaligaon D/C lines at 23:06 Hrs of 16.06.2024.	132 kV Dhaligaon-BTPS D/C lines
17	GD I	Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power System	16-06-2024 04:51	16-06-2024 07:38	02:47	7.5	7	0.29%	0.37%	2600	1896	Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power System were connected to NER Power system via 132kV Rongkhon - Nangalbibra line. At 04:51 Hrs of 16-06-2024, 132 kV Rongkhon - Nangalbibra line tripped. Due to tripping of this element, Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power System were isolated from NER Grid and collapsed due to load generation mismatch in these areas. Power supply was extended to Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power System by charging 132 kV Ronkhon-Nangalbibra line at 07:38 Hrs of 16.06.2024.	132 kV Rongkhon - Nangalbibra line
18	GD I	Rangia(220), Tangla, Sipajhar, Kamalpur, Nalbari, Nathkuchi and Barnagar areas of Assam Power system	17-06-2024 06:52	17-06-2024 07:26	00:34	0	53	0.00%	2.70%	2764	1960	Nalbari, Nathkuchi & part of Barnagar radially connected to 220 kV Rangia SS. Kamalpur, Sipajhar & Tangla radially connected to 132 kV Rangia SS. 220 kV Rangia SS radially connected via 220 kV BTPS-Rangia D/C only. At 06:52 Hrs of 17-06-2024 due to heavy downpour, 220kV BTPS- Rangia D/C lines tripped resulting in successful operation of SPS of the same. Due to which, 220kv Rangia S/S was isolated from 132kV Rangia S/S as well as Barnagar, Sipajhar, Kamalpur, , Nathkuchi, Tangla Substations were isolated from	220 kV BTPS- Rangia I&II

20.Details of Grid Events during the Month of June 2024 in North-East Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
19	GD I	Nangalbibra, Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power system	17-06-2024 07:03	17-06-2024 07:56	00:53	0	15	0.00%	0.78%	2767	1920	Nangalbibra, Rongkhon, Ampati areas and Ganol HEP of Meghalaya are connected with the rest of the grid through 132 kV Nangalbibra-Mendipathar line, 132 kV Agia-Nangalbibra line & 132 kV Nangalbibra-Nongstoin line. Prior to the event, 132 kV Nangalbibra-Nongstoin and 132kV Agia-Nangalbibra lines tripped at 07:02 hrs of 17-06-2024. At 07:03 Hrs of 17-06-2024, 132kV Agia-Nangalbibra line tripped. Due to tripping of this line, Nangalbibra, Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power System were isolated from NER Grid and	132kV Agia-Nangalbibra line
20	GD I	Leshka HEP of Meghalaya Power system	17-06-2024 20:30	17-06-2024 20:46	00:16	119	0	3.44%	0.00%	3457	2828	Leshka HEP of Meghalaya Power System was connected with rest of NER Grid via 132 kV Myntdu Leshka - Khleihriat D/C lines. At 20:30 Hrs of 17-06-2024, 132 kV Myntdu Leshka - Khleihriat D/C lines tripped. Due to tripping of these lines, Leshka HEP of Meghalaya Power System was isolated from NER Grid and collapsed due to loss of evacuation path. Power supply was extended to Leshka HEP of Meghalaya Power System by charging 132 kV Myntdu Leshka - Khleihriat 1 and 2 at 21:39 hrs and 20:46 hrs respectively of 17.06.2024.	132 kV Myntdu Leshka - Khleihriat D/C lines
21	GD I	Rangia(220), Tangla, Sipajhar, Kamalpur, Nalbari, Nathkuchi and Barnagar areas of Assam Power system	17-06-2024 21:17	17-06-2024 21:31	00:14	0	130	0.00%	4.68%	3269	2777	Nalabari, Nathkuchi & part of Barnagar radially connected to 220 kV Rangia SS. Kamalpur, Sipajhar & Tangla radially connected to 132 kV Rangia SS. 220 kV Rangia SS radially connected via 220 kV BTPS-Rangia D/C only. At 21:17 Hrs of 17-06-2024 due to heavy downpour, 220kV BTPS- Rangia D/C lines tripped resulting in successful operation of SPS of the same. Due to which, 220kv Rangia S/S was isolated from 132kV Rangia S/S as well as 132kV Dhaligaon, Barnagar, Sipajhar, Kamalpur, Nathkuchi, Tangla Subststions were isolated from Rangia System resulting in blackout of Nalbari. part load of Barnagar, Sipajhar, Kamalpur, Nathkuchi, Tangla areas of Assam Power System. Power supply was extended to the affected areas of Assam Power System by first charging 220kV BTPS-Rangia line I and II at 21:31 Hrs and 21:34 Hrs respectively of 17.06.2024	220 kV BTPS- Rangia D/C lines
22	GD I	Leshka HEP of Meghalaya Power system	17-06-2024 23:37	18-06-2024 00:33	00:56	42	0	1.42%	0.00%	2952	1923	Leshka HEP of Meghalaya Power System was connected with rest of NER Grid via 132 kV Myntdu Leshka - Khleihriat D/C lines. At 23:37 Hrs of 17-06-2024, 132 kV Myntdu Leshka - Khleihriat D/C lines tripped. Due to tripping of these lines, Leshka HEP of Meghalaya Power System was isolated from NER Grid and collapsed due to loss of evacuation path. Power supply was extended to Leshka HEP of Meghalaya Power System by charging 132 kV Myntdu Leshka - Khleihriat 1 and 2 at 00:33 hrs and 00:36 hrs respectively of 18.06.2024.	132 kV Myntdu Leshka - Khleihriat D/C lines
23	GD I	Nangalbibra, Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power system	18-06-2024 05:07	18-06-2024 05:41	00:34	0	15	0.00%	0.78%	2767	1920	Nangalbibra, Rongkhon, Ampati areas and Ganol HEP of Meghalaya are connected with the rest of the grid through 132 kV Nangalbibra-Mendipathar line, 132 kV Agia-Nangalbibra line & 132 kV Nangalbibra-Nongstoin line. Prior to the event, 132 kV Nangalbibra-Nongstoin and 132kV Agia-Nangalbibra lines tripped at 05:01 hrs of 18-06-2024. At 05:07 Hrs of 18-06-2024, 132kV Mendipathar – Nangalbibra line tripped. Due to tripping of this line, Nangalbibra, Rongkhon, Ampati areas and Ganol HEP of Meghalaya Power System were isolated from NER Grid and collapsed due to load generation mismatch in these areas. Power supply was extended to the affected areas of Meghalaya Power System by first charging 132kV Mendipathar - Nangalbibra line at 05:41 Hrs of 18.06.2024 and subsequently charged 132 kV Agia-Nangalbibra at 05:49 Hrs of 18.06.2024	132kV Mendipathar – Nangalbibra line
24	GD I	Rengpang area of Manipur Power system	20-06-2024 10:54	20-06-2024 15:23	04:29			0.00%	0.00%	2567	2128	Rengpang area of Manipur Power System was connected with rest of NER Grid through 132 kV Loktak – Rengpang line. 132 kV Jiribam-Rengpang line was under outage since 18:18 Hrs of 17.11.2023. At 19:52 Hrs of 19-04-2024, 132 kV Loktak – Rengpang line tripped. Due to tripping of this element, Rengpang area of Manipur Power System was isolated from NER Grid and collapsed due to no source available this areaAt 10:54 Hrs of 20-06-2024, 132kV Loktak-Rengpang line tripped. Power supply was extended to Rengpang area of Manipur Power System by charging 132 kV Loktak – Rengpang line at 15:23 Hrs on 20-06-2024	132 kV Loktak – Rengpang line
25	GD I	Sarupather Area of Assam Power system	20-06-2024 14:41	20-06-2024 14:55	00:14	0	3	0.00%	0.13%	2836	2285	Sarupathar area of Assam Power System was connected with connected to NER Power system via 132kV Golaghat-Sarupathar-Bokajan link. At 14:41 Hrs,132 kV Golaghat-Sarupathar and 132 kV Sarupathar-Bokajan lines tripped due to which Sarupathar area of Assam Power System was isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended by charging 132kV Golaghat – Sarupathar line at 14:55 Hrs of 20.06.2024	132 kV Golaghat- Sarupathar & 132 kV Sarupathar- Bokajan

20.Details of Grid Events during the Month of June 2024 in North-East Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
26	GD I	Gossaigaon and Gauripur area of Assam Power system	22-06-2024 11:58	22-06-2024 12:09	00:11	0	60	0.00%	2.34%	2675	2568	Gossaigon and Gauripur areas of Assam Power System were connected with connected to NER Power system via 132 kV Dhaligaon-Gossaigaon-Gauripur link (132 kV Gauripur-Bilasipara was under S/D for SF6 gas leakage issue at Bilasipara end). At 11:58 Hrs, 132kV Dhaligaon-Gossaigaon line tripped due to which Gosaigaon and Gauripur areas of Assam Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended by charging 132kV Dhaligaon–Gossaigaon line at 12:09 Hrs and then charging 132kV Gossaigaon-Gauripur at 12:15 Hrs of 22.06.2024	132kV Dhaligaon-Gossaigaon lin
27	GD I	Pasighat, Roing, Tezu, Namsai areas of Arunachal Pradesh power system and Chapakhowa area of Assam Power system	22-06-2024 12:03	22-06-2024 13:09	01:06	0	19	0.00%	0.72%	2573	2640	Pasighat, Roing, Tezu and Namsai areas of Arunachal Pradesh Power System and Chapakhowa area of Assam Power System was connected with rest of NER Grid by 132 kV Along -Pasighat line. (After tripping of 132kV Rupai – Chapakhowa line at 11:59 Hrs of 22-06-2024) At 12:03 Hrs of 22-06-2024, 132 kV Along - Pasighat tripped. Due to tripping of this element, Pasighat-Roing-Tezu-Namsai area of Arunachal Pradesh Power System and Chapakhowa area of Assam Power System was isolated from NER Grid and collapsed due to no source available in these areas. Power was extended to Chapakhowa area of Assam Power System, by charging 132 kV Chapakowa - Rupai line at 12:23 Hrs. Subsequently, Power was extended to Arunachal Pradesh power system by charging 132 kV Roing - Chapakhowa – I & II at 12:42 Hrs of 22-06-2024. Power was extended to Pasighat and Along areas by charging 132kV Roing – Pasighat at 12:59 Hrs and 132kV Pasighat – Along line at 13:09 Hrs respectively. Also, 132kV Roing – Tezu was charged at 12:45 Hrs and 132kv Tezu – Namsai was charged at 12:48 Hrs	132 kV Along - Pasighat
28	GD I	Ganol area of Meghalaya Power system	22-06-2024 12:03	22-06-2024 13:09	01:06	15	0	0.56%	0.00%	2677	2778	Ganol HEP of Meghalaya Power was connected with rest of NER Grid by 132 kV Rongkhon-Ganol line. At 14:03 Hrs of 22-06-2024, 132 kV Rongkhon-Ganol line tripped. Due to tripping of this element, Ganol HEP of Meghalaya Power System isolated from NER Grid and collapsed due to no evacuation path. Power supply was extended to Ganol HEP of Meghalaya Power System by charging 132 kV Rongkhon - Ganol line at 14:21 Hrs.	132 kV Rongkhon-Ganol line
29	GD I	Leshka HEP of Meghalaya Power system	23-06-2024 16:00	23-06-2024 16:09	00:09	119	0	4.51%	0.00%	2640	2721	Leshka HEP of Meghalaya Power System was connected with rest of NER Grid via 132 kV Myntdu Leshka - Khleihriat D/C lines. At 16:00 Hrs of 23-06-2024, 132 kV Myntdu Leshka - Khleihriat D/C lines tripped. Due to tripping of these lines, Leshka HEP of Meghalaya Power System was isolated from NER Grid and collapsed due to loss of evacuation path. Power supply was extended to Leshka HEP of Meghalaya Power System by charging 132 kV Myntdu Leshka - Khleihriat 1 at 16:09 hrs	132 kV Myntdu Leshka - Khleihriat D/C lines
30	GD I	Umiam area of Meghalaya Power system	24-06-2024 13:38	24-06-2024 14:30	00:52	0	10	0.00%	0.38%	3052	2601	Umiam area of Meghalaya Power System was connected with rest of NER Grid via 132 kV Umiam Stage I-Umiam and 132 kV Nehu-Umiam lines. At 13:38 Hrs of 24-06-2024, 132kV Umiam Stage I-Umiam and 132 kV Nehu-Umiam lines tripped. Due to tripping of these lines, Umiam S/S of Meghalaya Power System was isolated from NER Grid. Power supply was extended to Umiam area of Meghalaya Power System by charging 132kV NEHU-Umiam line at 14:30 hrs of 24-06-2024.	132kV Umiam Stage I-Umiam and 132 kV Nehu-Umiam lines
31	GD I	Gossaigaon area of Assam power system	25-06-2024 07:05	25-06-2024 07:10	00:05	0	2	0.00%	0.08%	3020	2580	Gossaigon area of Assam Power System is connected with connected to NER Power system via 132 kV Dhaligaon Gossaigaon-Gauripur line (132 kV Gauripur- Gossaigaon was kept open due to system requirement) At 07:05 Hrs of 25.06.2024, 132 kV Dhaligaon-Gossaigaon line tripped due to which Gosaigaon area of Assam Power System was isolated from NER Grid and collapsed due to no source available in the area. Power supply was extended by charging 132 kV Dhaligaon–Gossaigaon line at 07:10 Hrs of 25.06.2024.	132 kV Dhaligaon-Gossaigaon line
32	GD I	Kohima, Doyang, Sanis, Wokha, Chiephobozou, Zhadima and Karong areas	25-06-2024 13:15	25-06-2024 14:28	01:13	0	29	0.00%	1.06%	2946	2739	Kohima substation was connected with the rest of the grid via 132 kV Doyang-Sanis-Wokha- Chiephobozou-Zhadhima link while Doyang SS was connected with rest of the grid via 132 kV Doyang-Mokokchung and 132 kV Doyang-Sanis link while Karong was fed from 132 kV Kohima-Karong line. Prior to the event, 132 kV Dimapur-Kohima was under Planned shutdown, 132 kV Imphal-Karong line tripped at 13:02 Hrs of 25.06.2024, 132 kV Doyang-Dimapur D/C lines were under outage and 132 kV Meluri-Kohima line was under outage since 27-09-2023. At 13:15 Hrs of 25-06-2024, 132 kV Doyang-Mokokchung line tripped. Due to tripping of this element, Kohima, Sanis, Wokha, Chiephobozou, Zadhima areas of Nagaland, Karong area of Manipur and Doyang	132 kV Doyang-Mokokchung line

20.Details of Grid Events during the Month of June 2024 in North-East Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
33	GD I	Dhemaji and Silapathar areas of Assam Power System	26-06-2024 13:15	26-06-2024 13:26	00:11	0	37	0.00%	1.29%	3132	2869	Dhemaji and Silapathar areas of Assam Power System were connected with connected to NER Power system via 132 kV North Lakhimpur-Dhemaji line. At 13:15 Hrs of 26.06.2024, 132 kV North Lakhimpur-Dhemaji line tripped due to which Dhemaji and Silapathar areas of Assam Power System was isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Dhemaji and Silapathar areas of Assam Power System by charging 132 kV North Lakhimpur-Dhemaji line at 13:26 Hrs of 26.06.2024.	132 kV North Lakhimpur-Dhemaji line
34	GD I	Haflong area of Assam Power System	26-06-2024 16:59	26-06-2024 17:16	00:17	0	5	0.00%	0.18%	3126	2822	Haflong(PG) S/S and Haflong area of Assam Power System were connected with connected to NER Power system via 132 kV Haflong(PG)-Umrangshu line. 132 kV Haflong(PG)-Jiribam(PG) line was under outage since 20.06.2024. At 16:59 Hrs of 26.06.2024, 132 kV Haflong(PG)-Umrangshu line tripped due to which Haflong(PG) S/S and Haflong area of Assam Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Haflong(PG) S/S and Haflong area of Assam Power System by charging 132 kV Haflong(PG)-Umrangshu line at 17:16 Hrs of 26.06.2024.	132 kV Haflong(PG)-Umrangshu line
35	GD I	Leshka Generating Station of Meghalaya Power System	26-06-2024 12:01	26-06-2024 12:44	00:43	84	0	2.72%	0.00%	3088	2838	Leshka Generating Station of Meghalaya Power System was connected with connected to NER Power system via 132 kV Myntdu Leshka - Khleihriat D/C Lines. At 12:01 Hrs of 26.06.2024, 132 kV Myntdu Leshka - Khleihriat D/C Lines tripped due to which Myndtu Leshka Generating Station of Meghalaya Power System was isolated from NER Grid and collapsed due to loss of evacuation path. Power supply was extended to Myndtu Leshka Generating Station of Meghalaya Power System by charging 132 kV Myntdu Leshka -Khleihriat 1 Line at 12:44 Hrs of 26.06.2024.	132 kV Myntdu Leshka - Khleihriat D/C Lines
36	GD I	Lungmual, Melriat and Lunglei area of Mizoram Power System	26-06-2024 15:44	26-06-2024 21:45	06:01	0	22	0.00%	0.75%	3082	2941	Lungmual, Melriat and Lunglei areas of Mizoram Power System were connected with connected to NER Power system via 132 kV Aizawl - Luangmual Line. At 15:44 Hrs of 26.06.2024, 132 kV Aizawl - Luangmual Line tripped due to which Lungmual, Melriat and Lunglei areas of Mizoram Power System were isolated from NER Grid and collapsed due to no source available in these area. Power supply was extended to Lungmual, Melriat and Lunglei areas of Mizoram Power System by charging 132 kV Aizawl-Lungmual Line at 21:45 Hrs of 26.06.2024.	132 kV Aizawl - Luangmual Line
37	GD I	Mustem area of Meghalaya Power System	26-06-2024 12:28	26-06-2024 12:40	00:12	0	7	0.00%	0.24%	2987	2900	Mustem area of Meghalaya Power System is connected with connected to NER Power system via 132 kV Mawlyndep-Mustem line & 132 kV Khliehriat-Mustem line. Prior to the event, 132 kV Khliehriat-Mustem line tripped at 12:12 Hrs of 26.06.2024. At 12:28 Hrs of 26.06.2024, 132 kV Mawlyndep-Mustem line tripped due to which Mustem area of Meghalaya Power System was isolated from NER Grid and collapsed due to no source available in these area. Power supply was extended to Mustem area of Meghalaya Power System by charging 132kV Khliehriat-Mustem line at 12:40 Hrs of 26.06.2024.	132 kV Mawlyndep-Mustem line
38	GD I	Tuirial HEP of Mizoram Power System	26-06-2024 13:51	26-06-2024 14:05	00:14	45	0	1.42%	0.00%	3176	2883	Tuirial HEP of Mizoram Power System was connected NER Power system via 132kV Tuirial - Kolasib line. At 13:51 Hrs of 26.06.2024, 132 kV Tuirial - Kolasib line tripped due to which Tuirial HEP of Mizoram Power System was isolated from NER Grid and collapsed due to loss of evacuation path. Power supply was extended to Tuirial HEP of Mizoram Power System by charging 132kV Tuirial - Kolasib line at 14:05 Hrs of 26.06.2024.	132 kV Tuirial - Kolasib line
39	GD I	Wokha area of Nagaland Power System	26-06-2024 09:31	26-06-2024 10:08	00:37	0	5	0.00%	0.18%	3034	2768	Wokha area of Nagaland Power System was connected with connected to NER Power system via 132kV Wokha-Chiephobozou and 132 kV Sanis-Wokha lines. At 09:31 Hrs of 26.06.2024, 132 kV Wokha-Chiephobozou and 132 kV Sanis-Wokha lines tripped due to which Wokha area of Nagaland Power System was isolated from NER Grid and collapsed due to no source available in these area. Power supply was extended to Wokha area of Nagaland Power System by charging 132 kV Sanis-Wokha line at 10:08 Hrs of 26.06.2024.	132 kV Wokha-Chiephobozou and 132 kV Sanis-Wokha lines
40	GD I	Sanis area of Nagaland Power System	27-06-2024 03:54	27-06-2024 04:23	00:29	0	1	0.00%	0.04%	3177	2640	Sanis area of Nagaland Power System was connected with rest of NER Power system via 132 kV Doyang-Sanis and 132 kV Sanis Wokha lines. At 03:54 Hrs of 27.06.2024, 132 kV Doyang-Sanis and 132 kV Sanis-Wokha lines tripped due to which Sanis area of Nagaland Power System was isolated from NER Grid and collapsed due to no source available in these area. Power supply was extended to Sanis area of Nagaland Power System by charging 132 kV Doyang-Sanis line at 04:23 Hrs of 27.06.2024.	132 kV Doyang-Sanis and 132 kV Sanis-Wokha lines

20.Details of Grid Events during the Month of June 2024 in North-East Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
41	GD I	Along, Pasighat, Roing, Tezu, Namsai areas of Arunachal Pradesh Power System and Chapakhowa area of Assam Power System	28-06-2024 13:23	28-06-2024 13:46	00:23	0	24	0.00%	0.94%	2772	2564	Along, Pasighat, Roing, Tezu, Namsai areas of Arunachal Pradesh Power System and Chapakhowa area of Assam Power System was connected with rest of NER Power system via 132 kV Along - Basar Line & 132 kV Rupai-Chapakhowa line. Prior to theevent, 132 kV Rupai – Chapakhpwa line was under outage condition to facilitate shutdown of 132 kV Rupai-Tinsukia line. At 13:23 Hrs of 28.06.2024, 132 kV Along - Pasighat Line and 132 kV Along - Basar Line tripped due to which Along, Pasighat, Roing, Tezu, Namsai areas of Arunachal Pradesh Power System and Chapakhowa area of Assam Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Along, Pasighat, Roing, Tezu, Namsai areas of Arunachal Pradesh Power System and Chapakhowa area of Assam Power System by charging 132 kV Rupai – Chapakhpwa line at 13:46 Hrs of 28.06.2024.	132 kV Along - Pasighat Line and 132 kV Along - Basar Line
42	GD I	Pasighat, Roing, Tezu, Namsai areas of Arunachal Pradesh Power System and Chapakhowa area of Assam Power System	29-06-2024 09:25	29-06-2024 10:47	01:22	0	14	0.00%	0.60%	2873	2337	Pasighat, Roing, Tezu, Namsai areas of Arunachal Pradesh Power System and Chapakhowa area of Assam Power System are connected with rest of NER Power system via 132 kV Along - Pasighat Line and 132 kV Rupai – Chapakhowa line. At 09:25 Hrs of 29.06.2024, 132 kV Along - Pasighat Line, 132 kV Roing - Pasighat Line and 132 kV Rupai – Chapakhpwa line tripped due to which Pasighat, Roing, Tezu, Namsai areas of Arunachal Pradesh Power System and Chapakhowa area of Assam Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Chapakhowa area of Assam Power System by charging 132 kV Rupai – Chapakhpwa line at 09:42 Hrs of 29.06.2024. Subsequently power supply was extended to Roing, Tezu and Namsai areas of Arunachal Pradesh Power System by charging 132 kV Roing Chapakhowa D/C lines at 10:01 Hrs of 29.06.2024. Power supply was extended to Pasighat area of Arunachal Pradesh Power System by charging 132 kV Roing - Pasighat Line at 10:47 Hrs of 29.06.2024.	132 kV Along - Pasighat Line, 132 kV Roing - Pasighat Line and 132 kV Rupai – Chapakhpwa line
43	GI 1	Loktak	09-06-2024 10:32	09-06-2024 10:32	00:00	105	0	4.24%	0.00%	2476	2181	Loktak Unit-1,2,3 tripped at 10:32 Hrs on 09.06.2024 due to loss of evacuation path due to tripping of 132 kV Loktak-Jiribam line on Overcurrent from Loktak end.	Loktak Unit-1,2,3

21. NEW ELEMENTS COMMISSIONED* IN JUNE 2024

COMMISSIONING OF NEW GENERATING UNITS

REGION	S.NO.	ENTITY	UNIT NAME / LOCATION	ADD. CAPACITY (MW)	UNIT NO.	INS. CAPACITY (MW)	DATE@
WR	1	MASAYA SOLAR ENERGY PVT LTD	Masaya Solar Energy Pvt Ltd(Solar)/MP	3.1	-	300	21-06-2024
SR	2	JSW RENEW ENERGY TWO LTD	JSW Renew Energy Two Limited(Wind)/Tamilnadu	13.5	LOT-XIII	291.6	04-06-2024
	3	JSW RENEW ENERGY LTD	JSW Renew Energy Ltd(Wind)/Tamilnadu	18.9	LOT-VI	189	28-06-2024
	4	RENEW SURYA OJAS PRIVATE LTD	Renew Surya Ojas Private Limited(Wind)/Karnataka	6.93	LOT-VII	436.59	12-06-2024

@RE (Solar & Wind) generation FTC for complete capacity is in phases. Date mentioned is the date by which complete (Added) capacity synch./COD

COMMISSIONING OF NEW ICTs

REGION	S.NO.	ENTITY	SUB-STATION	VOLTAGE (kV)	ICT NO.	CAPACITY (MVA)	DATE
NR	1	UPPTCL	Gorakhpur	400/220/33	II	500	20-06-2024
	2	POWERGRID	Bhinmal	400/220/33	III	315	27-06-2024
WR	3	AGEL	Khavda PSS-3	400/33	III	330	22-06-2024
	4	AGEL	Khavda PSS-3	400/33	IV	330	22-06-2024
SR	5	POWERGRID	Arasur	400/220/33	IV	500	26-06-2024
	6	TNTRANSCO	Alamathi	400/110/33	VI	200	28-06-2024
NER	7	AEGCL	CTPS	132/33	I	25	07-06-2024
	8	AEGCL	Samaguri	132/33	II	25	13-06-2024

COMMISSIONING OF NEW TRANSMISSION LINES

REGION	S.NO.	ENTITY	LINE NAME	CKT NO.	VOLTAGE (kV)	DATE
NER	1	AEGCL	220kV Rangia - Amingaon	I	220	01-06-2024

LILO OF TRANSMISSION LINES

REGION	S.NO.	ENTITY	LINE NAME (LILOed LINE)	LILO AT	VOLTAGE (kV)	DATE
NIL						

COMMISSIONING OF BUS/LINE REACTORS

REGION	S.NO.	ENTITY	SUB-STATION / LINE (@SUB-STATION)	VOLTAGE (kV)	CAPACITY (MVAR)	DATE
NR	1	UPPTCL	Bus Reactor@Sahupuri	400	125	15-06-2024
SR	2	TSTRANSCO	Bus Reactor-I @Choutuppal	400	125	06-06-2024

COMMISSIONING OF HVDC/FACTS DEVICE/SYSTEM

REGION	S.NO.	ELEMENT NAME	SUB-STATION / CONSTITUENT	RATINGS	VOLTAGE (kV)	DATE
NIL						

*Includes first time charging (FTC) as well

22.Voltage Profile

विद्युत विभव रूपरेखा माह - जून 2024 VOLTAGE PROFILE - JUNE 2024

Date of reporting: 05-Jul-24

क्र.स. Sl. No.	क्षेत्र REGION	उपकेंद्र SUBSTATION	VOLTAGE < V(lower)* (V=380,728 kV)	V(lower) < VOLTAGE < V(upper)*	VOLTAGE > V(upper)* (V=420,800 kV)	Voltage Deviation Index (%age of time voltage is outside range)	Number of hours voltage was outside IEGC band during month	उच्चतम (कि.वो.) Maximum(kV)	निम्नतम(कि.वो.) Minimum(kV)	औसत (कि.वो.) Average(kV)
1	पूर्वी क्षेत्र ER	अंगुल ANGUL	0%	100%	0%	0%	0	791	755	772
2		दार्लिपली DARLIPALI	0%	100%	0%	0%	0	791	771	779
3		गया GAYA	0%	100%	0%	0%	0	791	748	769
4		जीरत JEERAT	0%	100%	0%	0%	1	801	737	771
5		झारसुगुड़ा JHARSUGUDA	0%	100%	0%	0%	0	802	776	787
6		मिदनापुर MEDINIPUR	0%	100%	0%	0%	1	803	746	775
7		रांची RANCHI	0%	100%	0%	0%	0	798	756	780
8		सासाराम SASARAM	0%	100%	0%	0%	0	786	744	766
1	उत्तरी क्षेत्र NR	आगरा AGRA	0%	100%	0%	0%	0	800	734	775
2		आगरा (फ़तेहाबाद) AGRA (FATEHABAD)	0%	100%	0%	0%	0	804	729	771
3		अजमेर AJMER	0%	99%	1%	1%	8	810	748	782
4		अलीगढ़ ALIGARH	0%	100%	0%	0%	0	807	725	770
5		अनपरा सी ANPARA-C	0%	100%	0%	0%	0	780	758	767
6		अनपरा डी ANPARA-D	0%	100%	0%	0%	0	777	756	764
7		अंता ANTA	0%	100%	0%	0%	0	801	740	776
8		बलिया BALLIA	0%	100%	0%	0%	0	786	734	762
9		बारा BARA	0%	100%	0%	0%	0	782	745	766
10		भाड़ला BHADLA	0%	100%	0%	0%	0	797	717	774
11		भाड़ला-2 BHADLA-2	0%	100%	0%	0%	0	797	737	778
12		भिवानी BHIWANI	0%	100%	0%	0%	1	813	717	775
13		बीकानेर BIKANER	0%	100%	0%	0%	0	799	725	776
14		बरेली BAREILLY	0%	99%	1%	1%	7	810	711	772
15		चित्तौड़गढ़ CHITTORGARH	0%	99%	1%	1%	8	804	762	786
16		फतेहगढ़-2 FATEHGARH-2	0%	100%	0%	0%	0	795	723	775
17		फ़तेहपुर FATEHPUR	0%	100%	0%	0%	0	796	738	772
18		ग्रेटर नोएडा GREATER NOIDA	0%	100%	0%	0%	2	809	724	775
19		हापुड़ HAPUR	0%	100%	0%	0%	1	806	719	769
20		जवाहरपुर JAWAHARPUR	0%	100%	0%	0%	0	799	731	771
21		झटिकरा JHATIKARA	0%	100%	0%	0%	1	804	708	767
22		कानपुर जीआईएस KANPUR GIS	0%	100%	0%	0%	0	796	731	770
23		खेतरी KHETRI	0%	100%	0%	0%	2	803	714	771
24		कोटेश्वर KOTESHWAR	0%	100%	0%	0%	2	807	723	775
25		ललितपुर LALITPUR	0%	100%	0%	0%	0	783	729	758
26		लखनऊ LUCKNOW	0%	100%	0%	0%	0	798	731	766
27		मैनपुरी MAINPURI	0%	100%	0%	0%	0	795	724	765
28		मेरठ MEERUT	0%	100%	0%	0%	3	808	714	775
29		मेरठ(UP) MEERUT (UP)	0%	100%	0%	0%	1	807	722	770
30		मोगा MOGA	0%	100%	0%	0%	1	810	717	776
31		ओबरा सी OBRA C	0%	100%	0%	0%	0	777	750	760
32		उरई ORAI	0%	100%	0%	0%	1	803	735	779
33		फागी PHAGI	0%	100%	0%	0%	1	810	701	777
34		रामपुर RAMPUR	0%	100%	0%	0%	0	804	722	771
35		उन्नाव UNNAO	0%	100%	0%	0%	0	790	735	758
36		वाराणसी VARANASI	0%	100%	0%	0%	0	789	743	772
1	पश्चिमी क्षेत्र WR	अकोला AKOLA	0%	100%	0%	0%	0	792	752	773
2		औरंगाबाद AURANGABAD	0%	100%	0%	0%	0	800	755	780
3		भोपाल (बीडीटीसीएल) BHOPAL (BDTCL)	0%	100%	0%	0%	0	787	747	771
4		भुज BHUJ	0%	100%	0%	0%	0	796	752	780
5		भुज-2 BHUJ-II	0%	100%	0%	0%	0	790	745	772
6		बिलासपुर BILASPUR	0%	100%	0%	0%	0	782	756	768
7		बीना BINA	0%	100%	0%	0%	0	797	742	779
8		बनसकंठा BANASKANTHA	0%	100%	0%	0%	1	802	755	785
9		चांपा CHAMPA	0%	99%	1%	1%	8	807	757	778
10		धुले (बीडीटीसीएल) DHULE (BDTCL)	0%	100%	0%	0%	0	793	755	777
11		धर्मजयगढ़ DHARAMJAIGARH	0%	100%	0%	0%	0	784	758	768
12		रायपुर पूलिंग RAIPUR POOLING	0%	100%	0%	0%	0	794	770	780
13		एकतूनी EKTUNI	0%	100%	0%	0%	0	791	745	773
14		गदरवाड़ा GADARWARA	0%	100%	0%	0%	0	801	764	786
15		ग्वालियर GWALIOR	0%	100%	0%	0%	0	802	736	777
16		इंदौर INDORE	0%	100%	0%	0%	0	791	749	771
17		जबलपुर JABALPUR	0%	100%	0%	0%	2	802	753	785
18		खंडवा KHANDWA	0%	100%	0%	0%	0	792	754	774
19		कोरडी KORADI	0%	100%	0%	0%	0	781	749	770
20		लकाडिया LAKADIYA	0%	100%	0%	0%	3	804	760	786
21		रायगढ़ पूलिंग RAIGARH POOLING	0%	100%	0%	0%	0	795	769	781
22		पड़घे PADGHE	0%	100%	0%	0%	2	803	746	779
23		परली PARLI	0%	97%	3%	3%	23	806	757	789
24		पुणे PUNE	0%	100%	0%	0%	2	803	741	777
25		राजनन्दगांव RAJNANDGAON	0%	99%	1%	1%	4	803	775	787
26		सासन SASAN	0%	100%	0%	0%	0	788	755	776
27		सतना SATNA	0%	100%	0%	0%	0	799	752	782
28		सिवनी SEONI	0%	100%	0%	0%	0	795	758	779
29		सीपत SIPAT	0%	100%	0%	0%	0	782	759	769
30		सोलापुर SOLAPUR	0%	97%	3%	3%	23	806	756	789
31		तिरोड़ा TIRORA	0%	100%	0%	0%	0	778	751	767
32		तामनार TAMNAR	0%	100%	0%	0%	0	795	769	781
33		वड़ोदरा VADODARA	0%	100%	0%	0%	1	801	759	784
34		विंध्याचल पूलिंग VINDHYACHAL PS	0%	100%	0%	0%	0	792	758	779
35		वर्धा WARDHA	0%	100%	0%	0%	3	803	765	783
36		वरोरा WARORA	0%	100%	0%	0%	0	801	765	785
1	दक्षिणी क्षेत्र SR	अरियलूर ARIYALUR	0%	100%	0%	0%	0	799	740	773
2		कड़पा CUDDAPAH	0%	93%	7%	7%	54	810	757	788
3		चिल्कलुरीपेट CHILAKALURIPETA	0%	99%	1%	1%	6	811	754	784
4		कर्नूल KURNOOL	0%	99%	1%	1%	5	806	756	782
5		महेश्वरम MAHESWARAM	0%	94%	6%	6%	45	812	766	789
6		निज़ामाबाद NIZAMABAD	0%	90%	10%	10%	71	814	762	792
7		नेल्लूर पूलिंग NELLORE PS	0%	100%	0%	0%	0	796	754	775
8		नॉर्थ चेन्नै पूलिंग NORTH CHENNAI PS	0%	100%	0%	0%	3	814	730	775
9		रायचूर RAICHUR	0%	100%	0%	0%	1	803	756	782
10		श्रीकाकुलम SRIKAKULAM	0%	99%	1%	1%	5	803	758	784
11		तिरुवलम THIRUVALEM	0%	100%	0%	0%	0	796	745	773
12		वेमगिरी VEMAGIRI	0%	97%	3%	3%	22	808	716	784
13		वारंगल WARANGAL	0%	96%	4%	4%	28	807	766	789
1	पूर्वोत्तर क्षेत्र NER	बिस्वनाथ चरिआलि BISWANATH CHARIALI (400 kV)	0%	100%	0%	0%	0	414	392	404
2		बोंगाईगांव BONGAIGAON (400 kV)	0%	100%	0%	0%	0	413	398	406
3		बोंगाईगांव टीपीएस BONGAIGAON TPS (400 kV)	0%	100%	0%	0%	0	414	399	407
4		इम्फाल IMPHAL (400 kV)	0%	100%	0%	0%	0	417	393	403
5		कामेंग KAMENG (400 kV)	0%	100%	0%	0%	0	410	392	402
6		अज़ारा AZARA (400 kV)	0%	100%	0%	0%	0	410	401	406
7		मीसा MISA (400 kV)	0%	100%	0%	0%	0	412	394	402
8		न्यू मरियानी NEW MARIANI (400 kV)	0%	100%	0%	0%	0	415	396	405
9		न्यू कोहिमा NEW KOHIMA (400 kV)	0%	100%	0%	0%	0	419	396	406
10		पालाटाना PALATANA (400 kV)	0%	100%	0%	0%	0	415	401	406
11		पूर्वकचनबारी PK BARI (400 kV)	0%	100%	0%	0%	0	410	393	401
12		रंगानदी RANGANADI (400 kV)	0%	100%	0%	0%	0	419	388	409
13		सिलचर SILCHAR (400 kV)	0%	100%	0%	0%	0	417	398	407
14		सुर्ज्यामणिनगर SURJYAMANINAGAR (400 kV)	0%	100%	0%	0%	0	411	369	401
15		थौबाल THOUBAL (400 kV)	0%	100%	0%	0%	0	415	384	401

All listed stations are 765 kV stations unless otherwise mentioned

***Percentage is calculated w.r.t. Time of one month.**

23.ALL TIME HIGHEST

30-06-2024

	Maximum Demand Met during the day (MW)	Demand Met during Evening Peak hrs(MW)	Energy Met (MU)	Hydro Gen. (MU)	Wind Gen. (MU)	Solar Gen. (MU)
NR	91215 19-06-2024	82152 17-06-2024	1990 18-06-2024	443 01-08-2023	86 07-08-2023	186 01-05-2024
WR	74974 01-06-2024	67482 30-05-2024	1648 06-05-2024	167 18-12-2014	310 28-05-2024	102.85 30-04-2024
SR	68735 13-03-2024	54798 29-04-2024	1416 05-04-2024	208 31-08-2018	298 21-07-2023	148.7 31-03-2024
ER	32531 10-06-2024	29695 29-05-2024	692 10-06-2024	157 14-09-2022	-	7 23-03-2023
NER	3712 04-09-2023	3544 24-05-2024	72 15-09-2023	43 27-06-2024	-	2 22-06-2022
All India	250070 30-05-2024	227354 29-05-2024	5466 30-05-2024	877 30-08-2022	619 28-05-2024	426 30-04-2024

Regions	States	Max. Demand Met during the day (MW)		Energy Consumption (MU)	
			As on date		As on date
NR	Punjab	15980	29-06-2024	357.2	26-06-2024
	Haryana	14394	19-06-2024	293.4	19-06-2024
	Rajasthan	17949	20-01-2024	379.1	30-05-2024
	Delhi	8568	18-06-2024	177.7	18-06-2024
	UP	30032	13-06-2024	658.8	17-06-2024
	Uttarakhand	2863	14-06-2024	62.1	14-06-2024
	HP	2235	20-01-2024	41.2	26-06-2024
	J&K(UT) and Ladakh(UT)	3636	29-06-2023	66.8	26-01-2024
	Chandigarh	443	13-06-2024	9.1	18-06-2024
	Railways_NR_ISTS	-	-	-	-
WR	Chhattisgarh	6347	05-04-2024	144.4	06-04-2024
	Gujarat	25502	07-06-2024	514.2	07-06-2024
	MP	17819	29-12-2023	335.9	10-11-2023
	Maharashtra	28969	07-02-2024	648.3	30-04-2024
	Goa	803	19-04-2024	17.6	18-05-2024
	DD & DNH	1353	25-06-2024	32.1	05-06-2024
	AMNSIL	1083	10-01-2024	21.0	31-05-2022
	Balco	-	-	-	-
SR	Andhra Pradesh*	13712	04-05-2024	263.8	16-06-2023
	Telangana*	15623	08-03-2024	311.0	13-03-2024
	Karnataka	17220	12-03-2024	336.9	05-04-2024
	Kerala	5797	02-05-2024	116.1	03-05-2024
	Tamil Nadu	20830	02-05-2024	443.6	30-04-2024
	Pondy	530	31-05-2024	11.8	31-05-2024
ER	Bihar	7588	12-06-2024	166.4	12-06-2024
	DVC	3751	18-04-2023	81.2	22-04-2022
	Jharkhand	2324	29-05-2024	50.4	10-06-2024
	Odisha	7104	13-07-2023	148.5	19-04-2024
	West Bengal	12756	30-04-2024	264.0	30-04-2024
	Sikkim	137	03-02-2022	2.5	28-01-2020
	Railways_ER_ISTS	-	-	-	-
NER	Arunachal Pradesh	198	26-05-2024	3.9	18-06-2024
	Assam	2498	16-09-2023	50.0	15-09-2023
	Manipur	258	31-01-2022	3.7	10-02-2022
	Meghalaya	424	29-06-2024	7.8	31-01-2022
	Mizoram	162	27-01-2024	2.6	02-03-2024
	Nagaland	184	07-06-2024	3.29	05-09-2023
	Tripura	386	04-05-2024	7.3	02-06-2024
*SR	Andhra Pradesh (Undivided)	13162	23-03-2014	284.8	22-03-2014

25. System Reliability Indices Report for the month of June 2024

Percentage (%) of times ATC was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	206	51.50	7.15
2	ER-NR	0	0.00	0.00
3	Import of NR	204	51.00	7.08
4	NEW-SR	0	0.00	0.00
5	NER Import	0	0.00	0.00

Percentage(%) of times (N-1) Criteria was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	73	18.25	2.53
2	ER-NR	0	0.00	0.00
3	Import of NR	40	10.00	1.39
4	NEW-SR	0	0.00	0.00
4	NER Import	8	2.00	0.28

Remarks: Flows crossing Total Transfer Capability (TTC) on interregional corridors has been worked out as a proxy for (N-1) violation.



POWER SYSTEM OPERATION CORPORATION LIMITED

पॉवर सिस्टम ऑपरेशन कारपोरेशन लिमिटेड

NATIONAL LOAD DESPATCH CENTRE – राष्ट्रीय भार प्रेषण केन्द्र

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